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Optimizing pulse for gate and decoherence for superconducting qubits

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ABSTRACT

Superconducting cavity–transmon architectures provide a powerful route to quantum information processing by combining long-lived cavity memories with the controllability of nonlinear ancillae. Their performance, however, is limited by two consequences of the same coupling mechanism: high-fidelity cavity control requires accurate dynamics in large Hilbert spaces, while ancilla noise is inherited by the cavity and degrades the bias structure needed for bosonic error correction.

This thesis develops practical methods to address both limitations. In the control setting, I first refvisit GRAPE with hard-coded gradients, eliminating the memory bottleneck of automatic differentiation while preserving runtime scaling and enabling optimal control in large cavity–transmon Hilbert spaces. I then use this framework to optimize ancilla pulses in an echoed conditional displacement protocol. these pulses suppress crosstalk errors by orders of magnitude and produce Bell-cat states with substantially lower infidelity than standard DRAG-based control.

In the noise setting, I show that cavity dephasing induced by 1/f flux noise in a flux-tunable transmon can be strongly suppressed by a weak off-resonant drive that creates driven flux-insensitive points. At these operating points, the drive-induced energy shift opposes the Lamb-shift fluctuations to first order, rendering the cavity transition frequency insensitive to transmon frequency noise. Monte Carlo simulations using realistic parameters predict an approximately twentyfold enhancement of cavity dephasing time, and the protocol extends naturally to bosonic encodings: a single drive simultaneously suppresses flux-noise sensitivity across all relevant transitions of a binomial code and reduces dephasing in both rails of a dual-rail qubit.

Taken together, these results provide a unified strategy for improving both operation fidelity and memory protection in cavity–transmon quantum processors, using hardware-efficient protocols compatible with existing experimental platforms.

ACKNOWLEDGEMENTS

A young scamp about fifteen years old, Isidore Duval by name, and called, for convenience, Zidore, took care of this pensioner, gave him his measure of oats and fodder in winter, and in summer was supposed to change his pasturing place four times a day, so that he might have plenty of fresh grass.

The animal, almost crippled, lifted with difficulty his legs, large at the knees and swollen above the hoofs. His coat, which was no longer curried, looked like white hair, and his long eyelashes gave to his eyes a sad expression.

When Zidore took the animal to pasture, he had to pull on the rope with all his might, because it walked so slowly.

LIST OF PUBLICATIONS

Chapters 3 and 4 are based largely upon work published in

> **Yunwei Lu**, Sandeep Joshi, Vinh San Dinh and Jens Koch, "Optimal control of large quantum systems: assessing memory and runtime performance of GRAPE", *Journal of Physics Communications* **8**, 025002 (2024).

Chapter 5 is based on the publication

> Xinyuan You, **Yunwei Lu**, Taeyoon Kim, Doğa Murat Kürkçüogl̆u, Shaojiang Zhu, David van Zanten, Tanay Roy, Yao Lu, Srivatsan Chakram, Anna Grassellino, Alexander Romanenko, Jens Koch, and Silvia Zorzetti, "Crosstalk-robust quantum control in multimode bosonic systems", *Phys. Rev. Applied* **22**, 044072 (2024).

Additionally, the author contributed to the quantum optimal control part of the following publication:

> Ziwen Huang, **Yunwei Lu**, Anna Grassellino, Alexander Romanenko, Jens Koch, and Shaojiang Zhu, "Completely Positive Map for Noisy Driven Quantum Systems Derived by Keldysh Expansion", *Quantum* **7**, 1158 (2023).

List of Symbols

$\bar{\omega}_b$ Lamb-shifted FTT frequency

$\bar{\omega}_c$ Lamb-shifted cavity frequency

χ dispersive shift

$\delta\omega_b(t)$ FTT-frequency fluctuation

$\delta\Phi(t)$ its fluctuation

Δ_m relevant detuning $\Delta_m \equiv \Delta_{bd} + m\chi$

Δ_{bc} detuning $\Delta_{bc} = \omega_b - \omega_c$

Δ_{bd} FTT–drive detuning

$\Gamma_{\phi,\mathrm{total}}$ total cavity dephasing rate

$\gamma_{\phi,n\to m}$ dephasing rate of cavity transitions induced by FTT-frequency fluctuations

$\gamma_{b\downarrow}$ FTT relaxation at rate $\gamma_{b\downarrow}$ (the FTT energy-relaxation rate)

$\gamma_{b\uparrow}^{1/f,\mathrm{un}}$ transverse-noise-induced excitation rate

$\gamma_{b\uparrow}^{\mathrm{sel}}$ FTT heating at a rate

$\gamma_{b\uparrow}^{\mathrm{un}}$ drive-induced FTT excitation

$\gamma_{c\phi}^{\mathrm{sel}}$ cavity pure dephasing at a rate

$\gamma_{c\phi}^{\mathrm{un}}$ cavity pure dephasing

κ Number of stored Hamiltonian matrix elements

μ Number of matrix-vector multiplications required for propagator-state product evaluation

Ω_0 amplitude of a coherent drive

ω_{ir} infrared cutoff of the $1/f$ noise spectrum

ω_d drive frequency

Φ static flux bias

ρ_d density matrix in the effective drive frame

ε_I, ε_Q Control envelopes in the I and Q quadratures

$\widetilde{\Delta}_{bd}$ shifted FTT detuning $\widetilde{\Delta}_{bd} \equiv \Delta_{bd} - \Omega_0^2/K$

$\widetilde{\gamma}_{b\downarrow}^{\mathrm{un}}$ drive-renormalized FTT relaxation

$\widetilde{\gamma}_{c\downarrow}^{\mathrm{un}}$ renormalized cavity decay

$\widetilde{\omega}_{c,m\to m+1}$ net cavity transition frequency between adjacent cavity levels

b annihilates an excitation in the FTT mode

c annihilates an excitation in the cavity mode

$D_{m\to m+1}$ susceptibility of the net cavity frequency to FTT-frequency fluctuation

g coupling strength

h_c Control operator coupling the field to the system

H_s Static system Hamiltonian

K FTT anharmonicity

K_q Transmon anharmonicity (self-Kerr)

$P_X(t)$ coherence signal

S_0 sets the flux-noise amplitude

$S_{1/f}(\omega)$ power spectral density of $1/f$ flux noise

T_2 dephasing time extracted from an exponential fit to $P_X(t)$

U_n Short-time propagator for time step n

$X(t)$ coherence operator

FTT Flux-Tunable Transmon

GRAPE Gradient Ascent Pulse Engineering

LINC Linear INductive Coupler

QEC quantum error-correction

SNAIL Superconducting Nonlinear Asymmetric Inductive eLement

To my family.

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CHAPTER 1
INTRODUCTION

Quantum computing matters because it addresses computational tasks for which classical methods appear fundamentally inadequate. Feynman argued that the state space of a generic quantum many-body system grows exponentially with system size, so that classical computers cannot efficiently simulate quantum physics in full generality, whereas a controllable quantum device should be able to do so more naturally [1, 2]. To appreciate the scale of this difficulty, note that fully describing the state of just 50 qubits already requires $2^{50} \approx 10^{15}$ complex amplitudes—more than the memory capacity of any existing classical computer. This prospect makes quantum computing especially compelling for quantum simulation. Anticipated applications in quantum simulations fall into three broad areas: (1) chemistry, including molecular electronic structure, ground-state energies, and reaction dynamics relevant to drug design [3, 4]; (2) physics, including strongly correlated materials, high-temperature superconductivity, and lattice gauge theories [5, 6]; and (3) computational biology and medicine, including protein folding, photosynthetic energy-transfer models, and quantum-assisted drug discovery [7, 8]. Beyond simulation, quantum algorithms can also provide rigorous advantages for certain classical problems. Shor's algorithm yields an exponential speedup for integer factorization and discrete logarithms, thereby threatening widely used public-key cryptosystems and motivating the ongoing transition to post-quantum cryptography [9–11]. Meanwhile, Grover's algorithm offers a quadratic query speedup for unstructured search [12, 13]. Recent surveys further identify finance as an important application area for fault-tolerant quantum algorithms, with proposed uses in portfolio optimization, Monte Carlo derivative pricing, and risk analysis [10, 14, 15]. Other directions, including quantum machine learning and quantum optimization, further broaden the field's scope, although their practical advantage remains an active research question [16, 17]. Together, these scientific and computational prospects have driven strong worldwide efforts, as well as substantial public and private investment, to build practical quantum computing platforms [18, 19].

A practical quantum computer must be realized in a physical system of effective two-level degrees of freedom, namely qubits, that can be initialized, coherently controlled, entangled, and measured with high fidelity, while maintaining coherence for times long compared with the duration of gate operations [20, 21]. Several hardware platforms satisfy these requirements to different degrees. Nitrogen-vacancy centers in diamond offer exceptionally long coherence times even at room temperature, but scalable fabrication and controlled interconnects remain challenging [22, 23]. Trapped ions provide some of the best coherence times and gate fidelities demonstrated so far, although gate operations are relatively slow and large-scale integration remains technically demanding [24, 25]. Neutral-atom processors have demonstrated large qubit arrays together with dynamically reconfigurable connectivity, making them attractive for both programmable quantum simulation and gate-based computation [26, 27]. Photonic platforms are especially natural for quantum communication and networking, but deterministic two-qubit gates remain difficult because photons interact only weakly in ordinary optical media [28, 29]. Superconducting circuits, by contrast, combine fast gate operations with scalable fabrication and mature microwave control techniques; crucially, they build on decades of microwave engineering and CMOS-compatible lithography, which makes fabrication of large-scale devices more tractable than for most competing platforms [30, 31]. No single platform has yet simultaneously optimized all of the requirements for large-scale quantum computation, and current hardware development is therefore shaped by trade-offs among coherence, connectivity, control speed, and manufacturability [20, 21]. Among gate-based implementations, superconducting circuits are currently one of the most mature platforms, as reflected both in major public roadmaps from IBM and Google and in recent large-scale error-correction demonstrations on superconducting processors [31–34]. This thesis therefore focuses exclusively on superconducting circuits.

With the platform choice made, the next question is which type of superconducting circuit to use as a qubit. Superconducting qubits encode information in the quantized energy levels of circuits containing Josephson junctions [30, 31]. A Josephson junction acts as a nonlinear inductor: unlike a capacitor or an ordinary inductor, it gives the circuit unevenly spaced energy levels—a property called *anharmonicity*. Anharmonicity is essential for a qubit, because without it a resonant drive meant to flip the qubit between its two lowest levels would also excite higher levels, scrambling the quantum state. The key engineering challenge is then to find a circuit where the energy levels are sufficiently nonlinear to be addressed individually yet stable enough to preserve coherence—these two requirements are in tension, and different qubit designs make different trade-offs.

Over the past two decades, a wide variety of designs has explored this trade-off. Early charge qubits (Cooper-pair boxes) first demonstrated coherent control but were severely limited by charge noise [35, 36]. Flux qubits offered strong anharmonicity through persistent-current superpositions at the cost of sensitivity to magnetic-flux noise [37, 38]. The transmon resolved the charge-noise problem by shunting the Cooper-pair box with a large capacitor, which exponentially suppresses charge sensitivity at the cost of reduced anharmonicity [39, 40]. Other designs seek to recover large anharmonicity through hardware-level noise protection, at the cost of greater fabrication complexity [41–44]. Among these, the transmon has become the workhorse of both academia and industry because it combines conceptual simplicity, reliable fabrication, high-fidelity control, and fast microwave readout in a scalable architecture [31, 45]. In present-day devices, transmons support coherence times from tens to hundreds of microseconds—meaning the quantum state survives long enough for hundreds to thousands of gate operations—and the platform has been scaled from

105-qubit-class processors such as Google's Willow chip to IBM's 1,121-qubit Condor processor [34, 46].

Transmon qubits are not the only useful superconducting degree of freedom. Alongside transmon-based processors, superconducting microwave cavities provide a distinct and complementary paradigm for quantum information processing. A cavity is a linear superconducting resonator—essentially a very high-quality electromagnetic box—that can store microwave photons with coherence times far exceeding those of typical transmon qubits [47, 48]. Early progress in three-dimensional circuit QED established the exceptional coherence [49, 50], and continued improvements in materials, fabrication, and surface treatment pushed 3D cavity photon lifetimes from the millisecond regime to tens of milliseconds and ultimately beyond $2\,\mathrm{s}$ in state-of-the-art niobium resonators at millikelvin temperatures [51–53]. Planar two-dimensional superconducting resonators generally suffer stronger dielectric and surface-interface losses and therefore do not yet match the best 3D cavities [54, 55]; nevertheless, recent on-chip quantum-memory designs have exceeded the millisecond scale [56, 57]. The underlying physical advantage of both 2D and 3D cavities is that, because the cavity mode does not involve a Josephson junction, it is immune to junction-parameter fluctuations such as charge noise and critical-current noise; photons can be lost through radiation and surface dissipation, but the quantum state of a surviving photon is not scrambled. Photon loss is therefore typically the primary error mechanism, while pure dephasing remains comparatively weak—a property known as *noise bias* [58, 59].

Having established why cavities are excellent quantum memories, we now turn to how logical qubits can be encoded within them. A single photon loss in a cavity is a well-defined, detectable event—the photon either escapes or it does not. Bosonic quantum error correction exploits this by encoding a logical qubit not in one physical two-level system but in a carefully chosen superposition of photon-number (Fock) states within a single oscillator mode. The key idea is that by spreading logical information across many Fock states, a single photon-loss event shifts the system into a distinguishable error subspace without destroying the encoded information; by measuring which subspace the system occupies—without measuring the logical information—the error can be detected and corrected [48, 59]. Because the cavity's noise is strongly biased toward photon loss, with dephasing playing a much smaller role, this correction can be highly efficient.

Among the major bosonic code families, cat codes encode information in superpositions of coherent states, for which single-photon loss changes the photon-number parity and appears as a logical bit-flip-type error [60, 61]. Binomial codes instead use carefully engineered finite superpositions of Fock states and can be designed to correct photon loss, photon gain, and dephasing [62, 63]. Gottesman–Kitaev–Preskill (GKP) codes encode a logical qubit in grid states of oscillator phase space and are tailored to correct small displacement errors in both quadratures [64, 65]. A complementary dual-rail encoding stores the logical qubit nonlocally across two cavities; in this case a photon-loss event maps the state outside the code space as an *erasure*-type error—an error whose occurrence is heralded—which is fundamentally easier to correct than an undetected depolarizing error [48, 66]. Experimentally, bosonic QEC in superconducting cavities has reached several milestones: (1) cat-code experiments achieved the first *break-even* demonstration—meaning the error-corrected logical qubit lived at least as long as the best uncorrected physical component—and exhibited extended logical lifetimes as well as exponentially suppressed bit-flip rates as the phase-space separation increased [59, 67–69]; (2) GKP states enabled the first full error-correction demonstrations in circuit QED and subsequently beyond-break-even real-time correction [65, 70, 71]; and (3) binomial codes achieved beyond-break-even real-time correction [59, 72]. This progress relies critically on noise bias: bosonic encodings are most powerful when photon loss remains the dominant cavity error, because then increasing the encoding size can strongly reduce logical failure rates [63, 73].

Storing quantum information in a cavity is only useful if one can also manipulate it—the next challenge is control. A superconducting cavity by itself is a harmonic oscillator with equally spaced energy levels, so a classical microwave drive cannot selectively address individual transitions or generate the nonlinear operations required for universal quantum control [74, 75]. In circuit QED, the standard solution is to couple the cavity to an anharmonic ancilla, typically a transmon, in the *dispersive* regime: the cavity and transmon are detuned far enough in frequency that they do not exchange energy, but the transmon's nonlinearity imprints a photon-number-dependent frequency shift on the ancilla. This shift, known as the dispersive coupling χ, means that the ancilla's transition frequency reveals how many photons are in the cavity, and conversely, a pulse on the ancilla can impart a phase that depends on the cavity's photon number [74, 76]. This interaction enables a rich toolbox for bosonic control.

The selective number-dependent arbitrary phase (SNAP) gate exploits dispersive coupling directly: because each photon-number state shifts the ancilla frequency by a different amount, one can address specific photon-number transitions by frequency and thereby apply independently programmable phases to chosen Fock states. Combined with displacement operations, SNAP gates provide universal single-cavity control [77, 78]. Recently, optimized SNAP-displacement sequences and improved cavity-control protocols have further reduced coherent errors and expanded the accessible family of bosonic states [79, 80]. However, SNAP relies on the ability to resolve individual photon-number transitions by frequency. In a multimode device, other modes shift these frequencies unpredictably (AC Stark shifts), making the spectral addresses unreliable and rendering SNAP increasingly vulnerable to crosstalk [75, 81]. Echoed conditional displacement (ECD) gates address the same problem differently: by combining conditional cavity displacements with ancilla echo pulses, they realize fast universal control even when the bare dispersive coupling is weak. Because the ancilla operation is not photon-number selective, ECD gates are generally less sensitive than SNAP to crosstalk, although not completely immune to it [81, 82]. More recently, conditional-NOT displacement (CNOD) has extended this philosophy to multimode architectures, showing that a single ancilla can efficiently control and entangle several weakly coupled oscillators [48, 83]. For two-cavity operations, dedicated parametric couplers—from SNAIL-based three-wave mixers to linear quantum couplers such as the LINC—provide fast mode-swapping interactions with high on/off ratios and low idle-time backaction on the memories [84–89]. Together, these developments define the modern control toolbox for bosonic circuit QED and have enabled beam-splitter-based logical operations, including dual-rail cavity encodings and error-detectable entangling gates between bosonic qubits [66, 90].

As bosonic encodings grow more powerful, they also grow more demanding to implement, and two challenges become acute.

The first challenge is computational. Large bosonic codes live in Hilbert spaces of hundreds of dimensions, and optimizing the microwave pulse sequences needed to prepare and manipulate logical states requires propagating and differentiating quantum dynamics through this large space. One systematic way to find such pulses is numerical quantum optimal control (QOC), in which experimental constraints—bandwidth limits, leakage suppression, crosstalk avoidance—are incorporated directly into a gradient-based optimization of the pulse shape. Current state-of-the-art QOC frameworks rely on automatic differentiation (AD) to compute the required gradients [91]. While AD is flexible and general-purpose, it works by recording every intermediate operation during the forward propagation of the quantum state and then replaying that record in reverse to accumulate gradients. In large Hilbert spaces this record becomes prohibitively expensive to store, and memory, rather than computation time, becomes the practical bottleneck. This thesis revisits the gradient computation underlying QOC and derives analytically hard-coded gradients that avoid storing the full propagation history, making large-system optimization tractable.

The second challenge is physical. To enable universal cavity control beyond what dispersive coupling alone provides, one can couple the cavity to a flux-tunable nonlinear element—such as a SNAIL or a tunable transmon coupler—that mediates parametric interactions between modes. However, making a circuit element flux-tunable opens a pathway through which magnetic-flux noise in the environment can modulate the coupler's parameters, which in turn dephases the cavity. This inherited dephasing erodes the noise-bias advantage that makes bosonic codes attractive in the first place: if the cavity acquires significant dephasing in addition to photon loss, error correction becomes much less efficient. This thesis introduces a driven sweet-spot protocol in which an off-resonant microwave drive on the transmon renders the cavity frequency first-order insensitive to flux fluctuations, substantially extending the cavity dephasing time without hardware modifications.

This thesis is organized around two complementary contributions. Part I (Chapters 2–5) develops a scalable framework for quantum optimal control of large bosonic systems, deriving hard-coded analytical gradients and benchmarking their performance against automatic-differentiation approaches. Part II (Chapters 6–8) identifies flux-noise-induced cavity dephasing as a limiting factor for cavity-based bosonic codes and introduces a dynamical suppression protocol with direct experimental implications. Both parts address bottlenecks that become acute as bosonic quantum processors move toward larger encodings and multimode architectures. The chapters are organized as follows.

Chapter 2 introduces the bosonic-control toolkit used throughout this thesis, including dispersive cavity–transmon coupling, ECD gates, and beam-splitting interaction enabled by a flux-tunable coupler. Chapter 3 introduces the GRAPE framework and derives hard-coded analytical gradients for key cost contributions used in state-transfer and gate-optimization problems. The chapter also discusses the numerical evaluation of these gradients using the scaling-and-squaring method and the auxiliary matrix method, culminating in a memory-efficient forward-backward propagation scheme. Chapter 4 presents a scaling analysis of runtime and memory usage for hard-coded gradients, full-AD, and semi-AD. Benchmark studies for state-transfer and gate-optimization problems illustrate the scaling behavior, and a decision tree is developed to guide the choice of the optimal numerical strategy. Chapter 5 turns to an application of GRAPE: the design of detuning-robust control pulses for transmon gates. The optimization procedure is detailed, and the resulting pulses are benchmarked against conventional DRAG pulses and pulses from Q-CTRL. Chapter 6 develops the theoretical framework for cavity dephasing inherited from flux noise in a flux-tunable transmon. Starting from the full circuit Hamiltonian, it derives the noise transduction pathway through the Lamb shift. The chapter introduces the driven sweet-spot mechanism, in which an off-resonant drive on the transmon renders the cavity frequency first-order insensitive to flux fluctuations. Analytical sweet-spot conditions are derived and confirmed by Floquet numerics. Chapter 7 analyzes the total cavity dephasing rate under sweet-spot driving by developing an effective Lindblad master equation in the drive frame. The competing dephasing channels are identified and their scalings derived analytically. Full Monte Carlo simulations validate the analytical predictions, and time-domain simulations demonstrate an approximately twentyfold enhancement of the cavity dephasing time in a representative example. Chapter 8 applies the driven sweet-spot protocol to bosonic quantum error correction. It shows that a single drive setting simultaneously suppresses flux-noise sensitivity across all relevant transitions of a binomial code and reduces dephasing in both rails of a dual-rail qubit, with direct improvements demonstrated via Wigner tomography and logical-subspace tomography. Chapter 9 summarizes the key findings of both parts of the thesis and discusses opportunities for future research.

CHAPTER 2

BOSONIC CONTROL WITH ANCILLA AND COUPLER

This chapter develops the control framework for bosonic modes in superconducting circuits. Throughout this chapter, we set $\hbar = 1$, so energies and angular frequencies are used interchangeably unless stated otherwise. The discussion begins with the quantized LC oscillator, which provides the basic description of a cavity mode and clarifies the limitations of purely linear dynamics. It then introduces the transmon as a weakly anharmonic ancilla and shows how its dispersive coupling to the cavity enables conditional control, including ECD gates and arbitrary single-cavity unitaries. Finally, it introduces flux-mediated three-wave mixing, which provides beam-splitter and two-mode-squeezing interactions for entangling multiple bosonic modes.

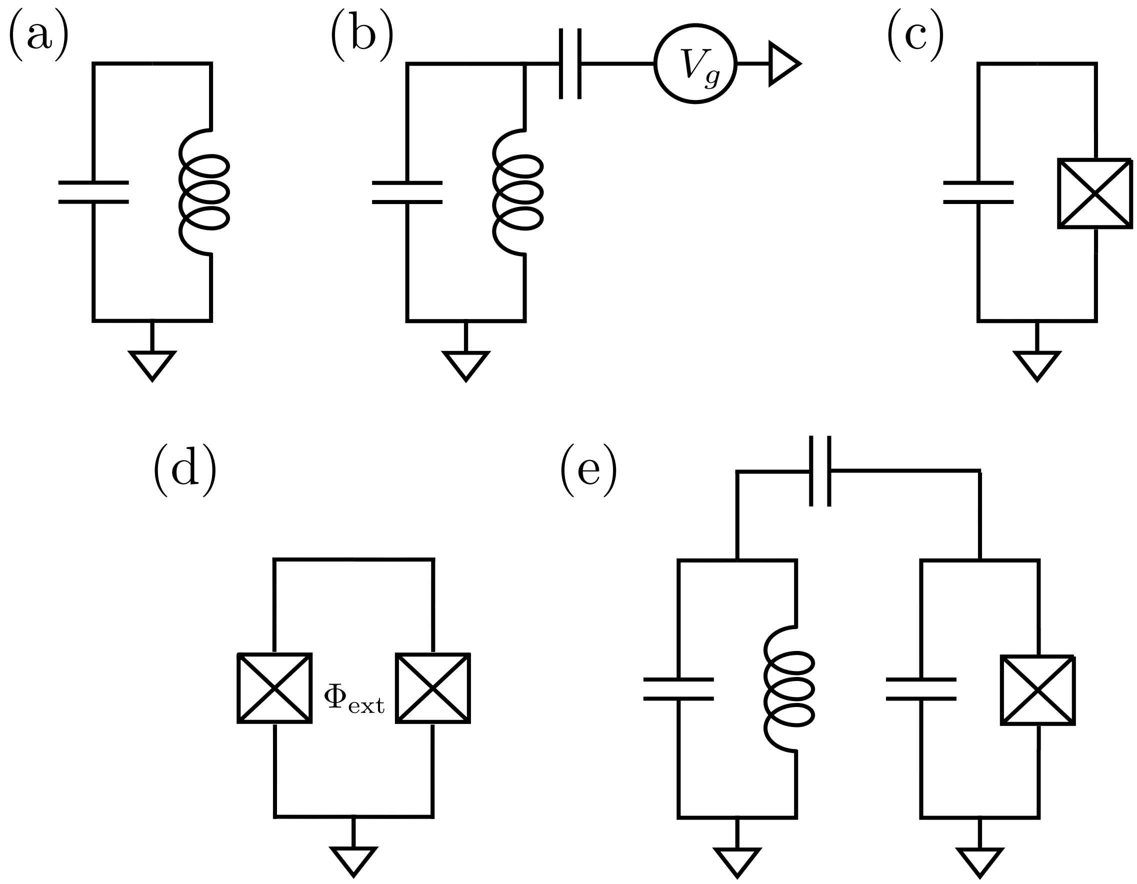

Figure 2.1: Circuit diagrams for the main building blocks of bosonic control in superconducting circuits. (a) A parallel LC oscillator, consisting of an inductor L and a capacitor C, whose quantization yields the harmonic Hamiltonian discussed in Section 2.1. (b) A capacitively driven LC oscillator, where an external voltage source $V_g(t)$ couples to the circuit through a coupling capacitor, producing the linear drive analyzed in Section 2.1. (c) A transmon qubit, formed by shunting a Josephson junction (cross) with a large capacitance C_Σ. The Josephson nonlinearity provides the anharmonic spectrum described in Section 2.2. (d) A SQUID loop, in which two Josephson junctions in parallel allow the effective Josephson energy to be tuned by the external magnetic flux Φ_{ext}, as discussed in Section 2.2. (e) A transmon–cavity composite system, where the transmon is capacitively coupled to a cavity mode. In the dispersive regime this coupling mediates qubit-state-dependent operations on the cavity, forming the basis of the ECD control scheme developed in Section 2.3.

2.1 Oscillator Circuit Hamiltonian

The simplest circuit is the lossless LC oscillator, which serves as the foundational circuit element of superconducting quantum circuits. Its importance is twofold. First, it provides a direct correspondence between circuit variables and canonical variables in quantum mechanics. Second, it illustrates a fundamental limitation: a purely linear oscillator, even when driven by classical fields, cannot generate non-Gaussian states such as Fock states from the vacuum. This limitation motivates the introduction of a nonlinear element in the next section.

2.1.1 Quantization of the LC oscillator

Consider an inductor L and a capacitor C connected in parallel [Fig. 2.1(a)]. The natural generalized coordinate is the node flux

$$\Phi(t) = \int_{-\infty}^{t} V(t')\, dt', \tag{2.1}$$

where $V(t)$ is the node voltage. The conjugate variable is the node charge Q, which is associated with the capacitor branch. In terms of the flux coordinate, the classical Lagrangian is

$$\mathcal{L}(\Phi, \dot{\Phi}) = \frac{C}{2}\dot{\Phi}^2 - \frac{1}{2L}\Phi^2. \tag{2.2}$$

The first term is the electrostatic energy stored in the capacitor, while the second term is the magnetic energy stored in the inductor.

The canonical momentum conjugate to Φ is

$$Q \equiv \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = C\dot{\Phi}. \tag{2.3}$$

This is precisely the capacitor charge. Performing the Legendre transform gives the Hamiltonian

$$H = Q\dot{\Phi} - \mathcal{L} = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}. \tag{2.4}$$

Thus, the LC circuit is formally identical to a mechanical harmonic oscillator, with the correspondence

$$\Phi \leftrightarrow x, \qquad Q \leftrightarrow p, \qquad C \leftrightarrow m, \qquad \frac{1}{L} \leftrightarrow m\omega^2. \tag{2.5}$$

The resonance frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}}. \tag{2.6}$$

Canonical quantization promotes Φ and Q to operators satisfying

$$[\Phi, Q] = i. \tag{2.7}$$

The quantum Hamiltonian is therefore

$$H_0 = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}. \tag{2.8}$$

It is convenient to introduce the characteristic impedance

$$Z = \sqrt{\frac{L}{C}}, \tag{2.9}$$

and the zero-point fluctuations

$$\Phi_{\mathrm{zpf}} = \sqrt{\frac{Z}{2}}, \qquad Q_{\mathrm{zpf}} = \sqrt{\frac{1}{2Z}}. \tag{2.10}$$

The annihilation and creation operators are then defined through

$$\Phi = \Phi_{\mathrm{zpf}}(a + a^{\dagger}), \qquad Q = -iQ_{\mathrm{zpf}}(a - a^{\dagger}), \tag{2.11}$$

with $[a, a^{\dagger}] = 1$. Substituting these expressions into H_0 yields

$$H_0 = \omega_0 \left(a^{\dagger}a + \frac{1}{2} \right), \tag{2.12}$$

which is the standard quantum harmonic oscillator Hamiltonian.

The eigenstates of this Hamiltonian are the Fock states $\{|n\rangle\}_{n=0}^{\infty}$, defined by

$$a^{\dagger}a\,|n\rangle = n\,|n\rangle\,. \tag{2.13}$$

These states have equally spaced energies,

$$E_n = \omega_0 \left(n + \frac{1}{2} \right). \tag{2.14}$$

Because the level spacing is uniform, the linear oscillator does not distinguish one transition from another. This spectral degeneracy is the central obstacle to selective state preparation using only linear control.

2.1.2 Classical voltage drive of a linear oscillator

A classical voltage source couples linearly to a linear LC oscillator [Fig. 2.1(b)]. In the circuit description, a drive voltage $V_d(t)$ applied through a capacitive coupling modifies the electrostatic energy and therefore adds a term linear in the node charge operator Q. After quantization, Q is proportional to a linear combination of a and $a^\dagger$, so the driven oscillator Hamiltonian can be written as

$$H(t) = \omega_0 a^\dagger a + f(t)a + f^*(t)a^\dagger, \tag{2.15}$$

up to a constant energy offset, which shifts all levels equally and can be absorbed into the zero of energy. The precise form of $f(t)$ depends on the drive amplitude and coupling geometry.

The dynamics generated by this Hamiltonian can be treated exactly by moving to a displaced frame. Define

$$\left|\tilde{\psi}(t)\right\rangle = D^\dagger[\alpha(t)]\,|\psi(t)\rangle\,, \qquad D(\alpha) = \exp\bigl(\alpha a^\dagger - \alpha^* a\bigr)\,. \tag{2.16}$$

Substituting this transformation into the Schrödinger equation and requiring that the terms linear in a and $a^\dagger$ cancel yields the equation of motion

$$\dot{\alpha}(t) = -i\omega_0\alpha(t) - if^*(t). \tag{2.17}$$

With this choice of $\alpha(t)$, the transformed Hamiltonian becomes

$$H_{\text{disp}}(t) = \omega_0 a^\dagger a + E(t), \tag{2.18}$$

where

$$E(t) = \omega_0|\alpha(t)|^2 + f(t)\alpha(t) + f^*(t)\alpha^*(t) - \frac{i}{2}\bigl(\alpha^*(t)\dot{\alpha}(t) - \dot{\alpha}^*(t)\alpha(t)\bigr) \tag{2.19}$$

is a real c-number. Being a c-number, it contributes only a global phase and can be ignored. The driven evolution is therefore exactly equivalent to the usual harmonic evolution generated by $\omega_0 a^\dagger a$, which produces phase-space rotation at angular frequency ω_0, together with a displacement set by $\alpha(t)$.

In particular, if the oscillator starts in the vacuum state and $\alpha(0) = 0$, then classical voltage drive maps vacuum only to a coherent state

$$|\psi(t)\rangle = e^{-i\phi(t)} |\alpha(t)\rangle . \tag{2.20}$$

This immediately explains why excited Fock states cannot be prepared from $|0\rangle$ by classical driving alone. In a perfectly harmonic oscillator, every adjacent transition has the same energy ω_0, so a resonant linear drive cannot isolate only the $|0\rangle \leftrightarrow |1\rangle$ transition without also coupling $|1\rangle \leftrightarrow |2\rangle$, $|2\rangle \leftrightarrow |3\rangle$, and so on. As a result, the drive produces phase-space motion rather than deterministic preparation of a specific number state.

This limitation motivates the introduction of anharmonicity. Once the level spacings become nonuniform, individual transitions can be spectrally resolved, and nonclassical states such as Fock states become accessible.

2.2 Transmon as a Weakly Anharmonic System

The linear LC oscillator provides a clean starting point for circuit quantization, but its equally spaced spectrum prevents selective preparation of non-Gaussian states. A superconducting qubit requires a circuit element that breaks exact harmonicity. This role is played by the Josephson junction. When a junction is shunted by a large capacitance, the resulting device—the transmon—forms a weakly anharmonic oscillator whose lowest two levels can be isolated and used as an effective qubit.

2.2.1 Josephson Hamiltonian

A Josephson junction [Fig. 2.1(c)] consists of two superconductors separated by a thin insulating barrier through which Cooper pairs tunnel coherently. Unlike a linear inductor, whose energy is quadratic in current, the tunneling energy depends periodically on the superconducting phase difference across the barrier. This nonlinear phase dependence is the origin of the anharmonicity that distinguishes a Josephson circuit from an LC oscillator. The critical current I_c sets the maximum dissipationless current that the junction can support.

The junction is characterized by the nonlinear current–phase relation

$$I = I_c \sin \varphi, \tag{2.21}$$

and by the Josephson energy

$$E_J = \frac{\Phi_0 I_c}{2\pi}, \tag{2.22}$$

where φ is the superconducting phase difference across the junction and $\Phi_0 = h/2e$ is the flux quantum. The superconducting phase φ and the Cooper-pair number n are conjugate variables, analogous to position and momentum, reflecting the number–phase uncertainty of the condensate. Upon quantization they become operators satisfying

$$[\varphi, n] = i. \tag{2.23}$$

The charging energy is

$$E_C = \frac{e^2}{2C_\Sigma}. \tag{2.24}$$

In general the circuit Hamiltonian includes an offset charge n_g,

$$H = 4E_C(n - n_g)^2 - E_J \cos\varphi. \tag{2.25}$$

Here n_g accounts for stray static charges that shift the effective island charge. In the transmon regime $E_J \gg E_C$, the resulting charge dispersion is exponentially suppressed, so to leading order one may neglect n_g and write

$$H = 4E_C n^2 - E_J \cos\varphi. \tag{2.26}$$

This Hamiltonian already shows the essential difference from the LC oscillator. The charging term is quadratic, but the Josephson potential is cosine rather than quadratic, and therefore introduces the nonlinearity required for anharmonic energy levels. The transmon is the most widely used realization of this circuit, obtained by operating in the regime where the Josephson energy greatly exceeds the charging energy.

2.2.2 Transmon limit and weak anharmonicity

The transmon operates in the regime

$$E_J \gg E_C, \tag{2.27}$$

where phase fluctuations are small and the cosine potential may be expanded around one of its minima, typically $\varphi = 0$:

$$-E_J \cos\varphi \approx -E_J + \frac{E_J}{2}\varphi^2 - \frac{E_J}{24}\varphi^4 + \mathcal{O}(\varphi^6). \tag{2.28}$$

Substituting this into Eq. (2.26) gives

$$H \approx 4E_C n^2 + \frac{E_J}{2}\varphi^2 - \frac{E_J}{24}\varphi^4 - E_J. \tag{2.29}$$

The first two terms define a harmonic oscillator, while the quartic correction produces a weak anharmonicity. To make this explicit, we introduce ladder operators through

$$\varphi = \varphi_{\mathrm{zpf}}(b + b^\dagger), \qquad n = i n_{\mathrm{zpf}}(b^\dagger - b), \tag{2.30}$$

where the zero-point fluctuations are fixed by requiring the quadratic Hamiltonian to take the standard harmonic-oscillator form. This gives

$$\varphi_{\mathrm{zpf}} = \left(\frac{2E_C}{E_J}\right)^{1/4}, \qquad n_{\mathrm{zpf}} = \left(\frac{E_J}{32E_C}\right)^{1/4}. \tag{2.31}$$

The quadratic part then becomes

$$H_{\mathrm{harm}} = \omega_p \left(b^\dagger b + \frac{1}{2}\right) - E_J, \tag{2.32}$$

with plasma frequency

$$\omega_p = \sqrt{8E_J E_C}. \tag{2.33}$$

This is the Josephson analogue of the LC resonance frequency $1/\sqrt{LC}$, with the identification

$$L_J = \frac{(\Phi_0/2\pi)^2}{E_J}, \qquad C = C_\Sigma. \tag{2.34}$$

The Josephson inductance L_J is the effective inductance obtained by linearizing the junction about $\varphi = 0$; unlike a geometric inductance, it is set by the junction parameters and can be tuned in flux-controlled circuits.

2.2.3 Flux-tunable transmon

Replacing a single Josephson junction by a SQUID loop [Fig. 2.1(d)] makes the effective Josephson energy tunable by external magnetic flux. A SQUID consists of two Josephson junctions connected in parallel around a superconducting loop. An external flux Φ_{ext} threading the loop imposes a phase bias between the two junctions, so the total Josephson tunneling amplitude is the coherent sum of two paths. As a result, the effective Josephson energy varies periodically with flux through constructive or destructive interference.

For junction energies E_{J1} and E_{J2}, the SQUID behaves as a single junction with flux-dependent effective Josephson energy

$$E_J(\Phi_{\text{ext}}) = E_{J\Sigma}\sqrt{\cos^2\left(\pi\frac{\Phi_{\text{ext}}}{\Phi_0}\right) + d^2\sin^2\left(\pi\frac{\Phi_{\text{ext}}}{\Phi_0}\right)}, \tag{2.35}$$

where

$$E_{J\Sigma} = E_{J1} + E_{J2}, \qquad d = \frac{E_{J1} - E_{J2}}{E_{J1} + E_{J2}}. \tag{2.36}$$

The parameter d measures the junction asymmetry. For $d = 0$, the SQUID is symmetric and the tunability is maximal. For $d > 0$, the effective Josephson energy no longer vanishes at half a flux quantum, so the tuning range is reduced but the spectrum retains a nonzero minimum frequency.

The transmon transition frequency therefore acquires flux dependence,

$$\omega_b(\Phi_{\text{ext}}) \approx \sqrt{8E_C E_J(\Phi_{\text{ext}})} - E_C. \tag{2.37}$$

At $\Phi_{\mathrm{ext}} = 0$, the Josephson energy is maximal and has zero slope with respect to flux, so the qubit frequency is first-order insensitive to small flux fluctuations. This operating point is the flux sweet spot and minimizes dephasing from low-frequency flux noise. Away from the sweet spot, the qubit becomes increasingly flux sensitive, which enables tunability but also increases susceptibility to noise.

In practice, flux tuning allows the qubit frequency to be varied over a range of several GHz, while the anharmonicity remains approximately $\alpha \approx -E_C$ because E_C is set primarily by the circuit capacitance and is largely independent of flux. External flux therefore provides direct control of the transmon spectrum: it can be used to bring the qubit into and out of resonance with other circuit elements, adjust detunings, and selectively activate interactions that are otherwise off resonant.

This flux tunability is also the basis of parametric control. By modulating Φ_{ext} in time, one can generate sideband interactions between the transmon and coupled cavity modes, including beam-splitter, as discussed in Section **??**.

2.3 ECD Scheme for Arbitrary Single-Cavity Unitary

Section 2.1 showed that a single linear cavity driven only by classical microwave fields remains confined to Gaussian dynamics. In particular, starting from vacuum, such control cannot produce Fock states or implement general non-Gaussian unitaries on the cavity mode. The transmon introduced in Section 2.2 provides the missing nonlinearity. When the cavity is dispersively coupled to the transmon, the qubit can mediate conditional phase-space operations on the cavity, and these operations promote the cavity from a Gaussian system to a controllable bosonic mode. To formulate this control scheme correctly, however, one should begin from the full weakly anharmonic transmon ladder and only then reduce to an effective two-level ancilla.

2.3.1 Dispersive cavity–transmon coupling

The transmon is not fundamentally a two-level system, but a weakly anharmonic oscillator with eigenstates $\{|j\rangle\}_{j=0}^{\infty}$, obtained by diagonalizing the Josephson Hamiltonian

$$H_{\mathrm{T}} = 4E_C n^2 - E_J \cos\varphi. \tag{2.38}$$

As discussed in Section 2.2, its spectrum forms a ladder of levels with transition frequencies

$$\omega_{j+1,j} = E_{j+1} - E_j \approx \omega_{01} + j\alpha, \tag{2.39}$$

where $\alpha < 0$ is the transmon anharmonicity. Because $|\alpha|$ is small compared with ω_{01}, the transmon remains close to a harmonic oscillator, but its level spacings are not exactly equal. This weak anharmonicity is sufficient to isolate the lowest transition under appropriate control conditions.

Now couple the transmon capacitively to a single cavity mode of frequency ω_c, described by the annihilation operator a [Fig. 2.1(e)]. In the transmon eigenbasis, the uncoupled Hamiltonian is

$$H_0 = \omega_c a^\dagger a + \sum_{j=0}^{\infty} \omega_j |j\rangle \langle j| , \tag{2.40}$$

where ω_j are the transmon eigenenergies. The capacitive interaction arises through the transmon charge operator n, whose dominant matrix elements connect neighboring transmon levels. Quantizing of the circuit Fig. 2.1(e) [92], the coupled cavity–transmon system is therefore described by the generalized Jaynes–Cummings Hamiltonian

$$H = \omega_c a^\dagger a + \sum_{j=0}^{\infty} \omega_j |j\rangle \langle j| + \sum_{j=0}^{\infty} g_j \left(a^\dagger |j\rangle \langle j+1| + a |j+1\rangle \langle j| \right), \tag{2.41}$$

where g_j is the cavity coupling matrix element for the $|j\rangle \leftrightarrow |j+1\rangle$ transition. In the transmon regime, these couplings approximately scale as

$$g_j \approx \sqrt{j+1}\, g_0, \tag{2.42}$$

reflecting the near-oscillator character of the device.

In the dispersive regime, define

$$\Delta_j = \omega_{j+1,j} - \omega_c \tag{2.43}$$

as the signed detuning of the $|j\rangle \leftrightarrow |j+1\rangle$ transition from the cavity. The dispersive condition is then

$$|\Delta_j| \gg g_j\sqrt{n+1}, \tag{2.44}$$

for all significantly occupied photon numbers n. In this limit, real exchange of excitations is suppressed, and the interaction can be treated perturbatively. To second order, the energy shift of the bare state $|j,n\rangle$ is

$$\delta E^{(2)}_{j,n} = \left[\frac{(n+1)g^2_{j-1}}{\Delta_{j-1}} - \frac{ng^2_j}{\Delta_j}\right], \tag{2.45}$$

with the convention $g_{-1} = 0$. Separating the photon-number-independent part as a level shift and the photon-number-dependent part as a cavity pull,

$$\delta E^{(2)}_{j,n} = \Lambda_j + n\,\chi_j, \tag{2.46}$$

where

$$\Lambda_j = \frac{g^2_{j-1}}{\Delta_{j-1}}, \qquad \chi_j = \frac{g^2_{j-1}}{\Delta_{j-1}} - \frac{g^2_j}{\Delta_j}, \tag{2.47}$$

one obtains the multilevel dispersive Hamiltonian

$$H_{\mathrm{disp}}^{(\mathrm{multi})} = \omega_c a^\dagger a + \sum_{j=0}^{\infty} (\omega_j + \Lambda_j) |j\rangle \langle j| + a^\dagger a \sum_{j=0}^{\infty} \chi_j |j\rangle \langle j|, \tag{2.48}$$

where Λ_j is the Lamb shift of transmon level j induced by its virtual coupling to the cavity. Equation (2.48) shows that each transmon level pulls the cavity by a different amount.

At this stage the ancilla is still a multilevel system. The reduction to an effective qubit is an approximation that must be justified. In the present setting, that approximation is valid for three reasons. First, the control pulses are designed to act resonantly on the $|g\rangle \equiv |0\rangle$ to $|e\rangle \equiv |1\rangle$ transition, while the next transition is detuned by the anharmonicity $|\alpha|$. Second, the qubit-drive Rabi rate and spectral bandwidth are chosen small enough that off-resonant excitation of $|f\rangle \equiv |2\rangle$ and higher states remains negligible. Third, the cavity population is kept below the critical photon number associated with the dispersive approximation, so that the cavity does not strongly hybridize with higher transmon levels. Under these conditions, population leakage out of $\{|g\rangle, |e\rangle\}$ is weak, and the transmon can be treated as an effective two-level ancilla for gate synthesis.

Projecting Eq. (2.48) onto the lowest two transmon levels gives

$$H_{\mathrm{disp}}^{(ge)} = \omega_c a^\dagger a + (\omega_g + \Lambda_g) |g\rangle \langle g| + (\omega_e + \Lambda_e) |e\rangle \langle e| + a^\dagger a (\chi_g |g\rangle \langle g| + \chi_e |e\rangle \langle e|). \tag{2.49}$$

Introducing

$$\sigma_z = |e\rangle \langle e| - |g\rangle \langle g|, \tag{2.50}$$

the dressed qubit frequency

$$\tilde{\omega}_q = (\omega_e + \Lambda_e) - (\omega_g + \Lambda_g), \tag{2.51}$$

the mean cavity pull

$$\bar{\chi} = \frac{\chi_e + \chi_g}{2}, \tag{2.52}$$

and the differential cavity pull

$$\chi = \chi_e - \chi_g, \tag{2.53}$$

one obtains

$$H_{\text{disp}}^{(ge)} = (\omega_c + \bar{\chi})a^\dagger a + \frac{\tilde{\omega}_q}{2}\sigma_z + \frac{\chi}{2}a^\dagger a\,\sigma_z. \tag{2.54}$$

The first term renormalizes the cavity frequency and can be absorbed into a redefinition of ω_c. The last term is the effective dispersive interaction used throughout the ECD protocol.

For a weakly anharmonic transmon, Eq. (2.54) reduces to the familiar transmon dispersive shift. Since $g_{-1} = 0$, one has

$$\chi_g = -\frac{g_0^2}{\Delta_0}. \tag{2.55}$$

For the excited state,

$$\chi_e = \frac{g_0^2}{\Delta_0} - \frac{g_1^2}{\Delta_1}. \tag{2.56}$$

Therefore,

$$\chi = \chi_e - \chi_g = \frac{2g_0^2}{\Delta_0} - \frac{g_1^2}{\Delta_1}. \tag{2.57}$$

Using $g_1 \approx \sqrt{2}\,g_0$ and $\Delta_1 = \Delta_0 + \alpha$, this becomes

$$\chi \approx 2g_0^2\left(\frac{1}{\Delta_0} - \frac{1}{\Delta_0 + \alpha}\right) = \frac{2g_0^2\alpha}{\Delta_0(\Delta_0 + \alpha)}. \tag{2.58}$$

In typical circuit-QED devices, $\chi/2\pi$ ranges from tens of kHz to several MHz, depending on the detuning Δ_0 and coupling g_0. Thus, the standard qubit–cavity dispersive Hamiltonian is not imposed by assumption, but emerges as the low-energy reduction of the full multilevel cavity–

transmon system.

The classical cavity and qubit drives used for bosonic control are then added as

$$H_{\text{ctrl}} = \epsilon^*(t)a + \epsilon(t)a^\dagger + \Omega^*(t)\sigma_- + \Omega(t)\sigma_+, \tag{2.59}$$

where $\epsilon(t)$ is the cavity-drive envelope and $\Omega(t)$ is the full complex qubit Rabi rate. Moving to a frame rotating at the dressed cavity frequency $\omega_c + \bar{\chi}$ and the dressed qubit frequency $\tilde{\omega}_q$, the free-evolution terms are removed and the controlled Hamiltonian becomes

$$H = \frac{\chi}{2}a^\dagger a\,\sigma_z + \epsilon^*(t)a + \epsilon(t)a^\dagger + \Omega^*(t)\sigma_- + \Omega(t)\sigma_+. \tag{2.60}$$

2.3.2 Echoed conditional displacement

The basic idea of ECD [70, 82] is to use the dispersive interaction to convert an ordinary cavity drive into a qubit-state-dependent displacement. Introduce a time-dependent displacement operator

$$D[\alpha(t)] = \exp\left[\alpha(t)a^\dagger - \alpha^*(t)a\right], \tag{2.61}$$

and choose $\alpha(t)$ so that the unconditional cavity drive is absorbed into the displaced frame. Using

$$D^\dagger(\alpha)aD(\alpha) = a + \alpha, \tag{2.62}$$

the dispersive term transforms as

$$D^\dagger(\alpha)\,a^\dagger a\,D(\alpha) = a^\dagger a + \alpha a^\dagger + \alpha^* a + |\alpha|^2. \tag{2.63}$$

The displaced-frame Hamiltonian becomes

$$\tilde{H} = \frac{\chi}{2} a^\dagger a \, \sigma_z + \frac{\chi}{2} \left(\alpha(t) a^\dagger + \alpha^*(t) a \right) \sigma_z + \frac{\chi}{2} |\alpha(t)|^2 \sigma_z + \frac{1}{2} \left[\Omega^*(t) \sigma_- + \Omega(t) \sigma_+ \right]. \tag{2.64}$$

Equation (2.64) contains three distinct qubit-dependent terms. The first term,

$$\frac{\chi}{2} a^\dagger a \, \sigma_z, \tag{2.65}$$

is an unwanted conditional phase rotation of the cavity. The third term,

$$\frac{\chi}{2} |\alpha(t)|^2 \sigma_z, \tag{2.66}$$

is an unwanted qubit Z rotation. The middle term,

$$\frac{\chi}{2} \left(\alpha(t) a^\dagger + \alpha^*(t) a \right) \sigma_z, \tag{2.67}$$

is the desired conditional linear drive on the cavity.

An echo sequence composed of two displacement segments separated by a qubit π pulse cancels the unwanted dispersive phase rotation and the qubit Z rotation while preserving the conditional linear drive. The resulting unitary is the echoed conditional displacement gate,

$$\mathrm{ECD}(\beta) = \exp\left[\frac{1}{2} \left(\beta a^\dagger - \beta^* a \right) \sigma_z \right], \tag{2.68}$$

where the complex amplitude β is determined by the integrated drive envelope. This gate displaces the cavity in opposite directions depending on the state of the ancilla and is therefore the basic non-Gaussian entangling primitive used for bosonic control.

2.3.3 Universal gate set for the joint qubit–cavity system

With the ECD gate defined in the previous subsection, we now show that alternating ECD operations and single-qubit rotations form a universal gate set for the joint qubit–cavity system. The ECD gate is combined with arbitrary single-qubit rotations

$$R(\theta, \phi) = \exp\left[-i\frac{\theta}{2}\left(\cos\phi\,\sigma_x + \sin\phi\,\sigma_y\right)\right]. \tag{2.69}$$

A generic control sequence then has the form

$$U_N = R(\theta_N, \phi_N)\,\mathrm{ECD}(\beta_N)\cdots R(\theta_2, \phi_2)\,\mathrm{ECD}(\beta_2)\,R(\theta_1, \phi_1)\,\mathrm{ECD}(\beta_1). \tag{2.70}$$

To make the universality claim precise, it is useful to work at the Hamiltonian level. Universal control of the oscillator alone means the ability to synthesize unitaries generated by Hamiltonians that are polynomial in the quadratures [82, 93]

$$q = \frac{1}{\sqrt{2}}(a^\dagger + a), \qquad p = \frac{i}{\sqrt{2}}(a^\dagger - a). \tag{2.71}$$

For the joint qubit–cavity system, the natural extension is control over Hamiltonians of the form

$$q^j p^k \sigma_i, \qquad j, k \in \mathbb{N}_0, \qquad \sigma_i \in \{I, \sigma_x, \sigma_y, \sigma_z\}. \tag{2.72}$$

Throughout this subsection, monomials $q^j p^k$ are written with all factors of q placed to the left of all factors of p. Since $[q, p] = i$, alternative orderings differ only by lower-order terms, so this convention does not affect the Lie-algebra argument. On a finite-dimensional truncation of the cavity Hilbert space, these operators span the full operator algebra of the joint system.

The key control-theoretic fact is that, on a finite-dimensional Hilbert space, a set of Hamiltonians is universal if the Lie algebra generated by the corresponding skew-Hermitian operators spans the full matrix algebra up to an overall phase. It therefore suffices to show that sums and commutators of the available generators produce all polynomial operators on the truncated joint space.

The role of the available gates is then straightforward. Arbitrary single-qubit rotations provide the generators σ_x and σ_y. Infinitesimal ECD pulses generate conditional displacements along arbitrary directions in phase space, and therefore provide Hamiltonians proportional to $q\sigma_z$ and $p\sigma_z$. The basic set of accessible generators is thus

$$\mathfrak{g}_0 = \{q\sigma_z,\ p\sigma_z,\ \sigma_x,\ \sigma_y\}. \tag{2.73}$$

The key point is that sums and commutators of available generators can themselves be synthesized. For sufficiently small δt,

$$e^{-iA\delta t}e^{-iB\delta t}e^{iA\delta t}e^{iB\delta t} = e^{-[A,B]\delta t^2 + O(\delta t^3)}, \tag{2.74}$$

and

$$e^{-iA\delta t/2}e^{-iB\delta t}e^{-iA\delta t/2} = e^{-i(A+B)\delta t + O(\delta t^3)}. \tag{2.75}$$

Thus, once two Hamiltonians A and B are available, one can generate both their sum and, up to a sign, their commutator to leading order; the opposite sign is obtained by reversing the pulse ordering. Repeated application of these identities closes the Lie algebra generated by $\mathfrak{g}_0$.

Starting from $\mathfrak{g}_0$, one immediately obtains the full set of linear qubit–oscillator generators. For example,

$$[q\sigma_z, \sigma_x] = 2i\, q\sigma_y, \qquad [q\sigma_z, \sigma_y] = -2i\, q\sigma_x, \tag{2.76}$$

and similarly

$$[p\sigma_z, \sigma_x] = 2i\, p\sigma_y, \qquad [p\sigma_z, \sigma_y] = -2i\, p\sigma_x. \tag{2.77}$$

Together with

$$[\sigma_x, \sigma_y] = 2i\, \sigma_z, \tag{2.78}$$

this shows that the Lie closure of the control set contains

$$\{\sigma_i,\ q\sigma_i,\ p\sigma_i\}, \qquad i \in \{x, y, z\}. \tag{2.79}$$

Quadratic generators follow from commutators of these linear terms. For instance,

$$[q\sigma_x, q\sigma_y] = q^2[\sigma_x, \sigma_y] = 2i\, q^2\sigma_z, \tag{2.80}$$

and likewise

$$[p\sigma_x, p\sigma_y] = 2i\, p^2\sigma_z. \tag{2.81}$$

Mixed quadratic terms are generated in the same way, for example through commutators such as

$$[q\sigma_x, p\sigma_y] = i(qp + pq)\sigma_z. \tag{2.82}$$

Since $[q, p] = i$, differences between symmetrized and ordered products only produce lower-order terms, which are already contained in the generated algebra. Hence all quadratic polynomial generators can be constructed.

This argument can then be iterated. Once polynomial generators of degree d are available, commutators with $q\sigma_i$ or $p\sigma_i$ generate degree-$(d+1)$ terms, again up to lower-order contributions fixed by $[q,p]=i$. By induction, the generated Lie algebra contains all operators of the form

$$q^j p^k \sigma_i, \qquad \sigma_i \in \{\sigma_x, \sigma_y, \sigma_z\}. \tag{2.83}$$

Terms without a Pauli operator are obtained as well. For example,

$$[q^{j+1}p^k\sigma_z,\ p\sigma_z] = [q^{j+1}p^k,\ p] = i(j+1)q^j p^k, \tag{2.84}$$

so polynomial operators proportional to the identity on the qubit are also generated.

Consequently, the Lie algebra generated by

$$\mathcal{G} = \{\mathrm{ECD}(\beta),\ R(\theta,\phi)\} \tag{2.85}$$

contains all polynomial operators on the joint qubit–cavity system. On any finite-dimensional truncation of the cavity Hilbert space, these operators span the full matrix algebra, and the corresponding unitaries can therefore be approximated arbitrarily well by sequences of ECD gates and single-qubit rotations. In this sense, $\mathcal{G}$ forms a universal gate set for the joint cavity–qubit Hilbert space.

This universality is stronger than the Gaussian controllability discussed in Section 2.1. There, the accessible Hamiltonians generate only affine transformations of phase space, meaning displacements and linear transformations of the quadratures that preserve the Gaussian character of any input state. Here, the qubit-mediated ECD interaction enlarges the Lie algebra to include nonlinear polynomial operators, which is precisely what enables universal control of the truncated joint system.

2.3.4 Synthesis of arbitrary cavity unitaries

To implement a target cavity unitary U_{cav}, one works in a truncated Fock space

$$\mathcal{H}_d = \text{span}\{|0\rangle, |1\rangle, \ldots, |d-1\rangle\}, \tag{2.86}$$

with d chosen so that Fock-state amplitudes outside $\mathcal{H}_d$ remain negligible throughout the protocol. In practice, d is taken comfortably above the largest photon number with appreciable occupation; for a code space with mean photon number $\bar{n}$, a common rule of thumb is $d \gtrsim 2\bar{n}$, with a larger cutoff required when the control sequence transiently populates higher Fock levels.

The control sequence in Eq. (2.70) is then optimized so that

$$U_N\big(|\psi\rangle \otimes |g\rangle\big) \approx \big(U_{\text{cav}}\,|\psi\rangle\big) \otimes |g\rangle, \qquad \forall\ |\psi\rangle \in \mathcal{H}_d. \tag{2.87}$$

Equation (2.87) expresses two requirements simultaneously: the cavity undergoes the desired unitary transformation for every input state in the truncated space, and the qubit is returned to its initial state so that no residual qubit–cavity entanglement remains.

A convenient way to express the optimization target is through the projected cavity map

$$M_N = (\mathbb{I}_d \otimes \langle g|)U_N(\mathbb{I}_d \otimes |g\rangle), \tag{2.88}$$

which describes the effective action of the sequence on the cavity when the qubit begins and ends in $|g\rangle$. The synthesis condition in Eq. (2.87) is satisfied if and only if $M_N = U_{\text{cav}}$ up to a global phase, which motivates minimizing the following infidelity

$$C_{\text{gate}} = 1 - \frac{\left|\operatorname{Tr}\left(U_{\text{cav}}^{\dagger} M_N\right)\right|^2}{d^2}. \tag{2.89}$$

When only a state-preparation task is required, one may instead minimize the state infidelity

$$C_{\text{state}} = 1 - \left|\langle\psi_{\text{tar}}| M_N |\psi_{\text{in}}\rangle\right|^2. \tag{2.90}$$

In practice, the parameters

$$\{\beta_k, \theta_k, \phi_k\}_{k=1}^{N} \tag{2.91}$$

are found by numerical minimization of the fidelity over the truncated Hilbert space. Because the gate set is universal, increasing the circuit depth N allows one, in principle, to approximate any desired single-cavity unitary to arbitrary accuracy on $\mathcal{H}_d$.

This synthesis strategy provides a concrete route from a small set of experimentally natural primitives to general bosonic control. Fock-state preparation, cat-state generation, logical-state encoding, and more general non-Gaussian cavity gates all arise as special cases of the same framework. The essential point is that the cavity is not controlled directly through a native cavity nonlinearity; rather, the nonlinearity is borrowed from the transmon and converted, through the dispersive interaction and echoed conditional displacements, into programmable phase-space control.

The remainder of this chapter uses this viewpoint throughout. Once the cavity–qubit system is equipped with the gate set $\{\mathrm{ECD}(\beta), R(\theta,\phi)\}$, single-mode bosonic control reduces to the problem of compiling the desired operation into an experimentally feasible sequence of echoed conditional displacements and qubit rotations.

2.4 Two-mode entanglement

The previous section showed that the single-cavity ECD scheme provides universal control of a bosonic mode. For the two-mode problem, there are likewise two complementary routes. One can extend ECD to two cavity modes and obtain universal control in the two-mode Hilbert space. Alternatively, in the cavity–flux-tunable-transmon–cavity architecture considered here, one can use flux modulation of the intermediate nonlinear element to activate an effective interaction between the two cavities. We discuss these two approaches in turn.

2.4.1 Two-mode ECD as a universal-control approach

The ECD framework extends naturally from a single bosonic mode to two bosonic modes. In that setting, conditional displacements and qubit-assisted control operations can be used to realize universal control over the two-mode Hilbert space and, in particular, to prepare and manipulate entangled cavity states. This provides a conceptually direct generalization of the single-mode protocol discussed in the previous section. A detailed treatment of the two-mode ECD protocol is given in Ref. [81].

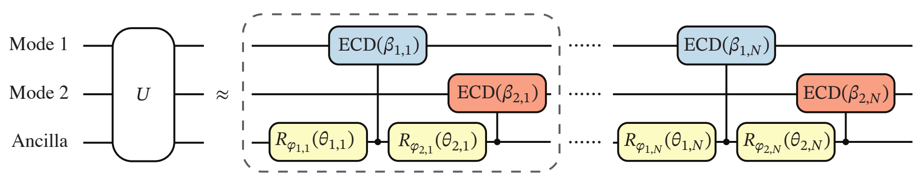

Figure 2.2: Decomposition of an arbitrary two-mode control sequence into modular blocks. Each block (enclosed by dashed lines) consists of a pair of ECD gates acting on the two cavity modes, interleaved with ancilla rotations. Reproduced from Ref. [81].

In the remainder of this section, however, we focus on a different route that is more closely tied to the present hardware: parametrically activated control in a cavity–flux-tunable-transmon–cavity system.

2.4.2 Parametrically activated two-mode interaction in a cavity–FTT–cavity system

To generate a coherent interaction between two cavity modes, the relevant hardware is not a directly coupled cavity pair, but a cavity–coupler–cavity system. Here we use a flux-tunable transmon (FTT) as the coupler. The two cavity modes do not couple directly to each other. Instead, each cavity couples to the same FTT, which mediates the effective interaction.

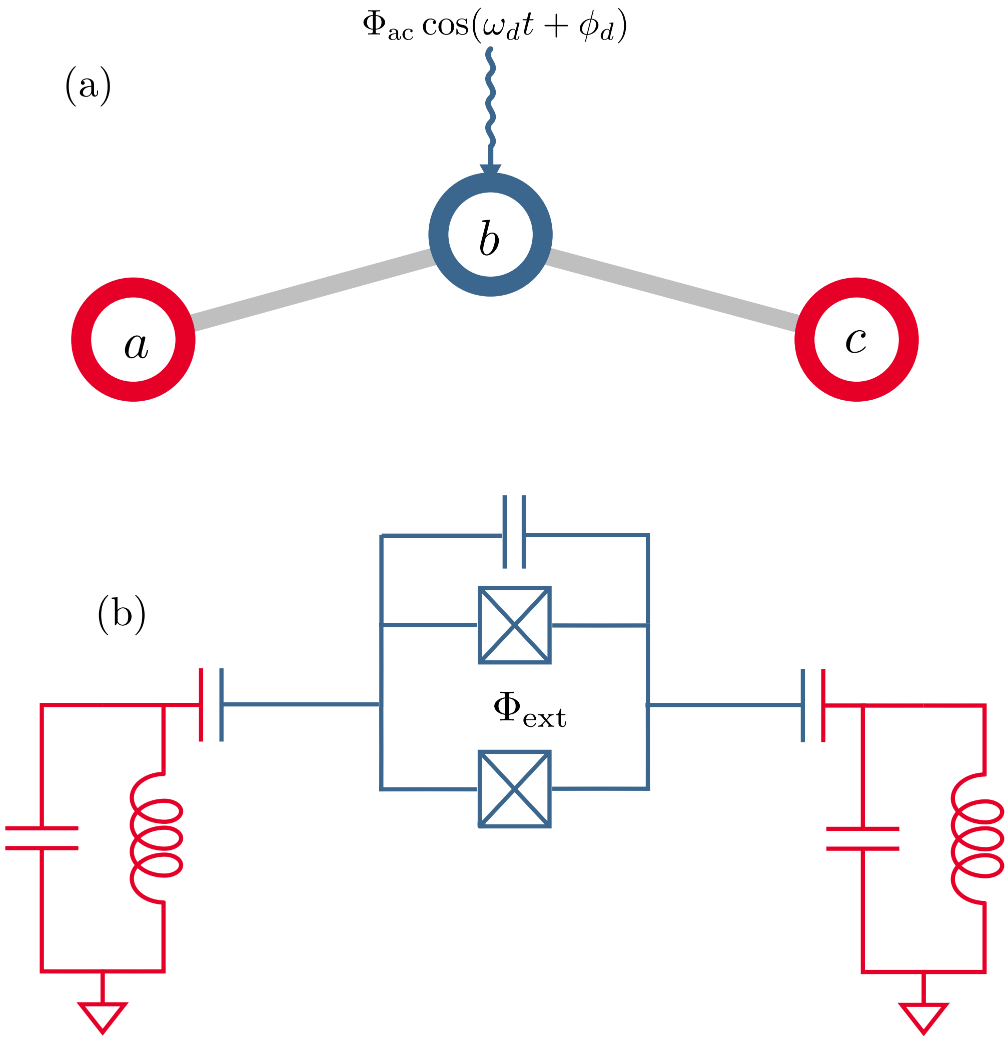

Figure 2.3: Parametric-drive model and circuit implementation. (a) Graph representation of the three-mode system, where the bare modes are coupled capacitively (light-gray edges) and mode b is a flux-tunable transmon whose frequency is modulated by a parametric flux drive. (b) Superconducting-circuit implementation of the model, in which modes a and c are cavity modes and mode b is realized as a flux-tunable transmon coupler.

2.4.2 *Starting Hamiltonian and flux-dependent transmon frequency*

We consider an FTT capacitively coupled to two cavities, as shown in Fig. 2.3. Let a and c denote the two linear cavity modes, and let b denote the FTT mode. The circuit Hamiltonian is

$$H = H_{\mathrm{lin}}(t) + H_{\mathrm{nl}}, \tag{2.92}$$

with

$$H_{\mathrm{lin}}(t) = \omega_a a^\dagger a + \omega_c c^\dagger c + \omega_b(\Phi_{\mathrm{ext}}(t))\, b^\dagger b + g_{ab}(a^\dagger b + ab^\dagger) + g_{bc}(c^\dagger b + cb^\dagger), \tag{2.93}$$

and

$$H_{\mathrm{nl}} = -\frac{K_b}{2} b^{\dagger 2} b^2. \tag{2.94}$$

Here the coupling terms have already been written in the rotating-wave approximation, so the linear part consists of the bare mode energies together with Jaynes–Cummings interactions between each cavity and the FTT. Importantly, there is no direct term of the form $g_{ac}(a^\dagger c + ac^\dagger)$.

The external flux is taken to be

$$\Phi_{\mathrm{ext}}(t) = \Phi_{\mathrm{dc}} + \Phi_{\mathrm{ac}} \cos(\omega_d t + \phi_d), \tag{2.95}$$

and the FTT frequency is

$$\omega_b(\Phi_{\mathrm{ext}}) \approx \sqrt{8 E_C E_J(\Phi_{\mathrm{ext}})} - E_C. \tag{2.96}$$

Expanding around the dc bias point gives

$$\omega_b(\Phi_{\mathrm{ext}}(t)) \approx \bar{\omega}_b + \delta\omega_b \cos(\omega_d t + \phi_d), \tag{2.97}$$

where

$$\bar{\omega}_b = \omega_b(\Phi_{\mathrm{dc}}), \qquad \delta\omega_b = \left.\frac{\partial \omega_b}{\partial \Phi}\right|_{\Phi_{\mathrm{dc}}} \Phi_{\mathrm{ac}}. \tag{2.98}$$

Thus, an AC flux drive produces an AC modulation of the FTT frequency.

2.4.2 *Black-box quantization of the static quadratic sector*

The black-box quantization strategy treats the linear circuit as a passive network whose normal modes are found first, while the nonlinear elements are then added perturbatively on top of these dressed modes [94]. In the present setting, it is therefore convenient to separate the Hamiltonian into a static quadratic part and a perturbation,

$$H = H^{(0)} + H^{(1)}(t), \tag{2.99}$$

with

$$H^{(0)} = \omega_a a^\dagger a + \omega_c c^\dagger c + \bar{\omega}_b b^\dagger b + g_{ab}(a^\dagger b + a b^\dagger) + g_{bc}(c^\dagger b + c b^\dagger), \tag{2.100}$$

and

$$H^{(1)}(t) = -\frac{K_b}{2} b^{\dagger 2} b^2 + \delta\omega_b \cos(\omega_d t + \phi_d)\, b^\dagger b. \tag{2.101}$$

Diagonalizing only the static sector is advantageous because the result is a time-independent basis in which the drive and nonlinearity can be treated perturbatively; the drive frequency ω_d then selects which normal-mode transition is activated.

Writing the normal-mode transformation as

$$\mu = \sum_{\nu=a,b,c} u_{\mu\nu}\, d_\nu^{(\mathrm{N})}, \qquad \mu = a, b, c, \tag{2.102}$$

where $d_\nu^{(\mathrm{N})}$ is the annihilation operator for normal mode ν, one obtains

$$H^{(0)} = \tilde{\omega}_a a_{\mathrm{N}}^\dagger a_{\mathrm{N}} + \tilde{\omega}_b b_{\mathrm{N}}^\dagger b_{\mathrm{N}} + \tilde{\omega}_c c_{\mathrm{N}}^\dagger c_{\mathrm{N}}. \tag{2.103}$$

Here $u_{\mu\nu}$ is an element of the inverse diagonalization matrix, expressing bare mode μ in terms of normal mode ν.

In the parameter regime of interest, two of these normal modes remain predominantly cavity-like and one remains predominantly transmon-like. For notational simplicity, we henceforth drop the subscript N and write a, b, and c for the dressed normal modes unless stated otherwise. In this basis,

$$b = \sum_{\nu=a,b,c} u_{b\nu}\, d_\nu, \tag{2.104}$$

where d_ν denotes the annihilation operator for normal mode ν (so $d_a \equiv a$, $d_b \equiv b$, and $d_c \equiv c$), so that

$$b^\dagger b = \sum_{\mu,\nu=a,b,c} u_{b\mu}^* u_{b\nu}\, d_\mu^\dagger d_\nu. \tag{2.105}$$

The perturbation therefore becomes

$$H^{(1)}(t) \approx -\frac{\tilde{K}_b}{2} b^{\dagger 2} b^2 + \delta\omega_b \cos(\omega_d t + \phi_d) \sum_{\mu,\nu=a,b,c} \eta_{\mu\nu}\, d_\mu^\dagger d_\nu, \tag{2.106}$$

with

$$\eta_{\mu\nu} = u_{b\mu}^* u_{b\nu}. \tag{2.107}$$

2.4.2 Drive-selected interaction and rotating-wave approximation

Equation (2.106) contains both diagonal frequency shifts and off-diagonal mode-conversion terms. For the cavity-like modes, the relevant off-diagonal coefficient is

$$\eta_{ac} = u_{ba}^* u_{bc}, \tag{2.108}$$

with $\eta_{ca} = \eta_{ac}^*$. The corresponding drive-induced interaction is therefore

$$H_{ac}^{(1)}(t) = \delta\omega_b \cos(\omega_d t + \phi_d) \left(\eta_{ac} a^\dagger c + \eta_{ac}^* a c^\dagger \right). \tag{2.109}$$

Writing

$$\eta_{ac} = |\eta_{ac}| e^{i\theta_{ac}}, \tag{2.110}$$

we define the real modulation amplitude

$$\delta g_{ac} = \delta\omega_b |\eta_{ac}|, \tag{2.111}$$

and absorb the hybridization phase θ_{ac} into an effective drive phase,

$$\phi_{\text{eff}} = \phi_d + \theta_{ac}. \tag{2.112}$$

Equation (2.109) can then be written as

$$H_{ac}^{(1)}(t) = \delta g_{ac} \cos(\omega_d t + \phi_{\text{eff}}) \left(a^\dagger c + a c^\dagger \right). \tag{2.113}$$

Moving to the interaction picture with respect to $\tilde{\omega}_a a^\dagger a + \tilde{\omega}_c c^\dagger c$, the interaction becomes

$$H_{I,ac}(t) = \frac{\delta g_{ac}}{2}\Big[a^\dagger c\, e^{-i(\tilde{\omega}_c-\tilde{\omega}_a-\omega_d)t+i\phi_{\rm eff}} + ac^\dagger\, e^{+i(\tilde{\omega}_c-\tilde{\omega}_a-\omega_d)t-i\phi_{\rm eff}}$$
$$+ a^\dagger c\, e^{-i(\tilde{\omega}_c-\tilde{\omega}_a+\omega_d)t-i\phi_{\rm eff}} + ac^\dagger\, e^{+i(\tilde{\omega}_c-\tilde{\omega}_a+\omega_d)t+i\phi_{\rm eff}}\Big]. \tag{2.114}$$

Choosing the drive frequency

$$\omega_d = \tilde{\omega}_c - \tilde{\omega}_a \tag{2.115}$$

makes the first pair of terms stationary, while the remaining terms oscillate at frequency $2\omega_d$ and are dropped under the rotating-wave approximation. The RWA is valid when

$$2\omega_d \gg G_{ac}, \tag{2.116}$$

so that the off-resonant terms average out on the interaction timescale. The effective Hamiltonian is then

$$H_{\rm BS} = G_{ac}\left(e^{+i\phi_{\rm eff}}a^\dagger c + e^{-i\phi_{\rm eff}}ac^\dagger\right), \tag{2.117}$$

with

$$G_{ac} = \frac{\delta g_{ac}}{2}. \tag{2.118}$$

This is the parametrically activated beam-splitter interaction between the two cavity-like modes. Its physical origin is explicit: the two cavities are not directly coupled; the interaction is mediated by the FTT, whose AC-flux-modulated frequency produces the resonant mode-conversion term after diagonalization of the static linear sector and application of the rotating-wave approximation.

CHAPTER 3
GRAPE BASED ON HARD-CODED GRADIENTS

Quantum optimal control (QOC) provides an alternative approach to bosonic state preparation and logical control. Rather than constructing the required operations analytically, QOC determines microwave pulses numerically by optimizing over the available control parameters. This framework is particularly useful for bosonic systems, where experimental constraints—such as bandwidth limits, leakage suppression, and crosstalk mitigation—can be incorporated directly into the pulse design.

One of the most widely used methods for quantum optimal control is Gradient Ascent Pulse Engineering (GRAPE). Its central objective is to optimize a set of control parameters so as to minimize the infidelity of a target state transfer or gate operation. In modern implementations of GRAPE, gradients are often computed using automatic differentiation (AD) [95], which has become an influential tool in quantum control as well [96–99]. By constructing a computational graph and applying the chain rule automatically, AD provides a flexible and convenient route to gradient evaluation.

This convenience, however, comes with a substantial memory cost, since the full computational graph must be stored during optimization. Semi-automatic differentiation (semi-AD) [100] alleviates part of this overhead relative to full AD, but the reduction is often insufficient for systems whose Hilbert space grows rapidly. In bosonic control, this growth arises for at least two reasons. First, stronger drives can stimulate multiphoton transitions, which must be captured by increasing the truncation dimension of the oscillator Hilbert space [101]. Second, the inclusion of additional cavity modes enlarges the joint Hilbert space even more dramatically.

These considerations motivate a return to the original GRAPE formulation of Ref. [102], which relies on hard-coded gradients (HG) rather than automatic differentiation. Because this approach avoids storing a full computational graph, it achieves the smallest possible memory overhead and is therefore especially well suited to optimal control problems in large bosonic Hilbert spaces. This chapter develops that viewpoint and applies it to scalable pulse optimization for bosonic systems.

This chapter introduces the GRAPE framework and derives analytical gradient expressions for key cost contributions. The numerical evaluation of these expressions—via the scaling-and-squaring method, the auxiliary matrix method, and a forward-backward propagation scheme—is then discussed in detail. The results of this chapter form the foundation for the scaling analysis and benchmarking presented in Chapter 4.

3.1 Brief sketch of GRAPE

I briefly sketch the basics of GRAPE [102], mainly to establish the notation used in subsequent sections. Consider a system described by the Hamiltonian

$$H(t) = H_s + a(t)\, h_c. \tag{3.1}$$

Here, H_s denotes the static system Hamiltonian and h_c is an operator coupling the classical control field $a(t)$ to the system.[1]Quantum optimal control aims to adjust $a(t)$ such that a desired unitary gate or state transfer is performed within a given time interval $0 \leq t \leq T$. To facilitate optimization, the total control time T is divided into N intervals of duration $\Delta t = T/N$, and the control field is specified by the amplitudes at the discrete times $t_n = n\Delta t$ ($n = 1, 2, \ldots, N$). The discretization interval Δt is chosen such that control amplitudes are approximately constant within

[1]For simplicity, I consider one control channel with real-valued $a(t)$. Generalizations to multiple control channels and complex a are straightforward.

each Δt. This treatment is close to modeling the actual drive output of an arbitrary waveform generator, where Δt can be chosen to align with the available resolution. I denote the time-discretized values by

$$a_n \coloneqq a(t_n), \tag{3.2}$$

and group these to form the vector $\mathbf{a} \in \mathbb{R}^N$. Under the influence of this piecewise-constant control field, the system's quantum state $|\psi_n\rangle \coloneqq |\psi(t_n)\rangle$ evolves by a single time step according to

$$|\psi_n\rangle = U_n|\psi_{n-1}\rangle. \tag{3.3}$$

Here, U_n is the short-time propagator

$$U_n \coloneqq U(t_n, t_{n-1}) = e^{-iH_n\Delta t} \qquad (\hbar = 1), \tag{3.4}$$

where $H_n = H_s + a_n\, h_c$. Optimization of the control field proceeds via minimization of a cost function C. Typically, C includes the state-transfer or gate infidelity, along with additional cost contributions employed to moderate pulse characteristics such as maximal power or bandwidth. To this end a composite cost function is constructed, $C(\mathbf{a}) = \sum_\nu \alpha_\nu C_\nu\,(\mathbf{a})$, where α_ν are empirically chosen weight factors and C_ν are individual cost contributions. GRAPE utilizes the method of steepest descent to minimize $C(\mathbf{a})$ by updating the control amplitudes according to the cost function gradient:

$$\mathbf{a}^{(j+1)} = \mathbf{a}^{(j)} - \eta_j\, \nabla_{\mathbf{a}} C(\mathbf{a}^{(j)}). \tag{3.5}$$

Here, j enumerates steepest-descent iterations and $\eta_j > 0$ is the learning rate governing the step size.

3.2 Analytical gradients

A critical ingredient for GRAPE is the computation of cost function gradients. One popular option is to leverage automatic differentiation for this purpose, such as implemented in `Tensorflow` [103] and `Autograd` [104]. A clear advantage of this strategy is the ease by which complicated cost function gradients are evaluated automatically at runtime. This convenience, however, is bought at the cost of significant memory consumption which can be prohibitive for large quantum systems. Therefore, I am motivated to revisit hard-coded gradients in a computationally efficient scheme for key cost contributions.

I discuss the analytical gradients of two key cost contributions and provide detailed expressions for the gradients of other cost contributions.

3.2.1 State-transfer infidelity

In state-transfer problems, a central goal of quantum optimal control is to minimize the state-transfer infidelity given by

$$C_1^{st} = 1 - |\langle\phi_T|\psi_N\rangle|^2\,, \tag{3.6}$$

where $|\phi_T\rangle$ is the target state and $|\psi_N\rangle$ is the state realized at final time t_N. The gradient of C_1^{st} takes the form

$$\frac{\partial C_1^{st}}{\partial a_n} = -\langle\phi_T|U_N\cdots\frac{\partial U_n}{\partial a_n}\cdots U_1|\psi_0\rangle\langle\psi_N|\phi_T\rangle - \mathrm{c.\,c.} = -\langle\phi_n|\frac{\partial U_n}{\partial a_n}|\psi_{n-1}\rangle\langle\psi_N|\phi_T\rangle - \mathrm{c.\,c.}, \tag{3.7}$$

where the state $|\phi_n\rangle$ obeys the recursive relation

$$|\phi_n\rangle = U_{n+1}^\dagger |\phi_{n+1}\rangle. \tag{3.8}$$

This can be interpreted as backward propagation.

3.2.2 Gate infidelity

In addition to state transfer, unitary gates are another operation of interest. Gate optimization can be achieved by minimizing the gate infidelity $C_1^g = 1 - \left|\operatorname{tr}\left(U_T^\dagger U_R\right)/d\right|^2$. Here, U_T is the target unitary gate and $U_R = U_N \cdots U_1$ is the actually realized gate. For a given orthonormal basis $\{|\psi_0^h\rangle\}_h$ (h enumerates the basis states), the gradient of C_1^g can be written as

$$\frac{\partial C_1^g}{\partial a_n} = -\frac{1}{d^2} \sum_h \langle \phi_n^h | \frac{\partial U_n}{\partial a_n} | \psi_{n-1}^h \rangle \sum_{h'} \langle \psi_N^{h'} | \phi_N^{h'} \rangle - \text{c. c.}, \tag{3.9}$$

where $|\psi_n^h\rangle = U_n \cdots U_1 |\psi_0^h\rangle$. The state $|\phi_n^h\rangle$ is obtained by applying the target unitary U_T to basis state $\{|\psi_0^h\rangle\}_h$ and backward propagating the result to time t_n,

$$|\phi_n^h\rangle = U_{n+1}^\dagger \cdots U_N^\dagger U_T |\psi_0^h\rangle. \tag{3.10}$$

3.2.3 Additional cost contributions

The cost contribution

$$C_2^{st} = \frac{1}{N} \sum_{n'=1}^{N} \langle \psi_{n'} | \Omega | \psi_{n'} \rangle \tag{3.11}$$

penalizes the integrated expectation value of the Hermitian operator Ω. The gradient of this cost contribution can be written as

$$
\begin{aligned}
\frac{\partial C_2^{st}}{\partial a_n} &= \frac{1}{N}\sum_{n'=1}^{N}\langle\psi_{n'}|\Omega\frac{\partial}{\partial a_n}\prod_{n''=1}^{n'}U_{n''}|\psi_0\rangle + c.c. \\
&= \frac{1}{N}\left(\sum_{n'=n}^{N}\prod_{n''=n+1}^{n'}\langle\psi_{n'}|\Omega U_{n''}\frac{\partial}{\partial a_n}U_n|\psi_{n-1}\rangle\right) + c.c. \\
&= \frac{2}{N}\,\mathrm{Re}(\langle\Phi_n|\frac{\partial}{\partial a_n}U_n|\psi_{n-1}\rangle),
\end{aligned}
\tag{3.12}
$$

with

$$
|\Phi_n\rangle := \sum_{n'=n}^{N} U_{n+1}^\dagger\cdots U_{n'}^\dagger\Omega|\psi_{n'}\rangle \tag{3.13}
$$

The vector $|\Phi_n\rangle$ obeys the recursion relation

$$
|\Phi_n\rangle = U_{n+1}^\dagger|\Phi_{n+1}\rangle + \Omega|\psi_n\rangle. \tag{3.14}
$$

Due to this relation, the expression in Eq. (3.12) can be evaluated with a forward-backward scheme analogous to the one discussed in Sec. 3.3.3.

The cost contribution $C_3^{st} = 1 - \frac{1}{N}\sum_{n'=1}^{N}|\langle\phi_T \mid \psi_{n'}\rangle|^2$ sums up infidelities from all time steps (as opposed to recording the final state infidelity only, as in C_1^{st}). This steers the system towards its target state more rapidly, thus helping to reduce the duration of state transfer. Similar to Eq. (3.12), the gradient of C_3^{st} is:

$$
\frac{\partial C_3^{st}}{\partial a_n} = -\frac{2}{N}\,\mathrm{Re}\left(\langle\Phi_{T,n}|\frac{\partial}{\partial a_n}U_n|\psi_{n-1}\rangle\right), \tag{3.15}
$$

with

$$
|\Phi_{T,n}\rangle := \sum_{n'=n}^{N} U_{n+1}^\dagger\cdots U_{n'}^\dagger|\phi_T\rangle\langle\phi_T|\psi_{n'}\rangle. \tag{3.16}
$$

The vector obeys the recursive relation

$$|\Phi_{T,n}\rangle = U_{n+1}^{\dagger}|\Phi_{T,n+1}\rangle + |\phi_T\rangle\langle\phi_T|\psi_n\rangle. \tag{3.17}$$

This allows the use of a forward-backward scheme to evaluate the gradient expression in Eq. (3.15).

For gate operations, the cost contribution analogous to C_3^{st} is given by

$$C_3^g = 1 - \sum_{n'=1}^{N} \left|\,\mathrm{tr}\left(U_T^{\dagger}U_{n',1}\right)\right|^2 / Nd^2, \tag{3.18}$$

where $U_{n',1} = U_{n'} \cdots U_1$ describes the evolution from time step 1 to n'. The gradient is

$$\frac{\partial C_3^g}{\partial a_n} = -\frac{2}{Nd^2}\sum_{h=1}^{d} \mathrm{Re}\left(\langle\Phi_{T,n}^h|\frac{\partial}{\partial a_n}U_n|\psi_{n-1}^h\rangle\right), \tag{3.19}$$

with

$$|\Phi_{T,n}^h\rangle := \sum_{n'=n}^{N} U_{n+1}^{\dagger}\cdots U_{n'}^{\dagger}U_T|\psi_0^h\rangle\,\mathrm{tr}(U_{n',1}U_T^{\dagger}). \tag{3.20}$$

Here, $\{|\psi_0^h\rangle\}$ forms an arbitrary orthonormal basis and $h = 1, 2, \ldots, d$ enumerates the basis states. The vector $|\Phi_{T,n}^h\rangle$ obeys the recursion relation

$$|\Phi_{T,n}^h\rangle = U_{n+1}^{\dagger}|\Phi_{T,n+1}^h\rangle + U_T|\psi_0^h\rangle\,\mathrm{tr}(U_{n,1}U_T^{\dagger}), \tag{3.21}$$

again enabling the evaluation of the gradient expression with a forward-backward propagation scheme.

Equations related to the gradient of cost functions such as (3.7) and (3.9) provide the analytical expressions needed for gradient descent. In the next two sections, I discuss how to numerically evaluate these expressions in a memory-efficient way.

3.3 Numerical implementation of hard-coded gradients

The calculation of gradients in equations including (3.7) and (3.9) requires numerical evaluation of expressions of the form $U_n|\Psi\rangle$ and $\frac{\partial U_n}{\partial a_n}|\Psi\rangle$. Here, the short-time propagator U_n is given by a matrix exponential e^A and $A = -iH_n\Delta t$.

Numerically evaluating $e^A|\Psi\rangle$ is not without challenge. As pointed out by Moler and Van Loan [105], several methods exist, yet none of them is truly optimal. Three methods ranked highly in Ref. [105] are based on solving ordinary differential equations, matrix diagonalization, and scaling and squaring (combined with series expansion). ODE solving, in this context, tends to be most expensive in runtime [100]. I focus mainly on scaling and squaring but comment on situations where matrix diagonalization is preferable in Sec. 4.3.

3.3.1 State propagation evaluation by scaling and squaring

Scaling and squaring [106] is based on the identity

$$e^A = \left(e^{A/s}\right)^s, \tag{3.22}$$

where the right-hand side matrix exponential now involves the matrix $B = A/s$ which has a norm that is reduced compared to $||A||$. This generally allows for a lower truncation order m of the exponential series[2],

$$e^B \approx e^B_m := \sum_{q=0}^{m} \frac{B^q}{q!}. \tag{3.23}$$

[2]While Chebyshev expansion has faster convergence than Taylor expansion, their numerical implementations have the same runtime scaling. Here, I focus on Taylor expansion due to its simplicity.

Typically, m is smaller than the one required for the original e^A. The goal is to approximate the state vector $e^A|\Psi\rangle \approx (e_m^B)^s|\Psi\rangle$ where

$$e_m^B|\Psi\rangle = \left(1 + B + \frac{B^2}{2!} + \cdots + \frac{B^m}{m!}\right)|\Psi\rangle = |\Psi\rangle + B|\Psi\rangle + \frac{B\left(B|\Psi\rangle\right)}{2} + \cdots + \frac{B\left(\cdots\left(B|\Psi\rangle\right)\right)}{m!}\,. \tag{3.24}$$

According to the last expression, approximation of $e^A|\Psi\rangle$ can thus proceed by repeated application of $B = A/s$ to a vector, without ever invoking matrix-matrix multiplication. This boosts runtime performance from $O(d^3)$ to $O(d^2)$.

To proceed with the evaluation of $e^A|\Psi\rangle$, the truncation order m and scaling factor s are determined in such a way that the error remains below the error tolerance τ,

$$\varepsilon_{A,|\Psi\rangle}(m,s) := \frac{\left|\left|e^A|\Psi\rangle - [\![(e_m^B)^s|\Psi\rangle]\!]\right|\right|}{||e^A|\Psi\rangle||} < \tau. \tag{3.25}$$

Here, $[\![\bullet]\!]$ denotes the floating point representation of $\bullet$. Since this error $\varepsilon_{A,|\Psi\rangle}$ is difficult and costly to evaluate, I instead derive an upper bound $E_A(m,s)$ and resort to the inequality

$$\varepsilon_{A,|\Psi\rangle}(m,s) < E_A(m,s) \leq \tau. \tag{3.26}$$

3.3.1 Determining the upper bound $E_A(m,s)$

The approximation of $e^A|\Psi\rangle$ required the determination of an upper bound $E_A(m,s)$ for the error $\varepsilon_{A,|\Psi\rangle}(m,s)$ [Eq. (3.25)]. Here, I show how to acquire such bound.

Considering the expression for the error [Eq. (3.25)], the denominator is simply 1. The error

$\varepsilon_{A,|\Psi\rangle}$ is bounded as follows:

$$\begin{aligned}\varepsilon_{A,|\Psi\rangle} &= \left|\left|e^A|\Psi\rangle - (e_m^B)^s|\Psi\rangle + (e_m^B)^s|\Psi\rangle - [\![(e_m^B)^s|\Psi\rangle]\!]\right|\right|_2 \\ &\leq \underbrace{\left|\left|e^A|\Psi\rangle - (e_m^B)^s|\Psi\rangle\right|\right|_2}_{\varepsilon_t} + \underbrace{\left|\left|(e_m^B)^s|\Psi\rangle - [\![(e_m^B)^s|\Psi\rangle]\!]\right|\right|_2}_{\varepsilon_r}.\end{aligned} \tag{3.27}$$

The error ε_t arises from the truncation of the Taylor series and rounding error ε_r is due to finite-precision arithmetics. I proceed to derive the upper bounds for these two errors separately.

Upper bound for the truncation error ε_t. The truncation error can be rewritten as

$$\varepsilon_t = \left|\left|e^A|\Psi\rangle - (e^B - R_m(B))^s|\Psi\rangle\right|\right|_2 = \left|\left|\sum_{i=1}^{s} \frac{s!}{i!(s-i)!}(-R_m(B))^i(e^B)^{s-i}|\Psi\rangle\right|\right|_2, \tag{3.28}$$

where $R_m(B) = \sum_{q=m+1}^{\infty} B^q/q!$ denotes the matrix-valued error of the truncated exponential series. Employing the triangle inequality and submultiplicativity [107], one finds the bound

$$\varepsilon_t \leq \sum_{i=1}^{s} \frac{s!}{i!(s-i)!}\big(R_m(||B||_2)\big)^i \left|\left|e^B\right|\right|_2^{s-i} |||\Psi\rangle||_2 = \sum_{i=1}^{s} \frac{s!}{i!(s-i)!}\big(R_m(||B||_2)\big)^i. \tag{3.29}$$

Here the second equality utilizes the fact that the state is normalized and e^B is unitary.

This upper bound must be evaluated numerically to determine m and s. However, computing the matrix 2-norm is inconvenient since it requires the use of an eigensolver. To mitigate this, I switch to the matrix 1-norm by exploiting[3]

$$||B||_2 \leq ||B||_1 . \tag{3.30}$$

Monotonicity of R_m leads to

$$\varepsilon_t \leq \sum_{i=1}^{s} \frac{s!}{i!(s-i)!} \left(R_m(||B||_1)\right)^i < sR_m(||B||_1) \frac{1-\left(sR_m(||B||_1)\right)^s}{1-sR_m(||B||_1)}. \tag{3.31}$$

Upper bound for the rounding error ε_r. The rounding error ε_r originates from the finite precision of the floating-point representation of real numbers as well as the finite accuracy of basic arithmetic operations. The floating-point representation $[\![\xi]\!]$ for the real number ξ can be written as $[\![\xi]\!] = \xi(1+\delta)$ where $\delta = \delta(\xi)$ is the relative deviation [107]. Its magnitude is always smaller than machine precision, i.e. $|\delta| < u$. Similarly, the complex number z obeys

$$[\![z]\!] = z(1+\delta) \tag{3.32}$$

with $|\delta| \leq u$. Arithmetic operations such as addition and multiplication with two complex numbers z_1, z_2 incur a relative deviation $\delta = \delta(z_1, z_2)$ such that

$$[\![z_1 + z_2]\!] = ([\![z_1]\!] + [\![z_2]\!])(1+\delta),\ |\delta| \leq u, \quad [\![z_1 z_2]\!] = [\![z_1]\!][\![z_2]\!](1+\delta),\ |\delta| \leq u', \tag{3.33}$$

where $u' := 2\sqrt{2}u/(1-2u)$ [107]. The relative deviation shown in Eqs. (3.32), (3.33) generally depend on both numbers involved as well as the operation in question. In the following, different δ are distinguished with subscript ν.

[3]Generally, $||B||_2 \leq \sqrt{||B||_\infty \, ||B||_1}$ [108] holds true for any B. When B is anti-Hermitian, $||B||_\infty = ||B||_1$ leads to the inequality in Eq. (3.30).

Using Eqs. (3.32), (3.33), I derive an upper bound for the rounding error in the evaluation of the dot product, which is the basic arithmetic operation in evaluating $e_m^B|\Psi\rangle$ according to Eq. (3.24).

Upper bound for the rounding error associated with evaluating the dot product. The rounding error incurred when evaluating the dot product is given by $|\Sigma_d - [\![\Sigma_d]\!]|$ with $\Sigma_d = \boldsymbol{w}^{\mathrm{T}}\boldsymbol{z}$ ($\boldsymbol{w}, \boldsymbol{z} \in \mathbb{C}^d$). Here, $\boldsymbol{w}^{\mathrm{T}}$ is a row of $B = i\delta t H/s$. Based on Eqs. (3.32) and (3.33), one has

$$|[\![\Sigma_1]\!] - \Sigma_1| = |[\![w_1 z_1]\!] - w_1 z_1| < |w_1||z_1|\Big|\prod_{\nu=1}^{3}(1+\delta_\nu) - 1\Big| < \gamma_3|w_1||z_1|, \tag{3.34}$$

where $\big|\prod_{\nu=1}^{n}(1+\delta_\nu) - 1\big| < \frac{nu'}{1-nu'}$ [107], and $\gamma_n := \frac{nu'}{1-nu'}$. For $d = 2$, the upper bound of the error is

$$|[\![\Sigma_2]\!] - \Sigma_2| = |([\![\Sigma_1]\!] + [\![w_2 z_2]\!])(1+\delta_1) - \Sigma_2| < \gamma_4 \sum_{i=1}^{2} |w_i||z_i|. \tag{3.35}$$

The general upper bound for arbitrary d is:

$$|[\![\Sigma_d]\!] - \Sigma_d| < \gamma_{d+2} \sum_{j=1}^{d} |w_j||z_j|. \tag{3.36}$$

Since $\boldsymbol{w}^{\mathrm{T}}$ is proportional to a row of the Hamiltonian matrix, sparsity of H implies sparsity of $\boldsymbol{w}^{\mathrm{T}}$. Vanishing entries in $\boldsymbol{w}^{\mathrm{T}}$ are special because they do not incur rounding errors. Thus, one can obtain a stronger upper bound

$$|[\![\Sigma_d]\!] - \Sigma_d| < \gamma_{\sigma+2} \sum_{j=1}^{d} |w_j||z_j| \tag{3.37}$$

where σ is the number of non-zero elements in $\boldsymbol{w}^{\mathrm{T}}$.

Upper bound for the rounding error of $e_m^B|\Psi\rangle$. The vector $e_m^B|\Psi\rangle$ is evaluated recursively via

$$b_j = \frac{B}{j}b_{j-1}, \qquad e_j^B b_0 = b_{j-1} + b_j \tag{3.38}$$

where $j = 1, 2, \cdots, m$ and b_0 is a vector representation of $|\Psi\rangle$ under a choice of orthonormal basis. Before evaluating the upper bound for the error $\big|\big|[\![e_m^B b_0]\!] - e_m^B b_0\big|\big|_2$, I first derive an upper bound for the error of a vector component $\big|[\![e_m^B b_0]\!]^{(i)} - (e_m^B b_0)^{(i)}\big|$.

For the case $m = 0$:

$$\big|[\![e_0^B b_0]\!]^{(i)} - (e_0^B b_0)^{(i)}\big| = \big|[\![b_0]\!]^{(i)} - b_0^{(i)}\big| < \gamma_2 |b_0^{(i)}|. \tag{3.39}$$

For $m = 1$:

$$\begin{aligned} |[\![e_1^B b_0]\!]^{(i)} - (e_1^B b_0)^{(i)}| &= \big|\big([\![b_0]\!]^{(i)} + [\![b_1]\!]^{(i)}\big)(1+\delta_1) - (e_1^B b_0)^{(i)}\big| \\ &= \Big|[\![b_0]\!]^{(i)}(1+\delta_1) + \frac{[\![Ab_0]\!]}{s}^{(i)}\prod_{\nu=1}^{3}(1+\delta_\nu)^3 - (e_1^B b_0)^{(i)}\Big| \overset{(3.37)}{<} \gamma_3|b_0^{(i)}| + \gamma_{\sigma_i+5}\sum_l |B^{(il)}||b_0^{(l)}|, \end{aligned} \tag{3.40}$$

where σ_i is the number of non-zero elements in the ith row of B. The general upper bound for arbitrary m is:

$$\big|[\![e_m^B b_0]\!]^{(i)} - (e_m^B b_0)^{(i)}\big| < \sum_{k=0}^{m} \gamma_{k(\sigma_i+2)+m+2}\Big(\frac{|B|^k}{k!}|b_0|\Big)^i, \tag{3.41}$$

where $|B|, |b_0|$ denote the matrix and the vector with elements $|B^{il}|$, $|b_0^i|$, respectively. Defining

$$\sigma' := \max_i \sigma_i \tag{3.42}$$

and using Eq. (3.41):

$$\big|\big|[\![e_m^B b_0]\!] - (e_m^B b_0)\big|\big|_2 = \big|\big|\big|[\![e_m^B b_0]\!] - (e_m^B b_0)\big|\big|\big|_2 < \sum_{k=0}^{m} \gamma_{k(\sigma'+2)+m+2}\big|\big|\frac{|B|^k}{k!}|b_0|\big|\big|_2$$
$$< \sum_{k=0}^{m} \gamma_{k(\sigma'+2)+m+2}\frac{\big|\big||B|\big|\big|_2^k}{k!}\big|\big||b_0|\big|\big|_2 \overset{(3.30)}{<} \sum_{k=0}^{m} \gamma_{k(\sigma'+2)+m+2}\frac{\big|\big||B|\big|\big|_1^k}{k!}\big|\big||b_0|\big|\big|_2 = \beta||b_0||_2 \quad (3.43)$$

with $\beta := \sum_{k=0}^{m} \gamma_{k(\sigma'+2)+m+2}\frac{||B||_1^k}{k!}$. Here, $\beta||b_0||_2$ is the desired upper bound for the rounding error of $e_m^B|\Psi\rangle$.

Upper bound for the error ε_r. The vector $(e_m^B)^s b_0$ is evaluated by the recursive relation

$$c_l = e_m^B c_{l-1}, \qquad l = 1, 2, \cdots, s, \quad (3.44)$$

where $c_0 = b_0$. I illustrate how to bound the error $\varepsilon_r = \big|\big|[\![(e_m^B)^s b_0]\!] - (e_m^B)^s b_0\big|\big|_2$ for $s = 1, 2$, and give the general result subsequently.

For $s = 1$:

$$\big|\big|[\![e_m^B b_0]\!] - e_m^B b_0\big|\big|_2 \overset{(3.43)}{<} \beta||b_0||_2 = \beta. \quad (3.45)$$

For $s = 2$:

$$\big|\big|[\![(e_m^B)^2 b_0]\!] - (e_m^B)^2 b_0\big|\big|_2 \leq \big|\big|[\![(e_m^B)^2 b_0]\!] - e_m^B[\![e_m^B b_0]\!]\big|\big|_2 + \big|\big|e_m^B[\![e_m^B b_0]\!] - (e_m^B)^2 b_0\big|\big|_2$$
$$< \big|\big|[\![e_m^B[\![e_m^B b_0]\!]]\!] - e_m^B[\![e_m^B b_0]\!]\big|\big|_2 + \beta\,\big|\big|e_m^B\big|\big|_2\,, \quad (3.46)$$

where the last step is enabled by Eq. (3.45). The matrix norm in the second term is bounded by

$$\big|\big|e_m^B\big|\big|_2 = \big|\big|e^B - R_m(B)\big|\big|_2 \leq 1 + R_m(||B||_2) \overset{(3.30)}{\leq} 1 + R_m(||B||_1) =: \alpha. \quad (3.47)$$

Since Eq. (3.43) holds for any complex vector b_0, the error in Eq. (3.46) is further bounded by:

$$\left|\left|[\![(e_m^B)^2 b_0]\!] - (e_m^B)^2 b_0\right|\right|_2 \overset{(3.43)}{<} \beta \left|\left|[\![e_m^B b_0]\!]\right|\right|_2 + \beta\alpha = \beta \left|\left|[\![e_m^B b_0]\!] - e_m^B b_0 + e_m^B b_0\right|\right|_2 + \beta\alpha \tag{3.48}$$

$$\overset{(3.45)}{<} \beta(\beta+\alpha) + \beta\alpha. \tag{3.49}$$

Generalizing this strategy further yields the expression for the upper bound of ε_r:

$$\begin{aligned}\varepsilon_r = \left|\left|[\![(e_m^B)^s b_0]\!] - (e_m^B)^s b_0\right|\right|_2 &\le \sum_{l=0}^{s-1} \left|\left|(e_m^B)^l [\![(e_m^B)^{s-l} b_0]\!] - (e_m^B)^{l+1} [\![(e_m^B)^{s-l-1} b_0]\!]\right|\right|_2 \\ &< \beta \sum_{l=0}^{s-1} (\alpha+\beta)^{s-l-1} \alpha^l = (\alpha+\beta)^s - \alpha^s.\end{aligned} \tag{3.50}$$

Combining the upper bounds for ε_r and ε_t, I return to Eq. (3.27) and obtain

$$\varepsilon_{A,|\Psi\rangle} < E_A(m,s) = sR_m(||B||_1) \frac{1 - \left(sR_m(||B||_1)\right)^s}{1 - sR_m(||B||_1)} + (\alpha+\beta)^s - \alpha^s, \tag{3.51}$$

where $||B||_1 = ||A||_1 / s$, $\alpha = \alpha(||A||_1, m, s)$ and $\beta = \beta(\sigma', ||A||_1, m, s)$.

3.3.1 Choosing scaling factor and truncation order

For a given A when computing state propagation by scaling and squaring discussed in Sec. 3.3.1, the inequality (3.26) generally has multiple solutions for m and s. As shown in Sec. 3.3.1.1, the evaluation requires ms matrix-vector multiplications, which are observed to dominate the runtime. Since systematic minimization of ms requires significant runtime itself, I instead: (1) select solutions to Eq. (3.26) with the smallest s, which is denoted by $\mathbb{s}$, and (2) among these pairs, choose the one with smallest m (denoted by $\mathbb{m}$).

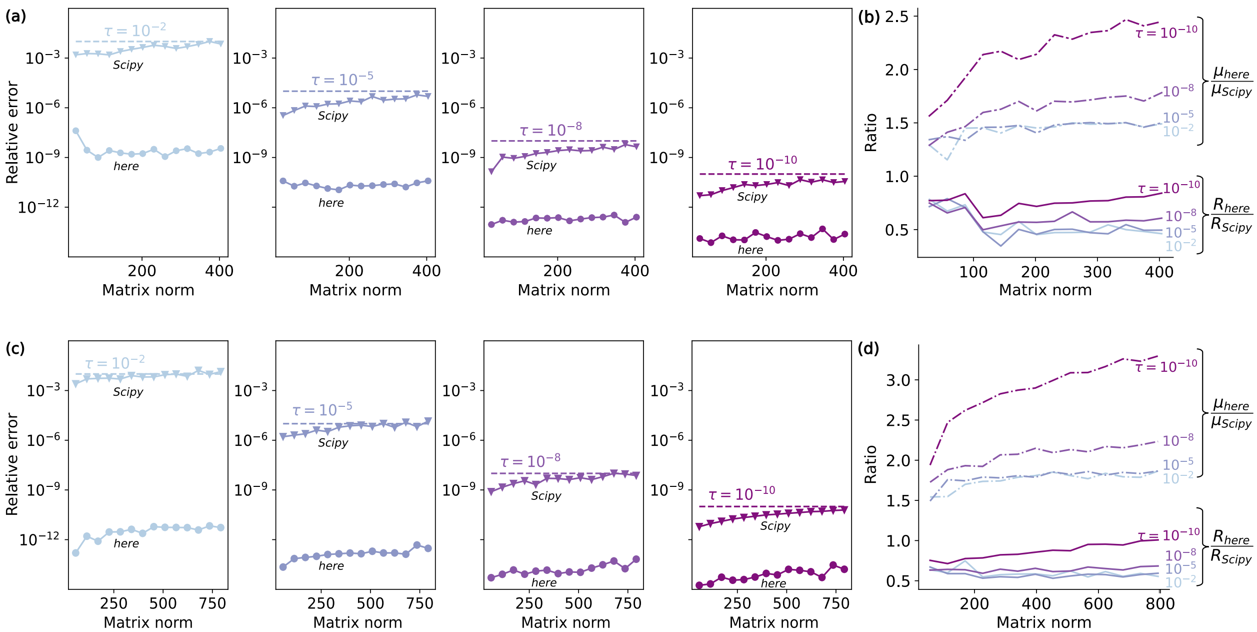

Figure 3.1: Comparison between scaling and squaring as implemented here and `Scipy`'s `expm_multiply`, considering relative error and runtime. (a), (c) Relative error versus matrix norm. (b), (d) Ratios of runtime and μ (i.e. number of matrix-vector multiplications) between the two implementations as functions of the matrix norm. The matrix norm $||-iH\Delta t||_2$ is varied by increasing $\Delta t \in [1, 10]$. In all panels, colors encode different error tolerances τ. The results of the first and second rows are obtained based on the cavity–transmon and three-coupled-transmons Hamiltonians, respectively. The system parameters of these two Hamiltonians are the same as those given in the captions of Figs. 4.1 and 4.2.

Thus, calculating $e^A|\Psi\rangle$ requires

$$\mu := ms \tag{3.52}$$

matrix-vector multiplications. As shown in the following, for two concrete examples, this method for evaluating $e^A|\Psi\rangle$ can achieve better runtime performance and accuracy than `Scipy`'s `expm_multiply` function [109].

3.3.1 Numerical comparison between scaling and squaring implementations

Scaling and squaring is used to evaluate the propagator-state product, $e^{-iH\Delta t}|\Psi\rangle$. I compare relative error and runtime between the implementation of scaling and squaring presented here and `Scipy`'s implementation, `expm_multiply` [109]. The relative errors follow the general expression in Eq. (3.25), but differ in the specific choice of m and s. As a proxy for the exact propagator-state product, I apply Taylor expansion to the matrix exponential with 400-digit precision by `Sympy`; the result of the product converges at 30 digits after the decimal point. The runtime is measured with the help of the `Temci` package [110].

As the basis for this comparison, I use the examples of Hamiltonians (4.1) and (4.2) and implement sparse storage for them. The relative errors are shown in Fig. 3.1(a) and (c) as a function of the matrix norm $||-iH\Delta t||_2$, respectively. In both examples, the relative errors for the implementation presented here are bounded by the error tolerance τ, while the relative errors for `Scipy`'s implementation tend to exceed the tolerance for large norms. There are two reasons for this tendency. First, rounding errors are not considered in the derivation of the upper bound for the errors [109]; second, the truncation of e_m^B [Eq. (3.23)] is merely based on a heuristic criterion.

Even though the relative errors of `Scipy`'s implementation are worse, it achieves a smaller number of matrix-vector multiplications μ. This is illustrated in Fig. 3.1(b) and (d): the ratio of μ between the two implementations is always larger than 1. Although `Scipy`'s implementation spends less time on matrix-vector multiplications, the total runtime of `Scipy`'s implementation is larger, see Fig. 3.1(b) and (d). This is due to the larger runtime overhead of `Scipy`'s implementation for determining (m, s).

In conclusion, the implementation presented here has better numerical stability and runtime than `Scipy`'s implementation in these numerical experiments.

3.3.2 Computing the propagator derivative acting on a state: auxiliary matrix method

Having addressed the challenges associated with numerically evaluating $e^{A}|\Psi\rangle$, I now turn to the calculation of $\frac{\partial U_n}{\partial a_n}|\Psi\rangle$, which is crucial for computing cost function gradients as in Eqs. (3.7) and (3.9). This evaluation is based on the approximation

$$\frac{\partial U_n}{\partial a_n}|\Psi\rangle \approx \frac{\partial (e_m^{B_n})^s}{\partial a_n}|\Psi\rangle, \tag{3.53}$$

where $B_n = -iH_n\Delta t/s$. I proceed by using the auxiliary matrix method based on the identity [111, 112]

$$(e_m^{\mathcal{A}_n/s})^s \begin{pmatrix} 0 \\ |\Psi\rangle \end{pmatrix} = \begin{pmatrix} \frac{\partial (e_m^{B_n})^s}{\partial a_n}|\Psi\rangle \\ (e_m^{B_n})^s|\Psi\rangle \end{pmatrix}. \tag{3.54}$$

Here, $\mathcal{A}_n$ is defined as

$$\mathcal{A}_n := \begin{pmatrix} -iH_n\Delta t & -ih_c\Delta t \\ 0 & -iH_n\Delta t \end{pmatrix}, \tag{3.55}$$

which is a 2×2 matrix of operators each acting on the Hilbert space with dimension d. Once a basis is chosen, $\mathcal{A}_n$ can be represented as a $2d \times 2d$ matrix of complex numbers. Numerical evaluation of the left-hand side of the identity (3.54) gives a $2d$-dimensional vector. The first d entries of the vector correspond to $\partial(e_m^{B_n})^s/\partial a_n|\Psi\rangle$, which is an approximation of $\frac{\partial U_n}{\partial a_n}|\Psi\rangle$.

3.3.3 Efficient evaluation of hard-coded gradients

The analytical gradient expressions, such as ∇C_1^{st} in Eq. (3.7), are generally complicated and involve a large number of contributing quantities. Depending on grouping and ordering of operations, these quantities may need to be stored simultaneously or computed repeatedly with critical implications for runtime and memory usage. The scheme originally proposed by Khaneja *et al.* [102] strikes a favorable compromise that avoids the proliferation of stored intermediate state

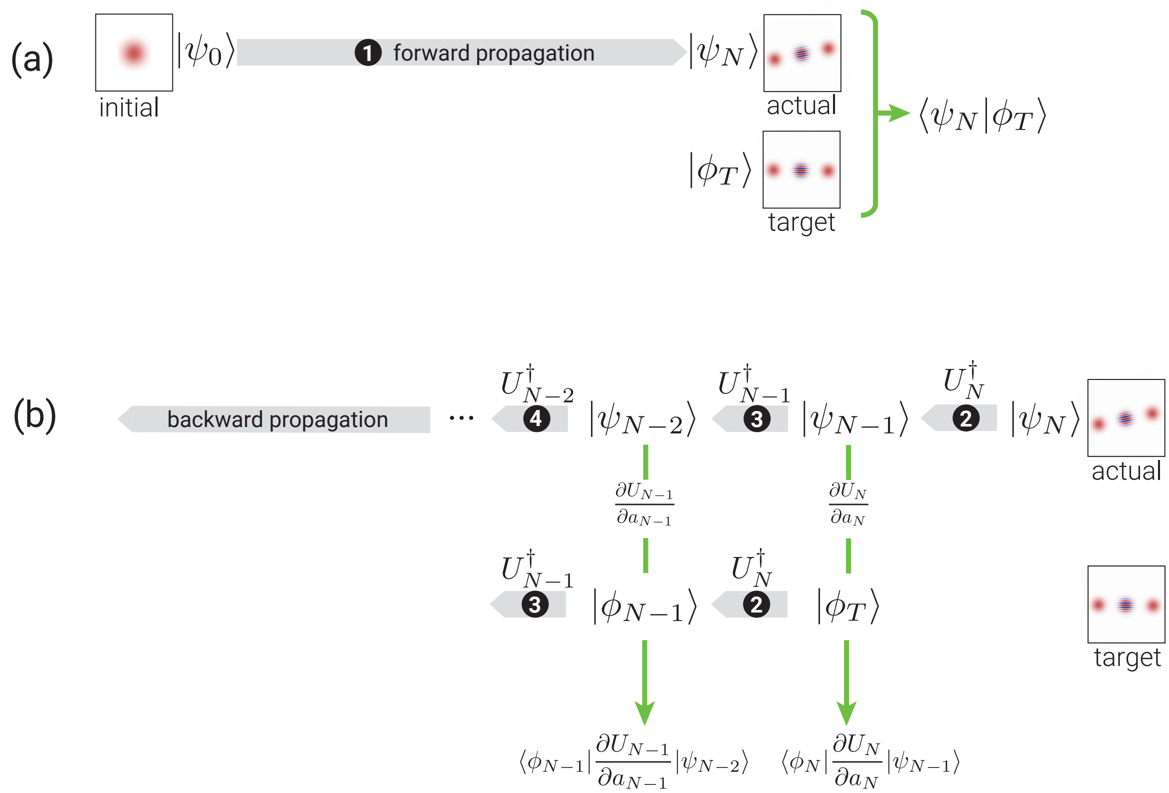

Figure 3.2: The forward-backward propagation scheme to compute the components of the gradient in Eq. (3.7). (a) Forward propagate the initial state by $U_N \cdots U_1$ to obtain the realized state $|\psi_N\rangle$ and calculate the overlap $\langle\psi_N|\phi_T\rangle$, where $|\phi_T\rangle$ is the target state. (b) Backward propagate $|\psi_N\rangle$ and $|\phi_T\rangle$ by the propagators $U_N^\dagger, U_{N-1}^\dagger, \cdots, U_1^\dagger$ iteratively, and compute $\langle\phi_n|\frac{\partial U_n}{\partial a_n}|\psi_{n-1}\rangle$ at each step.

vectors. More specifically, the final quantum state $|\psi_N\rangle$ is obtained by forward propagating $|\psi_0\rangle$. Then, backward propagation of $|\psi_N\rangle$ and $|\phi_T\rangle$ is carried out to calculate the matrix element $\langle\phi_n|\frac{\partial U_n}{\partial a_n}|\psi_{n-1}\rangle$ needed for the n-th component of the gradient, thus building up the gradient in descending order. For both propagation directions, only the current intermediate states are stored, leading to a memory cost of $\Theta(1)$.[4]

The forward-backward propagation scheme is illustrated in Fig. 3.2. This chapter has estab-

[4]Here, $f(x) = \Theta(g(x))$ is the usual Bachmann–Landau notation for $f(x)$ bounded above and below by $g(x)$ in the asymptotic sense.

lished the analytical and numerical machinery needed for evaluating cost function gradients in GRAPE without automatic differentiation. In the next chapter, I present a scaling analysis comparing the memory usage and runtime of hard-coded gradients with automatic differentiation approaches.

CHAPTER 4
SCALING ANALYSIS AND BENCHMARKING

Having established the analytical gradient expressions and their numerical evaluation in Chapter 3, this chapter presents a systematic comparison of three methods for computing gradients within GRAPE: evaluation of hard-coded gradients (HG), full automatic differentiation (full-AD) [96], and semi-automatic differentiation (semi-AD) [100]. To determine the method most appropriate for a given problem, I inspect the memory usage and runtime scaling of these three approaches. Following this analysis, I present benchmark studies of specific state-transfer and gate-operation problems to illustrate the scaling behavior.

4.1 Scaling analysis of runtime and memory usage

Runtime and memory usage can pose significant challenges when performing optimization tasks for quantum systems with large Hilbert space dimension (d) or for time evolution requiring a significant number of time steps (N). In this section, I analyze the scaling of memory usage and runtime with respect to both N and d, focusing on the cost contribution C_1 for state-transfer and unitary-gate optimization. The findings can be generalized to other cost contributions beyond C_1.

4.1.1 Hard-coded gradients via forward-backward propagation

I start by considering state-transfer problems. The *memory usage* for evaluating ∇C_1^{st} is mainly determined by the required storage of quantum states and the Hamiltonian. In forward-backward propagation, the number of states and instances of Hamiltonian matrices stored at any given time is of the order of unity. Each individual state vector of dimension d consumes memory $\sim \Theta(d)$.

Memory usage for the Hamiltonian depends on the employed storage scheme. In the "worst" case of dense matrix storage, the memory required for the Hamiltonian is $\Theta(d^2)$. For sparse matrix storage, memory usage generally depends on sparsity and is determined by the number of stored Hamiltonian matrix elements (here denoted as κ). For example, for a diagonal or tridiagonal matrix, memory usage is $\kappa \sim \Theta(d)$; for a linear chain of qubits with nearest-neighbor σ_z coupling, it is $\Theta(d \log_2 d)$. Combining the scaling relations above gives an overall asymptotic behavior of memory usage $\Theta(d + \kappa)$.

The *runtime* of the scheme is dominated by the required $\Theta(N)$ evaluations of propagator-state products and their derivative counterparts. Numerically approximating a single propagator-state product requires μ matrix-vector multiplications [see Eq. (3.52)], where the runtime of one instance of matrix-vector multiplication scales as $\Theta(\kappa)$. The implicit dependence of $\mu = \mu(d)$ on d is specific to each particular system and can be difficult to obtain analytically. Numerically, I find for a driven transmon coupled to a cavity that $\mu \sim \Theta(\sqrt{d})$; for a linear chain of coupled qubits, the scaling is $\Theta(\log_2 d)$; see Appendix A for more details. Once combined, the above individual scaling relations lead to an overall runtime asymptotic behavior $\Theta(N\mu\,\kappa)$. For example, in the case of the linear chain of coupled qubits, runtime scales as $\Theta(Nd\,(\log_2 d)^2)$.

For gate-optimization problems, the scaling behavior of memory usage and runtime is as follows. The gradient ∇C_1^g is calculated by sequentially applying forward-backward propagation to each of the d basis states. The memory usage is $\Theta(d + \kappa)$, the same as for ∇C_1^{st}. The runtime scaling is $\Theta(N\mu\kappa d)$, which is d times larger than the runtime scaling for ∇C_1^{st}. (Another way of calculating ∇C_1^g consists of parallelized forward-backward propagation of all basis states. The concrete scaling behavior then depends on the parallelization scheme; that discussion is beyond the scope of this thesis.)

Thanks to the similarities in the expressions and the existence of recursion relations, forward-backward schemes akin to Fig. 3.2 can also be applied to ∇C_2^{st}, ∇C_3^{st}, and ∇C_3^{g}. As a result, the scaling analysis leads to the same memory usage and runtime requirements as ∇C_1^{st} and ∇C_1^{g}.

4.1.2 Full automatic differentiation

Reverse-mode automatic differentiation extracts gradients by a forward pass and a reverse pass through the computational tree. With the forward pass the cost function C is computed, and the backward pass accumulates the gradient via the chain rule.

The *memory usage* for evaluating ∇C_1^{st} is determined by the requirement that all intermediate numerical results must be stored as part of the forward pass and act as input to the reverse pass. Specifically, as shown in Fig. 3.2, the forward pass evolves the initial state vector to the final state vector in N time steps. The total number of vectors that have to be stored along the way is $\Theta(N\mu)$, since computation of a single short-time propagator-state product via scaling and squaring necessitates μ iterations. In addition, considering the memory usage for the Hamiltonian, the full scaling is $\Theta(N\mu d + \kappa)$.

The *runtime* for evaluating ∇C_1^{st} combines forward and reverse passes. It is dominated by the evaluation of $\Theta(N)$ propagator-state products, each of which has the runtime scaling $\Theta(\mu\kappa)$, leading to an overall runtime scaling of $\Theta(N\mu\kappa)$.

For gate-optimization problems, C_1^g is evaluated by evolving not only a single initial state but a whole set of d basis vectors. Evaluating Nd propagator-state products requires $\Theta(N\mu d^2 + \kappa)$ memory, where the number of stored Hamiltonian matrix elements κ takes into account storage of the Hamiltonian. Since κ has the worst-case scaling $\Theta(d^2)$ in the case of a dense Hamiltonian matrix, the overall scaling is $\Theta(N\mu d^2)$. Assuming that the evolution of individual basis vectors is not parallelized, runtime also grows by a factor of d, so that runtime scaling for ∇C_1^g is $\Theta(N\mu\kappa d)$.

Despite the specific differences among the expressions of cost contributions C_2^{st}, C_3^{st}, and C_3^g,

Table 4.1: Memory usage and runtime scaling with respect to Hilbert space dimension d, number of time steps N, the number of stored Hamiltonian matrix elements $\kappa = \kappa(d)$, and the number of matrix-vector multiplications required for evaluating propagator-state products $\mu = \mu(d)$. (The scaling of μ and κ depends on the specific optimization task.) The table contrasts scaling for state transfer and gate operation for hard-coded gradients (HG), full automatic differentiation (full-AD), and semi-automatic differentiation (semi-AD).

	STATE TRANSFER		GATE OPERATION	
	Runtime	**Memory usage**	**Runtime**	**Memory usage**
HG	$\Theta(N\mu\kappa)$	$\Theta(d+\kappa)$	$\Theta(N\mu\kappa d)$	$\Theta(d+\kappa)$
Full-AD	$\Theta(N\mu\kappa)$	$\Theta(N\mu d+\kappa)$	$\Theta(N\mu\kappa d)$	$\Theta(N\mu d^2)$
Semi-AD	$\Theta(N\mu\kappa)$	$\Theta(Nd+\kappa)$	$\Theta(N\mu\kappa d)$	$\Theta(Nd^2)$

the scaling of runtime and memory usage is again set by the need to store intermediate vectors and repeatedly perform matrix-vector multiplications. Consequently, the scaling remains the same as for the cost contributions discussed above.

4.1.3 Semi-automatic differentiation

For gradients of cost contributions that are tedious to derive analytically, semi-automatic differentiation [100] offers an alternative approach in which some derivatives are hard-coded and others treated by automatic differentiation. Compared to full-AD, semi-AD has the benefit that the memory usage scaling is independent of μ while maintaining the same runtime scaling.

4.1.4 Comparison

The results of this scaling analysis are summarized in Table 4.1. For full-AD, memory usage grows linearly with respect to μ and number of time steps N. By contrast, memory usage for HG does not increase with μ and N, but remains constant. This advantage is not bought at the cost of worse runtime scaling. Therefore, HG is preferable over full-AD in the optimal control of large quantum systems where memory usage would otherwise be prohibitively large. Semi-AD presents an interesting compromise viable whenever the memory bottleneck is not too severe. Note that the runtime scaling $\Theta(N\mu\kappa d)$ of HG via scaling and squaring can in principle exceed the scaling $O(Nd^3)$ for matrix exponentiation based on matrix diagonalization (see Appendix B). In that case, the latter is the better option.

I illustrate the discussed scaling by benchmark studies for concrete optimization problems in the following section.

4.2 Benchmark studies

4.2.1 State-transfer benchmark

The first benchmark example studies the following state-transfer problem. Consider a setup in which a driven flux-tunable transmon is capacitively coupled to a superconducting cavity, and perform a state transfer from the cavity vacuum state to the 20-photon Fock state. The Hamiltonian in the frame co-rotating with the transmon drive is

$$H_1(t) = \Delta b^\dagger b - \frac{K_b}{2} b^{\dagger 2} b^2 + g(cb^\dagger + c^\dagger b) + a_z(t) b^\dagger b + a_x(t)(b + b^\dagger). \tag{4.1}$$

Here, the drive is both longitudinal and transverse; c and b are the annihilation operators for the cavity mode and the transmon mode, respectively, in the same sense as modes c and b in Fig. 2.3 and Eqs. (2.93)–(2.94). The transmon Kerr nonlinearity is written as in Eq. (2.94), with $K_b > 0$. The parameters K_b, g, and Δ denote the Kerr coefficient, interaction strength, and the detuning between the transmon and cavity frequencies.

Optimization proceeds by minimizing the infidelity C_1^{st} and limiting the occupation of higher transmon levels by including the cost contribution C_2^{st} [Eq. (3.11)]. The latter has the addi-

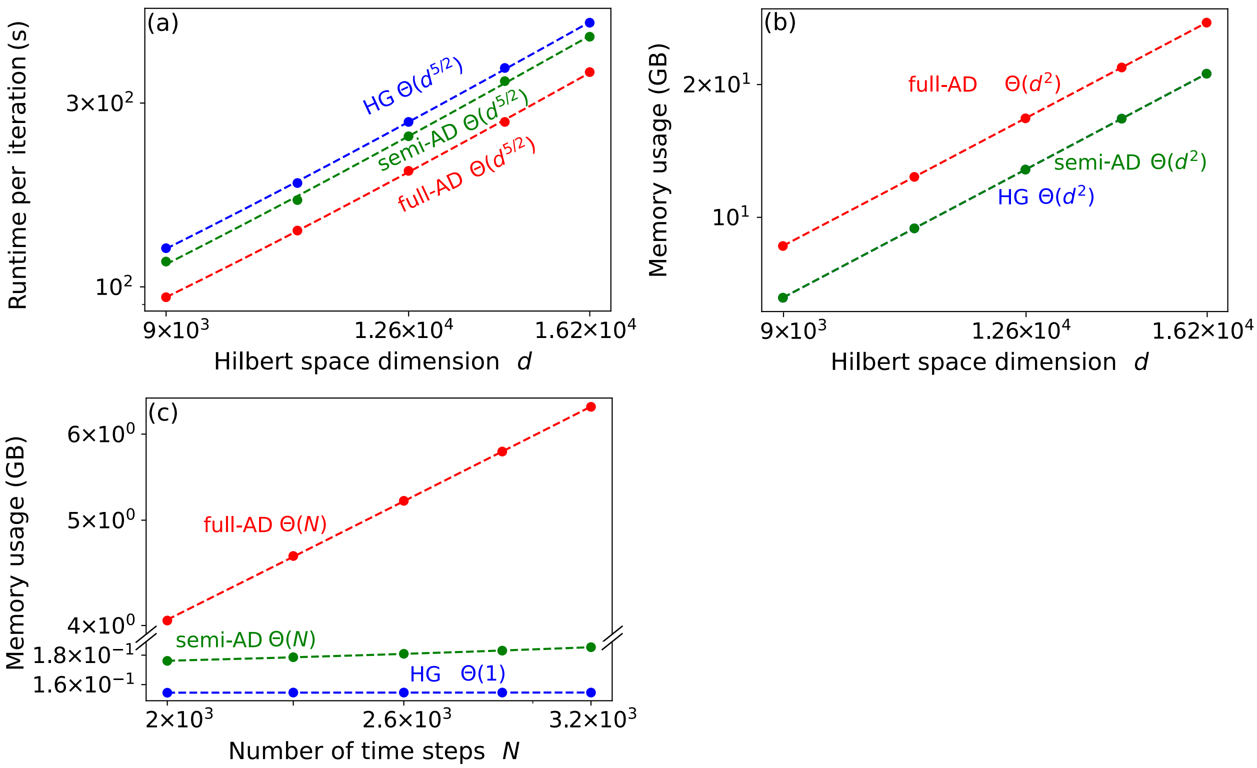

Figure 4.1: Benchmark results for Fock state generation in a system consisting of a cavity coupled to a transmon. (a) Runtime per iteration and (b) memory usage versus Hilbert space dimension d. The dimension is varied by increasing the cavity dimension while fixing the transmon dimension to 6. (c) Memory usage as a function of the number of time steps for $d = 240$. All plots are in log-log scale. Dashed lines are power-law fits. (Parameters: $\Delta/2\pi = 3\,$GHz, $g/2\pi = 0.1\,$GHz, $K_b/2\pi = 0.225\,$GHz, and constant controls $a_z/2\pi = a_x/2\pi = 0.1\,$GHz.)

tional benefit of enabling the truncation to only a few transmon levels. I measure runtime and peak memory usage of full-AD,[1] semi-AD, and HG for a single optimization iteration. The runtime is measured by the package `Temci` [110] and the peak memory usage is monitored by the `memory_profiler` package [113]. The optimization is performed on an AMD Ryzen Threadripper 3970X 32-Core CPU with 3.7 GHz frequency.

Benchmark results for the state-transfer optimization are shown in Fig. 4.1. I consider a single

[1] AD is implemented by the package `Autograd` [104] in this benchmark. This package does not support sparse linear algebra, so I implement dense matrix storage for the Hamiltonian in both full-AD and HG for fair comparison.

time step ($N = 1$) and measure the runtime versus d. The results in Fig. 4.1(a) confirm that full-AD, semi-AD, and HG have the same scaling of runtime per time step with respect to d (Table 4.1). The observed runtime scaling $\Theta(d^{5/2})$ is consistent with $\kappa \sim \Theta(d^2)$ due to dense matrix storage for H_1, and $\mu \sim \Theta(d^{1/2})$ (see Appendix A). The scaling of the required memory resources is illustrated in Fig. 4.1(b): memory usage scales with dimension d as $\Theta(d^2)$ for full-AD, semi-AD, and HG, consistent with $\kappa \sim \Theta(d^2)$ being the dominant contribution. As anticipated, the scaling of memory with N differs between full-AD, semi-AD, and HG [Fig. 4.1(c)]. For full-AD and semi-AD, the scaling is linear with N, while it is independent of N for HG. The slope of full-AD is much larger than the one of semi-AD, since the former has extra dependence on μ (Table 4.1), which is a constant ≈ 100 in this benchmark. The results confirm the favorable memory efficiency of HG and semi-AD compared to full-AD in state-transfer problems with large number of time steps.

4.2.2 Gate-operation benchmark

I next turn to an example for optimizing a unitary gate. This benchmark is performed for three driven transmons with nearest-neighbor coupling. The target gate consists of simultaneous Hadamard operations on each of the three transmons. For concreteness, I consider the case of transmons with the same frequency, driven resonantly. The full Hamiltonian in the rotating frame of the drive is

$$H_2(t) = \sum_{\nu=1}^{3} \left[-\frac{K_b}{2} b_\nu^{\dagger 2} b_\nu^2 + a^{(\nu)}(t)(b_\nu + b_\nu^\dagger) \right] + g(b_1 b_2^\dagger + b_2 b_3^\dagger + \text{h.c.}). \tag{4.2}$$

The goal of the optimization is to minimize the infidelity C_2^g. The setup for this benchmark is the same as in the previous example.

Fig. 4.2 records the benchmarks based on a single time step ($N = 1$). The observed runtime scaling with dimension d is $\Theta(d^{11/3})$ for full-AD, semi-AD, and HG, see Fig. 4.2(a). This power law arises from $\Theta(\mu\kappa d)$ for $\kappa \sim \Theta(d^2)$ and $\mu \sim \Theta(d^{2/3})$ [see Appendix A]. The memory usage scaling with d is illustrated in Fig. 4.2(b), confirming that full-AD has an unfavorable extra $\mu \sim \Theta(d^{2/3})$ dependence compared to HG and semi-AD (Table 4.1). The memory usage scaling with N is as expected [Fig. 4.2(c)]: for semi-AD, the scaling is linear with N; for HG, it is independent of N. (The memory usage of full-AD in this case exceeds available resources, and thus is not shown.) Throughout this benchmark, this linear dependence with N causes semi-AD to consume about 10 times more memory than HG. This underlines that HG is preferable whenever memory constraints become a limitation in optimization problems with large Hilbert space dimension and large number of time steps. This improvement is achieved without worsening the runtime scaling.

4.3 Choosing among different optimization strategies

There are several considerations which determine whether full-AD, HG, or semi-AD is the optimal strategy for a given optimization problem, see the decision tree in Fig. 4.3. Whenever the Hilbert space dimension d and the number of time steps N are sufficiently small, memory usage may not be a bottleneck. In this case, full-AD may be preferable, since it eliminates the need for hard-coded gradients. If the memory cost of full-AD is not affordable but analytical gradients are tedious to compute, semi-AD may be a viable compromise, though still less memory efficient than HG. In all other cases, HG is the method of choice, since it can handle optimization with larger N and d compared to full-AD and semi-AD.

The choice of implementing HG via scaling and squaring or matrix diagonalization should be informed by whether the Hamiltonian is sparse and κ scales better than $\Theta(d^2)$. In the latter case, scaling and squaring is preferable and leads to an overall memory usage $\sim \Theta(d+\kappa)$, which is better than the memory scaling $O(d^2)$ of matrix diagonalization. By contrast, for Hamiltonians with

$\kappa \sim \Theta(d^2)$, these two methods are equivalent in memory usage scaling, and the one with better runtime for a specific optimization task should be selected. While the runtime for optimization using matrix diagonalization scales as $O(d^3)$ [Appendix B], the runtime for optimization based on scaling and squaring differs between state-transfer and gate optimizations: (1) for state transfer, the runtime scales as $\Theta(\mu d^2)$; hence, whenever μ scales better than $\Theta(d)$, scaling and squaring wins over matrix diagonalization; (2) for gate optimization, the runtime scales as $\Theta(\mu d^3)$; hence matrix diagonalization is preferable.

The scaling analysis and benchmarks presented in this chapter confirm that hard-coded gradients provide the most memory-efficient approach for GRAPE-based optimization of large quantum systems. In the next chapter, I apply these insights to a concrete problem: designing detuning-robust control pulses for transmon gates.

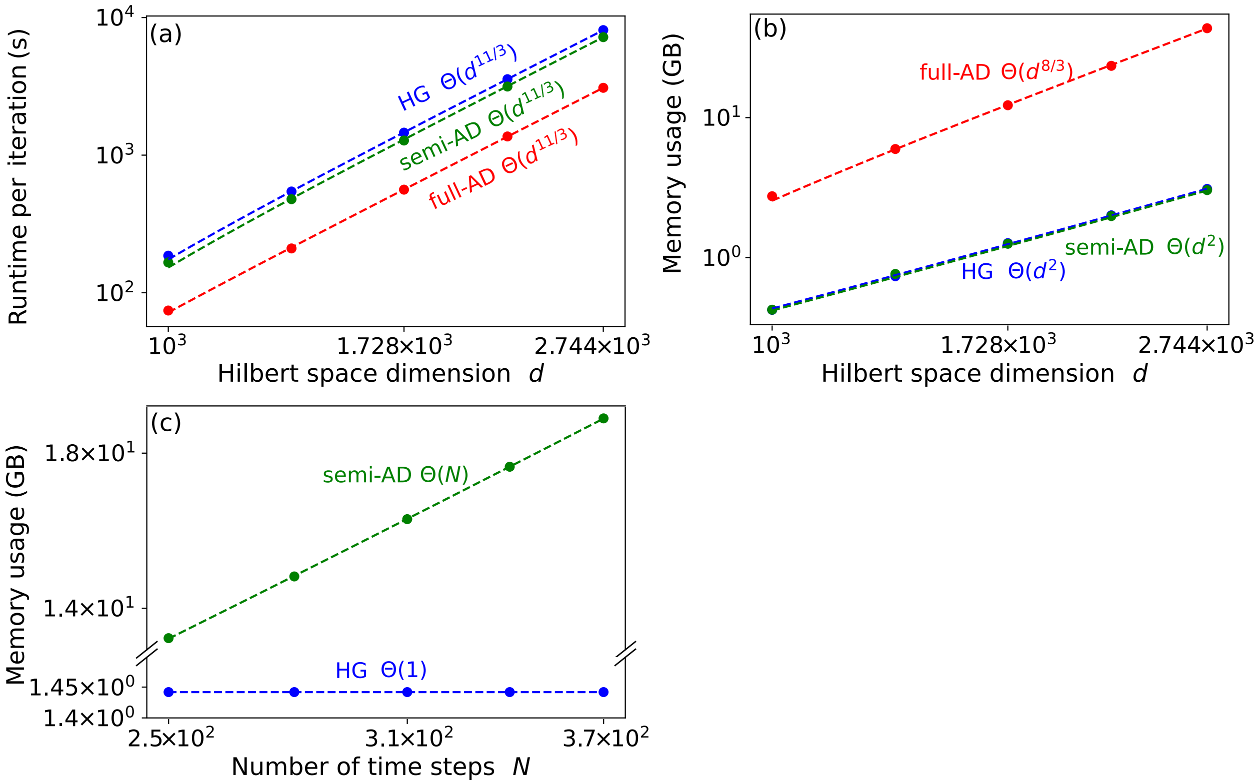

Figure 4.2: Benchmark results for unitary gate optimization in a system consisting of three coupled transmons. (a) Runtime per iteration and (b) memory usage versus Hilbert space dimension d. The total Hilbert space dimension d is increased by enlarging the cutoff of each transmon from 10 to 14 simultaneously. (c) Memory usage as a function of the number of time steps for $d = 12^3 = 1728$. Full-AD (not shown) would require more memory than available resources (estimated: 1 TB). All plots are in log-log scale. Dashed lines are power-law fits. (Parameters: $g/2\pi = 0.1$ GHz, $K_b/2\pi = 0.225$ GHz, and $a^{(\nu)}/2\pi = 0.1$ GHz.)

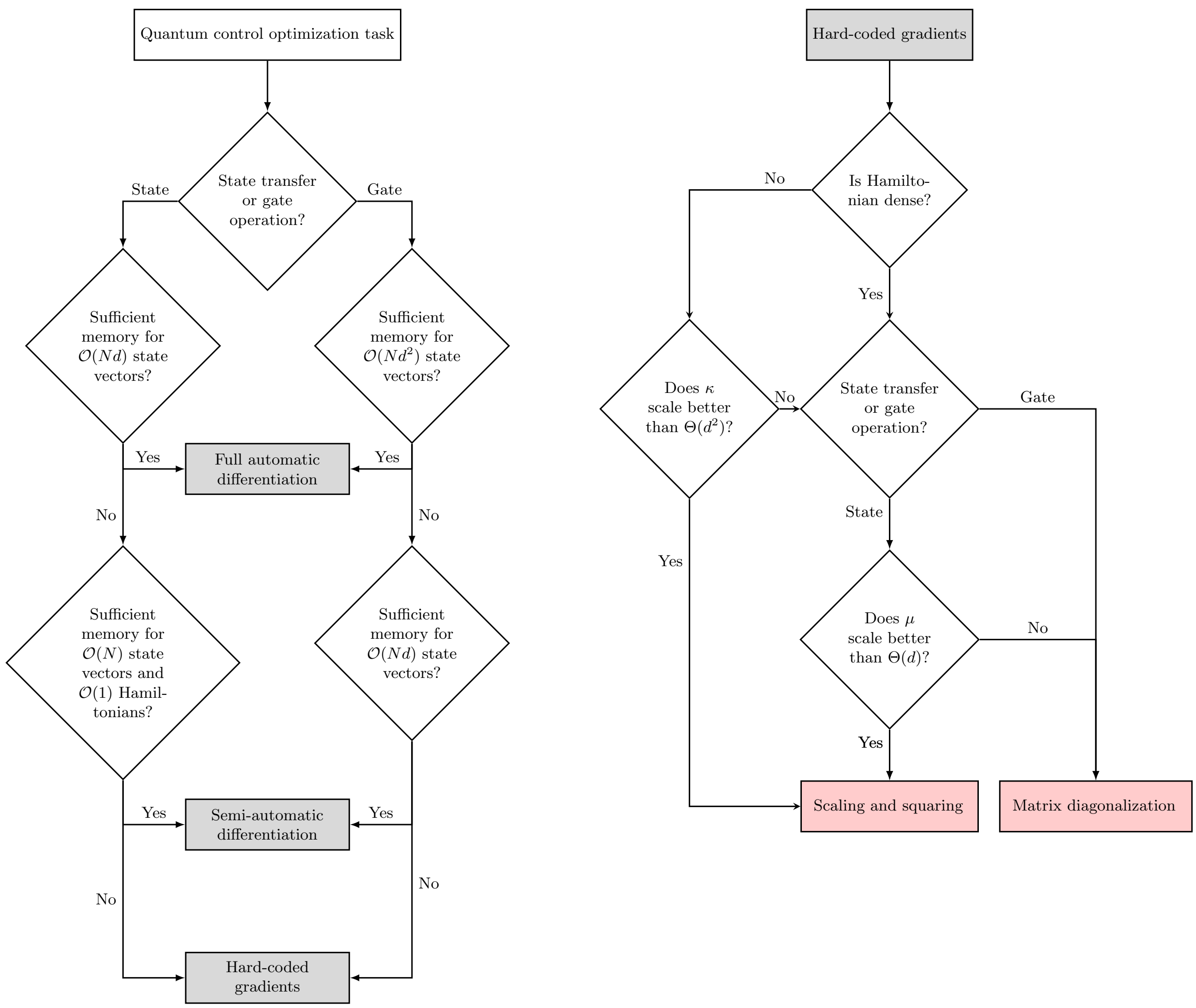

Figure 4.3: Decision trees for selecting the optimal numerical strategy. (a) Choosing among three numerical methods for evaluating gradients: full automatic differentiation, semi-automatic differentiation, hard-coded analytical gradients. (Here, $\mu \sim \Theta(d)$ is assumed; for different scaling see Table 4.1.) (b) Determining the appropriate strategy for evaluating propagator-state products.

CHAPTER 5

APPLICATION OF GRAPE: ROBUST CONTROL FOR TRANSMON

Quantum optimal control (QOC) provides a direct way to design microwave pulses that realize a desired state transfer or gate operation in bosonic qubits. An alternative approach is to synthesize the target operation from a decomposition into elementary control primitives, such as the ECD scheme discussed in Chapter 2. In this chapter, QOC is used as a complementary framework for constructing bosonic control protocols.

For quantum error correction, the integration of multiple bosonic modes is an important route toward universal quantum computation with protected logical qubits. In many hardware platforms for such multimode systems, a transmon ancilla is coupled directly to several cavity modes [114–117]. This architecture allows the ancilla to interact with all bosonic modes simultaneously and thereby enables both single-mode and multimode operations between arbitrary modes.

At the same time, this all-to-one architecture introduces significant crosstalk between the different bosonic modes. In the dispersive regime, the photon population in each cavity mode shifts the transition frequency of the transmon ancilla through the dispersive interaction. This becomes problematic because many standard control protocols rely on driving the ancilla near a specific resonant frequency. As a result, the occupation of one cavity mode can modify the ancilla frequency seen by the control pulse and thereby degrade the fidelity of operations acting on other modes.

In this chapter, we address the challenge of crosstalk by harnessing the power of quantum optimal control. Our approach involves the development and implementation of robust ancilla control with respect to detuning due to occupations in the cavity modes.

Detuning-robust control for two-level systems has been extensively investigated using various approaches, including reverse engineering [118, 119], space curve quantum control [120], pulse engineering [120–128], and the collocation method [129, 130]. However, protocols developed for two-level systems cannot be directly applied to weakly anharmonic transmons due to leakage out of the qubit subspace. One promising solution is to apply the DRAG scheme to robust single-quadrature pulses developed for two-level systems [131]. Additionally, Ref. [132] directly optimizes two-quadrature pulses for transmons. In the following, I introduce an alternative optimization-based procedure to design detuning-robust control for transmon X_θ gates and benchmark its performance.

5.1 Metrics of robust control

During the gate operation, the Hamiltonian of interest may depend on an unknown quantity δ. This uncertainty can cause the realized gate to deviate from the target gate, thus generally lowering gate fidelity $\mathcal{I}(\delta)$. To quantify the robustness of the infidelity against δ, I consider both the mean infidelity and its standard deviation as key metrics,

$$\mathcal{I}_\mu = \int_{\mathbb{D}} \mathrm{d}\delta\, p(\delta)\mathcal{I}(\delta), \quad \mathcal{I}_\sigma = \sqrt{\int_{\mathbb{D}} \mathrm{d}\delta\, p(\delta)\big[\mathcal{I}(\delta) - I_\mu\big]^2}. \tag{5.1}$$

Here, the probability distribution $p(\delta)$ satisfies $\int_{\mathbb{D}} \mathrm{d}\delta\, p(\delta) = 1$ for a specified range $\mathbb{D}$.

In the case of interest, the Hamiltonian governing the transmon qubit in the co-rotating frame of the drive frequency is

$$H_{\text{trans}}(t) = -\frac{K_b}{2} b^{\dagger 2} b^2 + \varepsilon_I(t)(b^\dagger + b) + i\varepsilon_Q(t)(b^\dagger - b) + \delta b^\dagger b. \tag{5.2}$$

Here b and $b^\dagger$ are the transmon ladder operators in the same convention as mode b in Sec. 2.4.2

(cf. Eq. (2.94)), with $K_b > 0$ the corresponding Kerr coefficient. The quantities $\varepsilon_I(t)$ and $\varepsilon_Q(t)$ describe control envelopes in the I and Q quadratures, and δ denotes the difference between the drive frequency and the unknown shifted qubit frequency.The closed-system gate infidelity follows

$$\mathcal{I}_{\mathrm{c}}(\delta) = 1 - \frac{1}{4}\left|\mathrm{Tr}\left[P_{0,1}U^{\dagger}_{\text{target}}U_{\delta}(t_{\mathrm{g}})\right]\right|^2, \tag{5.3}$$

where $P_{0,1}$ is a projector onto the computational subspace formed by the lowest two eigenstates. Here, U_{target} is the target unitary on the transmon, and $U_{\delta}(t_{\mathrm{g}}) = \mathcal{T}\exp\left[-i\int_0^{t_{\mathrm{g}}}\mathrm{d}\tau\, H_{\text{trans}}(\tau)\right]$ is the actual unitary achieved over a gate duration t_{g}, where $\mathcal{T}$ is the time-ordering operator.

In the presence of decoherence, the open-system infidelity follows [91]

$$\mathcal{I}_{\mathrm{o}}(\delta) = 1 - \frac{1}{4}\,\mathrm{Tr}\Big[\mathcal{P}_{0,1}\mathcal{L}^{\dagger}_{\text{target}}\mathcal{L}_{\delta}(t_{\mathrm{g}})\Big]. \tag{5.4}$$

Here, the density matrix is vectorized by stacking the elements row-by-row. $\mathcal{P}_{0,1} = P_{0,1}\otimes P_{0,1}$ is the superprojector onto the computational subspace, $\mathcal{L}_{\text{target}} = U_{\text{target}}\otimes U^{*}_{\text{target}}$ is the target superoperator, and $\mathcal{L}_{\delta}(t_{\mathrm{g}})$ is the superoperator realized by the Lindblad master equation. Note that in the absence of decoherence, $\mathcal{L}_{\delta}(t_{\mathrm{g}}) = U_{\delta}(t_{\mathrm{g}})\otimes U^{*}_{\delta}(t_{\mathrm{g}})$.

5.2 Optimization procedure

Quantum optimal control has been extensively used to develop robust control schemes by minimizing cost functions that encode robustness metrics. One approach is to penalize the derivative of the closed- or open-system fidelity with respect to detuning δ [131–133]. However, a more intuitive and pragmatic method is to penalize the average infidelity [134–138]

$$C_{\mathrm{o,c}} = \frac{1}{N}\sum_{i=1}^{N}\mathcal{I}_{\mathrm{o,c}}(\delta_i). \tag{5.5}$$

Here, δ_i denotes sample values of detunings within the specified range of $\mathbb{D}$. Minimization of the above cost function can suppress both mean infidelity and its standard deviation, as evidenced by the subsequent numerical results.

To minimize the cost function, I optimize the pulses, including both the envelopes $\varepsilon_{I,Q}(t)$ and the pulse duration t_{g}. An optimal duration is determined by balancing two competing factors: a shorter duration leads to unwanted leakage, whereas a longer duration results in decoherence errors. A direct approach to optimize pulses involves using an open-system optimizer for various durations and selecting the one that minimizes the cost function C_{o}. However, such optimization requires time-intensive open-system simulations. Instead, I propose a method that, while not guaranteed to be equivalent, yields promising results. The method consists of two steps: (1) employing closed-system optimization to minimize C_{c} for each potential duration, and (2) with given decoherence rates, evaluating C_{o} for the optimized pulses from the first step, then selecting the pulse with the lowest C_{o}. The approach is summarized in Fig. 5.1.

For the closed-system optimization, I use Gradient Ascent Pulse Engineering (GRAPE), a method widely employed for optimal control in various contexts [137, 139, 140]. The gradients required in GRAPE are calculated by using automatic differentiation, thereby eliminating the need for manually coding the analytical gradients of the cost function [96, 98, 141]. The initial pulse ansatz is

$$\begin{aligned}\varepsilon_I^{\text{initial}}(t) &= \frac{1}{t_{\text{g}}}\big[(a-\theta/2)\cos(2\pi t/t_{\text{g}}) + (b-a)\cos(4\pi t/t_{\text{g}}) \\ &\quad + (c-b)\cos(6\pi t/t_{\text{g}}) - c\cos(8\pi t/t_{\text{g}}) + \theta/2\big], \\ \varepsilon_Q^{\text{initial}}(t) &= \dot{\varepsilon}_I^{\text{initial}}(t)/K_b,\end{aligned} \tag{5.6}$$

where θ is the rotation angle for the target X_θ gate ($\theta = \pi, \pi/2$ in this study). This ansatz for the I-quadrature control has been used to obtain robust control for two-level systems [128]. The initial guess for the Q-quadrature is inspired by the DRAG pulse scheme [142] to suppress leakage. The pulse is then optimized starting from various initial guesses for parameters $a, b, c \in \{-10, -8, -6, \cdots, 10\}$, and the pulse with the lowest cost value C_{c} is selected. This process is accelerated by parallelizing with different initial values.

Throughout the closed-system optimization process, I ensure pulse smoothness by filtering out frequency components that exceed maximum allowed bandwidths of $5/t_{\text{g}}$ and $10/t_{\text{g}}$ for $\varepsilon_I(t)$ and $\varepsilon_Q(t)$, respectively. These empirical values effectively balance pulse smoothness and the level of robustness required.

5.3 Benchmarking and comparison

I benchmark the optimization procedure for a transmon with typical parameters: $K_b/2\pi = 200\,\text{MHz}$, $T_1 = 50\,\mu\text{s}$, and $T_\phi = 50\,\mu\text{s}$. The optimization results for the X_π gate are shown in Fig. 5.2. Figure 5.2(a) shows the average infidelities obtained for closed- and open-system simulations with varying pulse durations. Without decoherence, extending the pulse duration generally leads to a reduction in the cost value. In contrast, in the presence of decoherence, utilizing shorter pulses is more effective in reducing the infidelity. Given the particular transmon parameters under consideration, a pulse duration of 20 ns achieves the lowest cost. The temporal profiles and frequency components of the optimized 20-ns pulse are shown in Fig. 5.2(b) and (c), respectively. In Fig. 5.2(d), the infidelity as a function of detuning is studied, comparing the performance between the optimized and the DRAG pulses. In both closed- and open-system simulations, the optimized pulse consistently demonstrates substantially reduced infidelities across almost all detunings. Consequently, the optimized pulse exhibits enhanced robustness against frequency detuning in the transmon ancilla. The open-system infidelities obtained from these two pulses are limited by distinct factors: the infidelity derived from the QOC pulse is dominated by decoherence errors, while the one generated from the DRAG pulse is limited by coherent error.

I then compare the robustness of the pulses derived from this optimization procedure with selected recent literature. First, following the methodology outlined in Ref. [132], I obtain robust pulses using the `Q-CTRL` package [126, 143]. To distinguish between these results and those generated by the optimization procedure presented here, I use the label "Q-CTRL" for the former and "QOC" for the latter. The averaged infidelities [Eq. (5.5)] of the closed system as a function of pulse duration are shown in Fig. 5.3(a). While the average infidelities for Q-CTRL (red) and QOC (blue) are similar for shorter durations, QOC demonstrates orders of magnitude lower infidelity for longer durations. To illustrate the robustness details of the Q-CTRL results, I examine the 20-ns case, which exhibits the lowest infidelity. The I and Q quadratures of the pulse envelopes and their corresponding Fourier components are shown in Fig. 5.3(b) and (c). The infidelity as a function of detuning is presented in Fig. 5.3(d). For a closed system, the Q-CTRL results (red dashed line) show lower infidelity near resonance but exhibit higher infidelity off-resonance ($\delta/2\pi > 5\,\mathrm{MHz}$) compared to the QOC results (blue dashed line). This difference likely arises from the different cost functions used in the two optimizations: Q-CTRL penalizes the derivative of the infidelity at zero detuning, whereas QOC minimizes the average infidelity over a sampling of detunings ($\delta_i/2\pi \in$

$\{\pm 1, \pm 3, \pm 4, \pm 7, \pm 10\}$ MHz in this example). Notably, in the presence of ancilla decoherence ($T_1 = 50\,\mu$s, $T_\phi = 50\,\mu$s), the infidelity at small detuning is dominated by incoherent error. This makes QOC the overall better choice within the considered range of detuning for robust pulses.

The results of this chapter demonstrate that the GRAPE-based optimization procedure developed here generates detuning-robust transmon pulses that outperform both conventional DRAG pulses and Q-CTRL pulses across the relevant detuning range.

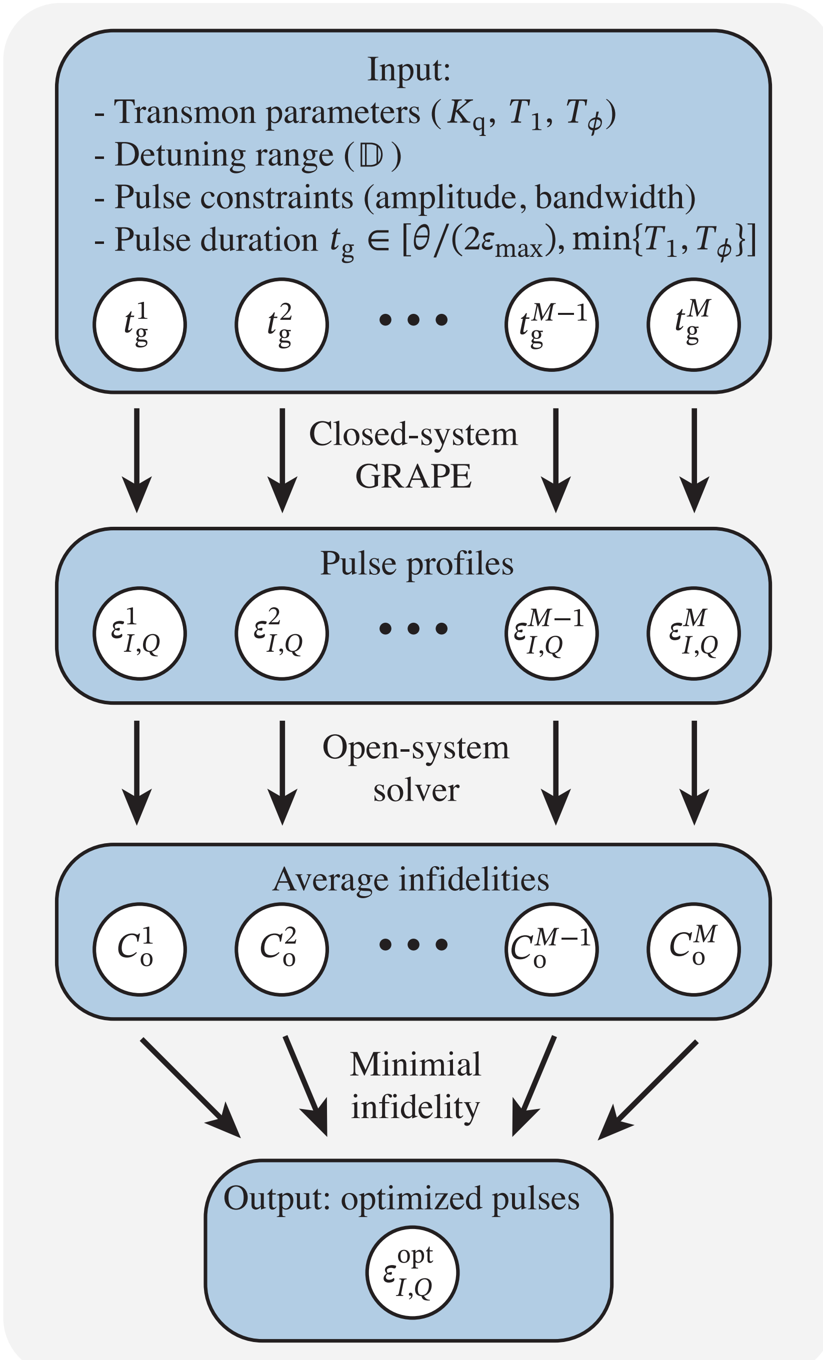

Figure 5.1: Optimization procedure to obtain detuning-robust X_θ gates in transmons. Input includes pulse constraints, transmon parameters, and a set of M pulse durations. The choice of duration should be shorter than the decay time T_1 and longer than $1/\varepsilon_{max}$, where ε_{max} is the maximum pulse amplitude. Closed-system GRAPE is performed for each duration to obtain the corresponding pulse profiles. Subsequently, an open-system solver evaluates the cost function C_o for pulses with varying durations, selecting the pulse that minimizes C_o.

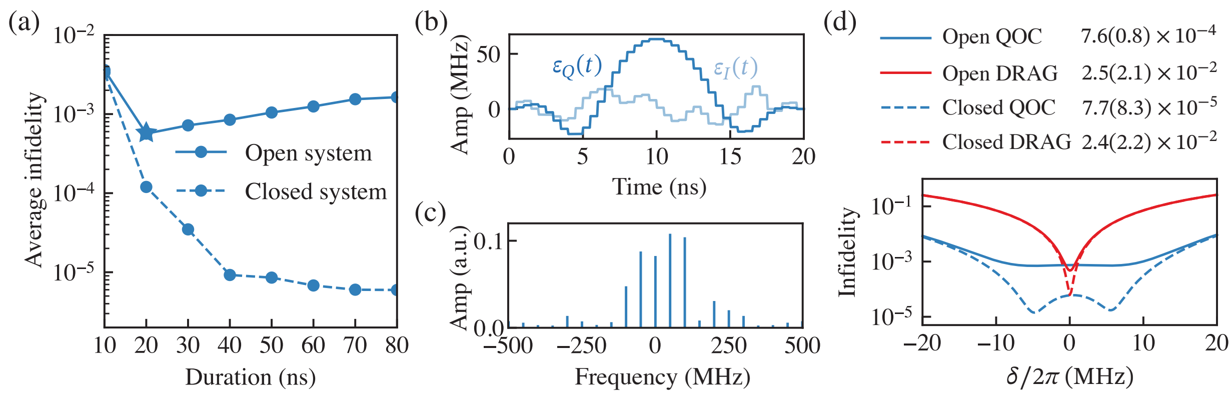

Figure 5.2: Optimization results for the X_π gate. (a) Average infidelities in Eq. (5.5) versus pulse durations for both closed (dashed line) and open (solid line) systems. The optimal duration selected from the range considered for this optimization is 20 ns (indicated by a star). (b) I and Q quadratures of the pulse envelopes for the selected 20-ns pulse from (a). (c) The frequency components of the pulse envelopes, corresponding to the 20-ns pulse. (d) Comparative analysis of the 20-ns pulses, from both QOC (blue) and DRAG (red) pulses, in terms of their infidelities as a function of detuning. The average and standard deviation of the infidelities are tabulated correspondingly by considering a uniform distribution of $\rho(\delta)$ for simplicity. The result shows that the optimized pulse is more robust than the commonly used DRAG pulse. [Parameters: $\delta_i/2\pi \in \{\pm1, \pm3, \pm4, \pm7, \pm10\}$ MHz, $K_b/2\pi = 200\,$MHz, $T_1 = 50\,\mu$s, $T_\phi = 50\,\mu$s.]

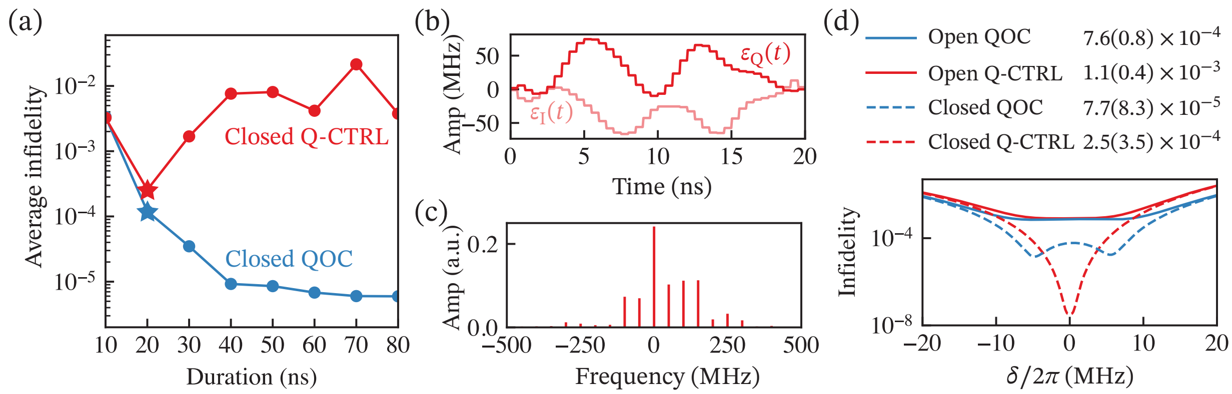

Figure 5.3: Comparison of optimization results between Q-CTRL and QOC for the X_π gate. (a) Average infidelities of a closed system [Eq. (5.5)] versus pulse durations derived from Q-CTRL (red) and QOC (blue). The optimal duration selected for Q-CTRL from the range considered is 20 ns (indicated by a red star). (b) I and Q quadratures of the Q-CTRL pulse envelopes for the selected 20-ns pulse from (a). (c) Frequency components of the Q-CTRL pulse envelopes corresponding to the 20-ns pulse. (d) Comparative analysis of the 20-ns pulses from both QOC (blue) and Q-CTRL (red) in terms of their infidelities as a function of detuning. The average and standard deviation of the infidelities are tabulated considering a uniform distribution of $\rho(\delta)$ for simplicity. [Parameters: $\delta_i/2\pi \in \{\pm1, \pm3, \pm4, \pm7, \pm10\}$ MHz, $K_b/2\pi = 200$ MHz, $T_1 = 50\,\mu$s, $T_\phi = 50\,\mu$s.]

CHAPTER 6
CAVITY DEPHASING INHERITED FROM TRANSMON FLUX NOISE

6.1 Introduction

Superconducting microwave cavities are a leading platform for quantum information processing because they enable information to be encoded in harmonic-oscillator modes with exceptionally long coherence times [51–53, 144, 145]. Their linear character suppresses many parasitic nonlinear loss channels, so photon damping is the dominant error mechanism. At the same time, the large oscillator Hilbert space enables hardware-efficient bosonic encodings. These features make cavities attractive as quantum memories and logical qubits, with bosonic quantum error-correction (QEC) schemes to protect information against dominant physical errors [60, 68, 68, 70, 71, 146–154]. Many protocols rely not only on long coherence, but also on a strong bias in the noise: photon loss is far more likely than dephasing. This biased error landscape is central to cavity encodings such as dual-rail qubits, whose fault-tolerance advantages derive from converting dominant single-photon loss into a detectable erasure channel [155]. Preserving the photon-loss-dominated bias is therefore as important as extending coherence itself.

Universal control of cavity modes generally requires dispersive coupling to a nonlinear element [81, 156–159]. A family of flux-tunable circuits such as the Superconducting Nonlinear Asymmetric Inductive eLement (SNAIL) [158], the Flux-Tunable Transmon (FTT) [160], and the Linear INductive Coupler (LINC) [159] enable beam-splitter interactions through three-wave mixing, providing a practical route to high-fidelity bosonic gates. Activating these interactions often requires biasing the nonlinear element away from its sweet spot, where flux noise causes fluctuations of the nonlinear-element frequency. In the dispersive regime, these fluctuations produce Lamb-shift fluctuations, leading to cavity-frequency fluctuations and thus cavity dephasing. This inherited dephasing is especially harmful because it introduces an error channel that directly competes with the favorable photon-loss-dominated bias. As the coupling strength between the cavity and the nonlinear element is increased to speed up gates, this inherited dephasing can become a leading decoherence mechanism, degrading the bias structure that bosonic QEC and dual-rail encodings rely on for their advantage.

A commonly used strategy is to idle the nonlinear element at a flux sweet spot and tune it away from the sweet spot only when operations are needed. In three-dimensional architectures, fast flux tuning can be implemented using magnetic hoses [161, 162] or on-chip flux lines [163, 164]. However, these schemes increase hardware complexity and can slow the operation of interest. Moreover, tuning flux might risk leakage. Thus, although sweet-spot biasing suppresses flux sensitivity, fast on-demand bosonic control still relies on flux excursions away from the sweet spot.

This work introduces an alternative method to suppress inherited cavity dephasing without changing the dc flux bias. An off-resonant drive applied to the nonlinear mode induces a photon-number-dependent AC-Stark shift of the cavity transition frequency. With suitable drive parameters, this shift renders the cavity frequency first-order insensitive to flux noise, thereby creating a driven sweet spot. This mechanism differs from the dynamical sweet spots studied for qubits in [165, 166], where a resonant or near-resonant drive on the qubit itself engineers the sweet spot. Here, by contrast, the drive acts on the nonlinear mode rather than on the protected cavity mode, and it introduces only a small correction to the cavity decay rate, while substantially suppressing inherited dephasing. For a representative parameter set, Monte Carlo simulations show a $\sim 20\times$ improvement in the cavity dephasing time, from ~ 100 μs to ~ 2 ms with negligible effect on decay time. Our scheme is also different from dynamical decoupling [167]. The scheme does not require cavity gate sequences and therefore avoids the $\sim 1\ \mu$s overhead of typical cavity gates [82, 168].

The paper is organized as follows. Section 6.5 introduces the drive-induced suppression mechanism and derives the condition for first-order insensitivity to flux fluctuations. Section 7.1 develops a theory of the total cavity dephasing rate, including relaxation and $1/f$ flux noise, and validates the analytical predictions numerically. Section 8.1 applies the framework to bosonic encodings, emphasizing implications for binomial and dual-rail architectures. Section 8.2 summarizes the conclusions, with technical details deferred to the Appendices. While the analysis focuses on the FTT–cavity system as a concrete example, the underlying mechanism extends broadly to flux-tunable nonlinear elements.

6.2 Hamiltonian Derivation and Noise Analysis

This appendix provides an analytical approximation showing that, to leading order, the dominant effect of flux noise is a fluctuation of the FTT frequency.

The full Hamiltonian of the system shown in Fig. 6.1(a) is

$$H_{\text{full}}(t) = 4E_C(n-n_g)^2 - E_{J,\text{eff}}(\Phi+\delta\Phi(t))\cos[\varphi-\varphi_0(\Phi+\delta\Phi(t))] + \omega_c c^\dagger c + g' n(c^\dagger + c) + 2\Omega_0'\cos(\omega_d t)\, n. \tag{6.1}$$

Here E_C is the charging energy, n is the charge operator, and n_g is the offset charge. The operator c annihilates photons in the cavity mode of frequency ω_c. The charge–charge coupling strength is g', and the final term describes a charge drive applied to the FTT with amplitude Ω_0' and frequency ω_d.

Both the effective Josephson energy $E_{J,\text{eff}}$ and the potential phase φ_0 depend on flux:

$$E_{J,\text{eff}}(\Phi + \delta\Phi(t)) = (E_{J1} + E_{J2}) \cos\left[\frac{\pi\,(\Phi + \delta\Phi(t))}{\Phi_0}\right],$$
$$\varphi_0(\Phi + \delta\Phi(t)) = d\,\tan\left[\frac{\pi\,(\Phi + \delta\Phi(t))}{\Phi_0}\right], \quad (6.2)$$

where E_{J1} and E_{J2} are the Josephson energies of the two junctions, $\Phi_0 = h/2e$ is the superconducting flux quantum, and $d \equiv (E_{J2} - E_{J1})/(E_{J1} + E_{J2})$ is the SQUID-loop asymmetry.

Flux noise in these systems is typically observed to follow a $1/f$ characteristic. Accordingly, the power spectral density of these Gaussian fluctuations is

$$S_{1/f}(\omega) = \int_0^t \langle \delta\Phi(0)\,\delta\Phi(\tau)\rangle e^{i\omega\tau}\,d\tau = \frac{A_0^2}{|\omega/2\pi|}, \quad (6.3)$$

where A_0 is the flux-noise spectral-density amplitude.

For small flux fluctuations $\delta\Phi(t) \ll \Phi_0$, a Taylor expansion of H_{full} gives

$$H_{\text{full}}(\Phi + \delta\Phi(t)) \approx H_{\text{sys}}(\Phi) + \underbrace{\delta\Phi(t)\frac{\partial H_{\text{sys}}}{\partial \Phi}}_{H'_{\text{noise}}} + \mathcal{O}\big((\delta\Phi)^2\big), \quad (6.4)$$

where the system Hamiltonian is evaluated at the static bias point Φ,

$$\begin{aligned} H_{\text{sys}}(\Phi) &= 4E_C(n - n_g)^2 - E_{J,\text{eff}}(\Phi)\cos[\varphi - \varphi_0(\Phi)] \\ &\quad + \omega_c c^\dagger c + g' n(c^\dagger + c) + 2\Omega_0' \cos(\omega_d t)\, n. \end{aligned} \quad (6.5)$$

The dependence on Φ is dropped below for compactness.

In the transmon regime ($E_C \ll E_{J,\text{eff}}$), the wavefunction is localized in the potential well, and the Hamiltonian can be expressed using ladder operators:

$$
\begin{aligned}
\varphi - \varphi_0 &= \varphi_{\text{ZPF}} \left(b + b^\dagger\right), \\
n &= \frac{-i\left(b - b^\dagger\right)}{2\varphi_{\text{ZPF}}}, \\
\varphi_{\text{ZPF}} &= \left(\frac{2E_C}{E_{J,\text{eff}}(\Phi)}\right)^{\frac{1}{4}}.
\end{aligned}
\tag{6.6}
$$

Expanding the cosine potential to fourth order and applying the rotating wave approximation (RWA) yields

$$
\begin{aligned}
H_{\text{sys}}(t) \approx{}& \omega_b b^\dagger b + \frac{K}{2} b^{\dagger 2} b^2 + \omega_c c^\dagger c \\
&+ g(b + b^\dagger)(c + c^\dagger) + 2\Omega_0 \cos(\omega_d t)(b + b^\dagger),
\end{aligned}
\tag{6.7}
$$

where the FTT frequency, anharmonicity, renormalized drive amplitude and coupling strength are $\omega_b = \sqrt{8E_C E_{J,\text{eff}}} - E_C$, $K = -E_C$, $\Omega_0 = \Omega_0'/2\varphi_{\text{ZPF}}$ and $g = g'/2\varphi_{\text{ZPF}}$ respectively.

The noise Hamiltonian in the oscillator basis defined at the static flux bias Φ is

$$
\begin{aligned}
H'_{\text{noise}}(t) &= \delta\Phi(t) \frac{\partial H_{\text{sys}}}{\partial \Phi} \\
&= -\delta\Phi(t) \Bigg[\frac{\partial E_{J,\text{eff}}}{\partial \Phi} \cos(\varphi - \varphi_0) \\
&\qquad + E_{J,\text{eff}} \frac{\partial \varphi_0}{\partial \Phi} \sin(\varphi - \varphi_0) \Bigg] \\
&\approx -\delta\Phi(t) \Bigg[\frac{\partial E_{J,\text{eff}}}{\partial \Phi} \left(1 - \frac{(\varphi - \varphi_0)^2}{2}\right) \\
&\qquad + E_{J,\text{eff}} \frac{\partial \varphi_0}{\partial \Phi} (\varphi - \varphi_0) + \mathcal{O}\big((\varphi - \varphi_0)^3\big) \Bigg] \\
&\approx -\delta\Phi(t) \Bigg[\frac{\partial E_{J,\text{eff}}}{\partial \Phi} \left(1 - \frac{\varphi_{\text{ZPF}}^2}{2} (b + b^\dagger)^2\right) \\
&\qquad + E_{J,\text{eff}} \frac{\partial \varphi_0}{\partial \Phi} \varphi_{\text{ZPF}} (b + b^\dagger) \Bigg].
\end{aligned}
\tag{6.8}
$$

This expression contains a term linear in $(b + b^\dagger)$ and a quadratic contribution proportional to $(b+b^\dagger)^2$. The linear term drives transitions near the oscillator frequency ω_b, with rates governed by $S_{1/f}(\omega_b)$. Expanding $(b+b^\dagger)^2 = 2b^\dagger b+1+b^2+b^{\dagger 2}$ shows that the squeezing terms (b^2 and $b^{\dagger 2}$) drive transitions near $2\omega_b$, with rates proportional to $S_{1/f}(2\omega_b)$. Because $S_{1/f}(\omega)$ is strongly suppressed at these high frequencies ($\omega_b/2\pi \sim$ GHz) compared to the low-frequency noise ($\omega/2\pi \to 0$ Hz) responsible for pure dephasing, both the linear and squeezing transition channels are neglected. Retaining only the number-conserving contribution ($\propto b^\dagger b$) from the quadratic term gives

$$H'_{\text{noise}}(t) \approx \frac{\partial E_{J,\text{eff}}}{\partial \Phi}\, \varphi_{\text{ZPF}}^2\, \delta\Phi(t)\; b^\dagger b, \tag{6.9}$$

up to an additive constant. Using

$$\frac{\partial \omega_b}{\partial \Phi} = \frac{\partial \omega_b}{\partial E_{J,\text{eff}}} \frac{\partial E_{J,\text{eff}}}{\partial \Phi} = \varphi_{\text{ZPF}}^2\, \frac{\partial E_{J,\text{eff}}}{\partial \Phi}, \tag{6.10}$$

one obtains

$$H'_{\text{noise}}(t) \approx \frac{\partial \omega_b}{\partial \Phi}\, \delta\Phi(t)\; b^\dagger b, \tag{6.11}$$

which is the noise Hamiltonian in Eq. (6.49).

6.3 Obtain H_d by a Schrieffer–Wolff transformation

Starting from the rotating-frame Hamiltonian (three-level truncation of the FTT),

$$\begin{aligned} H_R = {}& \Delta_{bd}\, |e\rangle\langle e| + \big(2\bar{\omega}_b + K\big)\, |f\rangle\langle f| + \bar{\omega}_c\, c^\dagger c \\ & + \chi\big(|e\rangle\langle e| + 2|f\rangle\langle f|\big)c^\dagger c \\ & + \Omega_0\Big(|g\rangle\langle e| + \sqrt{2}\, |e\rangle\langle f| + \text{h.c.}\Big), \end{aligned} \tag{6.12}$$

the weak-drive regime $\Omega_0 \ll K$ allows the decomposition

$$H_R = H_0 + V, \tag{6.13}$$

where H_0 is the Ω_0-independent part of Eq. (6.12) (diagonal in the bare basis) and V is the drive term.

A Schrieffer–Wolff transformation uses an anti-Hermitian generator $D_d = -D_d^\dagger$ to obtain

$$H_d' = e^{-D_d} H_R e^{D_d}, \tag{6.14}$$

which is diagonal in the bare basis. To first order,

$$\begin{aligned} D_d = \\ \sum_m \left[-\frac{\Omega_0}{\Delta_m} |e,m\rangle\langle g,m| + \frac{\sqrt{2}\,\Omega_0}{\Delta_m + K} |f,m\rangle\langle e,m| - \text{h.c.} \right] \\ \approx \sum_m \left[-\frac{\Omega_0}{\Delta_m} |e,m\rangle\langle g,m| + \frac{\sqrt{2}\,\Omega_0}{K} |f,m\rangle\langle e,m| - \text{h.c.} \right], \end{aligned} \tag{6.15}$$

where the approximation keeps the leading order in $\mathcal{O}(\Delta_m/K)$.

Keeping terms up to second order in the SWT gives

$$\begin{aligned} H_d' \approx H_R + \sum_m \Omega_0^2 \Bigg[-\frac{1}{\Delta_m} |g,m\rangle\langle g,m| + (\frac{1}{\Delta_m} \\ - \frac{1}{K})|e,m\rangle\langle e,m| + \frac{1}{K} |f,m\rangle\langle f,m| \Bigg]. \end{aligned} \tag{6.16}$$

Because the FTT is initialized in $|g\rangle$ and $\Omega_0 \ll K$, population of $|f\rangle$ remains negligible;

projecting onto the $\{|g\rangle, |e\rangle\}$ subspace yields

$$\begin{aligned} H_d = \widetilde{\Delta}_{bd}\sigma^+\sigma^- + \bar{\omega}_c\, c^\dagger c - \sum_m \frac{\Omega_0^2}{\Delta_m}\, |m\rangle\langle m| \\ + \sigma^+\sigma^-\Bigg[\chi\, c^\dagger c + \sum_m \frac{2\Omega_0^2}{\Delta_m}\, |m\rangle\langle m|\Bigg], \end{aligned} \tag{6.17}$$

where $\widetilde{\Delta}_{bd} = \Delta_{bd} - \frac{\Omega_0^2}{K}$ is the shifted FTT frequency.

6.4 Transformation

This section derives H_d by transforming $H(t)$ and then expresses the master equation in Eq. (6.31) in the corresponding effective drive frame.

6.4.1 Transformation of Hamiltonian

In the regime $|g| \ll |\omega_c - \omega_b| \ll |\omega_c + \omega_b|$, counter-rotating terms can be neglected in the Hamiltonian H_{sys} (Eq. (6.31)), giving

$$\begin{aligned} H(t) \approx\, & \omega_b b^\dagger b + \omega_c c^\dagger c + \frac{K}{2} b^{\dagger 2} b^2 \\ & + g\,(bc^\dagger + b^\dagger c) + 2\Omega_0 \cos(\omega_d t)\,(b + b^\dagger). \end{aligned} \tag{6.18}$$

The static Hamiltonian H_{static} consists of the first four terms. To remove the interaction term $g(bc^\dagger + b^\dagger c)$, a second-order Schrieffer–Wolff transformation is applied with $U_{\text{disp}} = e^{D_{\text{disp}}}$,

$$H_{\text{static}}^{\text{disp}} = e^{-D_{\text{disp}}} H_{\text{static}} e^{D_{\text{disp}}}. \tag{6.19}$$

The generator D_{disp} is chosen to eliminate off-diagonal terms at first order,

$$D_{\text{disp}} = \sum_n \sqrt{n+1}\,\frac{g}{\omega_b + nK - \omega_c}\,|n+1\rangle\langle n|\,c - \text{h.c.} \tag{6.20}$$

Expanding in the detuning Δ_{bc} yields

$$D_{\text{disp}} \approx \frac{g}{\Delta_{\text{bc}}} b^\dagger c - \frac{gK}{\Delta_{\text{bc}}^2}(b^\dagger b - 1)\, b^\dagger c + \mathcal{O}\big((nK/\Delta_{\text{bc}})^2\big)\,. \tag{6.21}$$

Applying the same transformation to the full (driven) Hamiltonian gives

$$\begin{aligned} H'_{\text{disp}} &= e^{-D_{\text{disp}}} H(t)\, e^{D_{\text{disp}}} \\ &\approx H_{\text{disp}} + \Omega_0 \cos(\omega_d t)\, \Big[e^{-D_{\text{disp}}}(b + b^\dagger) e^{D_{\text{disp}}}\Big] \\ &\approx H_{\text{disp}} + \Omega_0 \cos(\omega_d t)\, \Big[b^\dagger + [D_{\text{disp}}, b] + \text{h.c.}\Big], \end{aligned} \tag{6.22}$$

where $H_{\text{disp}} \equiv e^{-D_{\text{disp}}} H_{\text{static}} e^{D_{\text{disp}}}$. Using the expansion above,

$$[D_{\text{disp}}, b] \approx -\frac{g}{\Delta_{\text{bc}}}\, c - \frac{Kg}{\Delta_{\text{bc}}^2}\, b\, c^\dagger + \frac{2Kg}{\Delta_{\text{bc}}^2}\, b^\dagger b\, c. \tag{6.23}$$

These terms rotate at frequencies set by detunings involving the b–c manifold and the bare c mode. Because ω_d is far detuned from these transitions, the rotating-wave approximation justifies neglecting them, leaving

$$H'_{\text{disp}} \approx H_{\text{disp}} + \Omega_0 \cos(\omega_d t)\, (b + b^\dagger). \tag{6.24}$$

The explicit time dependence of the drive is removed by moving to the rotating frame defined by $U_R = e^{i\omega_d t\, b^\dagger b}$. The transformed Hamiltonian is

$$\begin{aligned} H_R' &= U_R^\dagger \left(H_{\text{disp}}' - i\frac{\partial}{\partial t} \right) U_R \\ &\approx H_R + \left(\Omega_0 e^{-i\omega_d t} b + \text{h.c.} \right), \end{aligned} \tag{6.25}$$

and a final RWA removes counter-rotating contributions at frequency $\pm 2\omega_d$, yielding the effective Hamiltonian H_R.

The subsequent transformation with U_d (Appendix 6.3) maps H_R to H_d.

6.4.2 Transformation of H_{noise}

We consider the flux-noise Hamiltonian $H_{\text{noise}} = \frac{\partial \omega_b}{\partial \Phi} \delta\Phi(t) b^\dagger b$. After applying the dispersive transformation $U_{\text{disp}} = e^{D_{\text{disp}}}$, the number operator becomes

$$\begin{aligned} e^{-D_{\text{disp}}} b^\dagger b e^{D_{\text{disp}}} &= e^{-D_{\text{disp}}} b^\dagger e^{D_{\text{disp}}} e^{-D_{\text{disp}}} b e^{D_{\text{disp}}} \\ &\approx \left(b^\dagger + [D_{\text{disp}}, b^\dagger] \right) \left(b + [D_{\text{disp}}, b] \right) \\ &\approx b^\dagger b + [D_{\text{disp}}, b^\dagger b] + [D_{\text{disp}}, b^\dagger][D_{\text{disp}}, b] \\ &\overset{(6.20)}{\approx} b^\dagger b + \frac{g^2}{\Delta_{bc}^2} c^\dagger c - \frac{4Kg^2}{\Delta_{bc}^3} b^\dagger b c^\dagger c \\ &\quad + \left(-\frac{g}{\Delta_{bc}} b c^\dagger + \frac{Kg}{\Delta_{bc}^2} b^\dagger b c^\dagger + \text{h.c.} \right). \end{aligned} \tag{6.26}$$

The non-number-conserving contributions (e.g., terms proportional to $bc^\dagger$ and $b^\dagger b\, c^\dagger$, together with their Hermitian conjugates) drive transitions at frequencies near $\Delta_{bc}/2\pi \sim$ GHz. For $1/f$ flux noise, $S_{1/f}(\omega)$ is negligible at such frequencies, so these rapidly oscillating terms are discarded.

The resulting dispersively transformed noise Hamiltonian is

$$H_{\text{noise}}^{\text{disp}} \approx \frac{\partial \omega_b}{\partial \Phi}\delta\Phi \left(b^\dagger b + \frac{g^2}{\Delta_{bc}^2}c^\dagger c - \frac{4Kg^2}{\Delta_{bc}^3}b^\dagger b c^\dagger c\right). \tag{6.27}$$

In the rotating frame defined by U_R, the remaining terms are diagonal in the relevant number operators, so the form is unchanged: $H_{\text{noise}}^{R} = H_{\text{noise}}^{\text{disp}}$. Under the drive transformation $U_d = e^{D_d}$, the noise Hamiltonian is expanded to second order,

$$\begin{aligned} {H_{\text{noise}}^{d}}' &= U_d^\dagger H_{\text{noise}}^{R} U_d \\ &\approx H_{\text{noise}}^{R} + [D_d, H_{\text{noise}}^{R}] + \frac{1}{2}[D_d, [D_d, H_{\text{noise}}^{R}]], \end{aligned}$$

where we truncate the FTT Hilbert space to three levels in the regime of interest $|\Omega_0/K| \ll 1$,

$$\begin{aligned} H_{\text{noise}}^{R} \approx \frac{\partial \omega_b}{\partial \Phi}\delta\Phi \Bigg(&|e\rangle\langle e| + 2|f\rangle\langle f| + \frac{g^2}{\Delta_{bc}^2}c^\dagger c \\ &- \frac{4Kg^2}{\Delta_{bc}^3}\left(|e\rangle\langle e| + 2|f\rangle\langle f|\right)c^\dagger c\Bigg). \end{aligned} \tag{6.28}$$

Evaluating the required commutators:

$$\begin{aligned} &\left[D_d,\, b^\dagger b\right] = -\sum_m \big[\frac{\Omega_0}{\Delta_m}|g,m\rangle\langle e,m| \\ &+ \frac{\sqrt{2}\Omega_0}{K}|e,m\rangle\langle f,m| + \text{h.c.}\big] \\ &\frac{1}{2}\left[D_d,\left[D_d, b^\dagger b\right]\right] = \sum_m \Bigg[2\left(\frac{\Omega_0}{\Delta_m}\right)^2|g,m\rangle\langle g,m| \\ &- 2\left(\frac{\Omega_0}{\Delta_m}\right)^2|e,m\rangle\langle e,m| - \frac{4\Omega_0^2}{K^2}|f,m\rangle\langle f,m|\Bigg] \\ &[D_d, c^\dagger c] = 0 \end{aligned} \tag{6.29}$$

gives

$$H^{d}_{\text{noise}}{}' \approx \frac{\partial \omega_b}{\partial \Phi}\delta\Phi \left(\frac{g^2}{\Delta_{bc}^2} c^\dagger c + \left(\frac{\Omega_0}{\Delta_m}\right)^2 |m\rangle\langle m| \right.$$
$$+ \sigma^+\sigma^- - \sigma^+\sigma^- \left(\frac{4Kg^2}{\Delta_{bc}^3} c^\dagger c + 2\left(\frac{\Omega_0}{\Delta_m}\right)^2 |m\rangle\langle m| \right)$$
$$\left. - \sum_m \left[\frac{\Omega_0}{\Delta_m} \sigma^- |m\rangle\langle m| + \text{h.c.} \right] \right), \quad (6.30)$$

with a two-level truncation in the FTT.

6.4.3 Transformation of Lindbladian

Following Ref. [74], in the dispersive regime the reduced dynamics are described by the Lindblad master equation

$$\dot{\rho}_{disp} = -i[H_{\text{disp}}, \rho_{disp}] + \gamma_{b\downarrow}\mathcal{D}[b]\rho_{disp} + \gamma_{c\downarrow}\mathcal{D}[c]\rho_{disp}. \quad (6.31)$$

Here $\gamma_{b\downarrow}$ is the FTT relaxation rate, and $\gamma_{c\downarrow} = \frac{g^2}{\Delta_{bc}^2}\gamma_{b\downarrow}$ is the Purcell-induced cavity decay rate. The cavity relaxation is assumed to be Purcell-limited in state-of-the-art hardware designs [145].

In the rotating frame defined by U_R, the equation retains the same Lindblad structure:

$$\dot{\rho}_R = -i[H_R, \rho_R] + \gamma_{b\downarrow}\mathcal{D}[b]\rho_R + \gamma_{c\downarrow}\mathcal{D}[c]\rho_R, \quad (6.32)$$

where $\rho_R = U_R^\dagger \rho_{disp} U_R$.

Effective dissipation in the driven frame is obtained by transforming the jump operators with U_d. The FTT operator is first truncated to the three lowest levels ($|g\rangle, |e\rangle, |f\rangle$), so that $b \approx |g\rangle\langle e| + \sqrt{2}|e\rangle\langle f|$, and the drive transformation $U_d = e^{D_d}$ is applied. Projecting the result to the two-level

FTT subspace then yields

$$\begin{aligned}
e^{-D_d} b e^{D_d} &\approx b + [D_d, b] + \frac{1}{2}[D_d, [D_d, b]] \\
&\approx \sigma^- + \sum_m \Omega_0 \Bigg(-\frac{1}{\Delta_m} |m\rangle\langle m| \\
&\quad + \left[\frac{2}{\Delta_m} - \frac{2}{K}\right] \sigma^\dagger \sigma |m\rangle\langle m| \Bigg) \\
&\quad + \sum_m \Omega_0^2 \Bigg[\left(\frac{1}{K\Delta_m} - \frac{1}{\Delta_m^2}\right) \sigma^+ |m\rangle\langle m| \\
&\quad + \left(\frac{2}{K\Delta_m} - \frac{1}{\Delta_m^2}\right) \sigma^- |m\rangle\langle m| \Bigg],
\end{aligned} \tag{6.33}$$

where we truncate FTT to two levels. Similarly, the cavity operator c transforms under the influence of the drive:

$$\begin{aligned}
&e^{-D_d} c e^{D_d} \\
&\approx c + \sum_m \frac{\sqrt{m+1}\,\Omega_0 \chi}{\Delta_m \Delta_{m+1}} \big((\sigma^+ - \sigma^-)|m\rangle\langle m+1| \\
&\quad + \sum_m \sqrt{m+1}\,\Omega_0^2 \left(\frac{1}{\Delta_{m+1}^2} - \frac{1}{\Delta_m^2}\right) |m\rangle\langle m+1| \\
&\quad - \sum_m \sqrt{m+1} \frac{\Omega_0^2(\chi^2 + \Delta_m \chi)}{(\Delta_m \Delta_{m+1})^2} \sigma^+ \sigma^- |m\rangle\langle m+1|
\end{aligned} \tag{6.34}$$

In this driven frame, the Lindblad master equation becomes

$$
\begin{aligned}
\dot{\rho}_d = &-i[H_d, \rho_d] + \gamma_{b\downarrow}\mathcal{D}\left[e^{-D_d} b e^{D_d}\right]\rho_d \\
&+ \gamma_{c\downarrow}\mathcal{D}\left[e^{-D_d} c e^{D_d}\right]\rho_d \\
&\approx -i[H_d, \rho_d] + \\
&\gamma_{b\downarrow}\mathcal{D}\left[\sigma^- + \sum_m \Omega_0^2 \left(\frac{2}{K\Delta_m} - \frac{1}{\Delta_m^2}\right)\sigma^- |m\rangle\langle m|\right]\rho_d \\
&\quad + \gamma_{b\downarrow}\mathcal{D}\left[-\sum_m \frac{\Omega_0}{\Delta_m}|m\rangle\langle m|\right]\rho_d \\
&\quad + \gamma_{b\downarrow}\mathcal{D}\left[\sum_m \Omega_0^2 \left(\frac{1}{K\Delta_m} - \frac{1}{\Delta_m^2}\right)\sigma^+ |m\rangle\langle m|\right]\rho_d \\
&\quad + \sum_m \gamma_{c\downarrow,m}\mathcal{D}\big[|m\rangle\langle m+1|\big]\rho_d \\
&+ \mathcal{O}\left(\frac{g^2}{\Delta_{bc}^2}\frac{(m+1)\Omega_0^2\chi^2}{(\Delta_m\Delta_{m+1})^2}\right),
\end{aligned}
\tag{6.35}
$$

where $\rho_d = U_d^\dagger \rho_R U_d$. The effective cavity decay rate $\gamma_{c\downarrow,m}$ is perturbatively modified by the drive:

$$
\gamma_{c\downarrow,m} \approx (m+1)\left(1 - \Omega_0^2\left[\frac{1}{\Delta_{m+1}^2} - \frac{1}{\Delta_m^2}\right]\right)\gamma_{c\downarrow}. \tag{6.36}
$$

Equation (6.35) uses the secular approximation, valid when the relevant detunings and mode frequencies exceed the decay rates ($\widetilde{\Delta}_{bd}, \omega_c \gg \gamma_{b\downarrow}$). In the interaction picture, the transformed jump operator decomposes into spectral components $\sum_k L_k$ oscillating at distinct frequencies ω_k. When $\gamma \ll |\omega_j - \omega_k|$, rapidly oscillating cross-terms can be neglected, so that dissipation channels decouple:

$$
\gamma\mathcal{D}\left[\sum_k L_k\right]\rho \approx \sum_k \gamma\mathcal{D}[L_k]\rho. \tag{6.37}
$$

6.4.4 Further approximation under two regimes

We have transformed the Lindbladian in Eq. (6.31) into an effective drive frame, yielding the dissipators in Eq. (6.35) and the noise Hamiltonian in Eq. (6.30). We then derive simplified expressions for the Lindbladian and H^d_{noise} in the selective and unselective regimes discussed in Sec. 6.5.2.

Unselective regime Retain up to second order of $\mathcal{O}((\Delta_{bd}/\chi)^2)$, dissipators in Eq. (6.35) and the noise Hamiltonian in Eq. (6.30) reduce to

$$\begin{aligned}\dot{\rho} = &-i[H_d + H^d_{\text{noise}}, \rho] + \widetilde{\gamma}_{b\downarrow}\mathcal{D}\left[\sigma^-(1-\lambda_\downarrow c^\dagger c)\right]\rho \\ &+ 2\gamma_{c\phi}\mathcal{D}\left[c^\dagger c\right]\rho + \gamma_{b\uparrow}\mathcal{D}\left[\sigma^+(1-\lambda_\uparrow c^\dagger c)\right]\rho + \widetilde{\gamma}_{c\downarrow}\mathcal{D}[c]\rho,\end{aligned} \tag{6.38}$$

with the noise operator

$$\begin{aligned}H^d_{\text{noise}} \approx \delta\omega_b\Bigg[&\left(\frac{g^2}{\Delta_{bc}^2} - \frac{2\Omega_0^2\chi}{\Delta_{bd}^3}\right)c^\dagger c \\ &- 2\left(\frac{\Omega_0}{\Delta_{bd}}\right)^2\sigma^+\sigma^- + \left(\frac{4\Omega_0^2\chi}{\Delta_{bd}^3} - \frac{4Kg^2}{\Delta_{bc}^3}\right)\sigma^+\sigma^- c^\dagger c \\ &- \frac{\Omega_0}{\Delta_{bd}}\sigma_x + \frac{\Omega_0\chi}{\Delta_{bd}^2}\sigma_x c^\dagger c\Bigg],\end{aligned} \tag{6.39}$$

and normalized FTT relaxation rate, pure cavity dephasing rate, FTT excitation rate, normalized cavity decay rate respectively:

$$\widetilde{\gamma}_{b\downarrow} \approx \gamma_{b\downarrow}\left[1 + 2\left(\frac{2\Omega_0^2}{K\Delta_{bd}} - \frac{\Omega_0^2}{\Delta_{bd}^2}\right)\right] + \mathcal{O}\left(\frac{\Omega_0^4}{\Delta_{bd}^4}\right) \tag{6.40}$$

$$\gamma_{c\phi} = \gamma_{b\downarrow}\left(\frac{\Omega_0\chi}{\Delta_{bd}^2}\right)^2/2 \tag{6.41}$$

$$\gamma_{b\uparrow} = \gamma_{b\downarrow}\left[\Omega_0^2\left(\frac{1}{K\Delta_{bd}} - \frac{1}{\Delta_{bd}^2}\right)\right]^2 \tag{6.42}$$

$$\widetilde{\gamma}_{c\downarrow} \approx \gamma_{c\downarrow}\left(1 + \frac{2\chi\Omega_0^2}{\Delta_{bd}^3} + \mathcal{O}\left(\frac{\chi^2}{\Delta_{bd}^2}\right)\right). \tag{6.43}$$

When small parameters

$$\lambda_\downarrow \approx \frac{2\chi}{\Delta_{bd}}\Omega_0^2\left(\frac{1}{K\Delta_{bd}} - \frac{1}{\Delta_{bd}^2}\right) + \mathcal{O}\left(\frac{\Omega_0^4}{\Delta_{bd}^4}\right) \tag{6.44}$$

$$\lambda_\uparrow = \frac{\chi}{\Delta_{bd}}\left(\frac{2K - \Delta_{bd}}{K - \Delta_{bd}}\right). \tag{6.45}$$

The cavity-photon dependence of the decoherence rates originates from the drive-induced nonlinearity. For higher cavity occupations, the AC-Stark-shifted detuning Δ_m increases, which suppresses the FTT heating rate and drives the relaxation rate toward $\widetilde{\gamma}_{b\downarrow}$. In the main text, we neglect this photon-number dependence to keep the discussion focused on the leading scaling. Using Eq. (6.62), we then recover the simplified master equation quoted in subsection 7.1.1.

Selective regime. In the selective regime, the condition $|\Delta_{m^\star}/\chi| \ll 1$ holds. Combining the relation $\chi \approx 2\frac{g^2}{\Delta_{bc}^2}K$ with the hierarchy of energy scales gives

$$|\Omega_0| \ll |\Delta_{m^\star}| \ll |\chi| \ll |K|. \tag{6.46}$$

Accordingly, drive-induced corrections to the dissipators are retained up to $\mathcal{O}(\Omega_0/K)$. Terms of the form $\Omega_0/(\Delta_{m^\star}+m\chi)$ in Eq. (6.35) scale as $\mathcal{O}(\Omega_0/\chi)$ for $m \neq m^*$ (since $|\Delta_{m^\star}+m\chi| \sim |m\chi|$), whereas the resonant contribution scales as $\Omega_0/\Delta_{m^\star}$ for $m = m^*$. To leading order of $\Omega_0/\Delta_{m^\star}$, the master equation reduces to

$$\begin{aligned}
\dot{\rho} &\approx -i\,[H_d + H^d_{\rm noise},\, \rho] \\
&+ \gamma_{b\downarrow}\,\mathcal{D}\!\left[\sigma^- - \frac{\Omega_0^2}{\Delta_{m^\star}^2}\,\sigma^-|m^\star\rangle\langle m^\star|\right]\rho + 2\gamma_{c\phi}^{\rm sel}\,\mathcal{D}[|m^\star\rangle\langle m^\star|]\,\rho \\
&+ \gamma_{b\uparrow}^{\rm sel}\,\mathcal{D}\big[\sigma^+|m^\star\rangle\langle m^\star|\big]\,\rho + \sum_m \widetilde{\gamma}^{\rm sel}_{c\downarrow,m}\,\mathcal{D}[|m\rangle\langle m+1|]\,\rho,
\end{aligned} \tag{6.47}$$

and the noise Hamiltonian is

$$\begin{aligned}
H^d_{\rm noise} \approx \delta\omega_b(t)\Bigg[&\frac{g^2}{\Delta_{bc}^2}\,c^\dagger c + \frac{g^2}{\Delta_{bc}^2}\,|m^\star\rangle\langle m^\star| \\
&+ \sigma^+\sigma^-\left(\frac{4Kg^2}{\Delta_{bc}^3}\,c^\dagger c + 2\frac{g^2}{\Delta_{bc}^2}\,|m^\star\rangle\langle m^\star|\right) \\
&- \frac{g}{\Delta_{bc}}\,\sigma^+|m^\star\rangle\langle m^\star| + \text{h.c.}\Bigg].
\end{aligned} \tag{6.48}$$

Here, the selective-regime sweet-spot condition Eq. (6.64) has been used. At this operating point, the longitudinal component of $H^d_{\rm noise}$ produces no dephasing between $|g, m^\star\rangle$ and $|g, m^\star+1\rangle$. Meanwhile residual cavity dephasing from AC-Stark-shift fluctuations induced by $\delta\omega_b(t)$ is negligible for the dynamics of interest because the FTT remains near its ground state when initialized in $|g\rangle$. The transverse component instead induces depolarization; within a Markovian approximation, the corresponding excitation rate due to $1/f$ noise is $\gamma_{b\uparrow}^{1/f,\rm sel} = (\partial\omega_b/\partial\Phi)^2\,(g/\Delta_{bc})^2\,S_{1/f}(\Delta_{bd})$.

The dissipative processes in the effective drive frame can be summarized as follows: (i) FTT relaxation with a cavity-state-dependent correction, together with cavity decay at a renormalized rate $\gamma^{\text{sel}}_{c\downarrow,m} = \gamma^{\text{sel}}_{c\downarrow}\left(1 + \delta_{m^*,m} g^2/\Delta^2_{bc}\right)$, where $\gamma^{\text{sel}}_{c\downarrow} = (g^2/\Delta^2_{bc})\gamma_{b\downarrow}$ is the Purcell decay rate; (ii) FTT heating at rate $\gamma^{\text{sel}}_{b\uparrow} = (g^4/\Delta^4_{bc})\gamma_{b\downarrow}$ when the cavity occupies $|m^\star\rangle$; and (iii) pure dephasing between $|m^\star\rangle$ and $|m^\star + 1\rangle$ with rate $\gamma^{\text{sel}}_{c\phi} = (g^2/\Delta^2_{bc})\gamma^{\text{sel}}_{b\downarrow}/2$.

Dephasing between $|g, m^\star\rangle$ and $|g, m^\star + 1\rangle$ is therefore limited by cavity pure dephasing at rate $\gamma^{\text{sel}}_{c\phi} \propto g^2/\Delta^2_{bc}$, and by FTT heating at rate $\gamma^{\text{sel}}_{b\uparrow} \propto 1/\Delta_{bd}$ due to photon shot noise in the selective regime.

6.5 Dephasing mitigation by drive

6.5.1 System setup

The Hamiltonian of the FTT–cavity system is

$$H(t) = H_{\text{static}} + H_{\text{noise}}, \tag{6.49}$$

$$H_{\text{static}} = \omega_b\, b^\dagger b + \frac{K}{2}\, b^{\dagger 2} b^2 + \omega_c\, c^\dagger c + g\,(b + b^\dagger)(c + c^\dagger), \tag{6.50}$$

$$H_{\text{noise}} = \delta\omega_b(t)\, b^\dagger b. \tag{6.51}$$

Here, b (c) annihilates an excitation in the FTT (cavity) mode; K is the FTT anharmonicity; and g is the coupling strength. Flux noise can cause both depolarization and dephasing. Considering flux noise as small-amplitude $1/f$ noise, we neglect depolarization at GHz-frequency transitions triggered (see Appendix 6.2 for more details). Thus, the noise dominantly causes an FTT-frequency fluctuation $\delta\omega_b(t) = \delta\Phi(t)\, \partial\omega_b/\partial\Phi$, where Φ is the static flux bias and $\delta\Phi(t)$ is its fluctuation.

In the dispersive frame [74], the effective Hamiltonian for H_{static} takes the form

$$H_{\text{disp}} = \bar{\omega}_b\, b^\dagger b + \frac{K}{2}\, b^{\dagger 2} b^2 + \bar{\omega}_c\, c^\dagger c + \chi\, b^\dagger b\, c^\dagger c, \tag{6.52}$$

where χ is the dispersive shift and $\bar{\omega}_b$ and $\bar{\omega}_c$ are the Lamb-shifted FTT and cavity frequencies, which depend on the detuning $\Delta_{bc} = \omega_b - \omega_c$:

$$\chi \approx 2\frac{g^2}{\Delta_{bc}^2}K, \quad \bar{\omega}_b \approx \omega_b + \frac{g^2}{\Delta_{bc}}, \quad \bar{\omega}_c \approx \omega_c - \frac{g^2}{\Delta_{bc}}. \tag{6.53}$$

The noise Hamiltonian H_{noise} primarily induces dephasing between eigenstates of H_{disp} (see Appendix 6.4.2). In particular, because Δ_{bc} depends on $\bar{\omega}_c$, FTT-frequency fluctuations $\delta\omega_b(t)$ lead to cavity-frequency fluctuations

$$\delta\bar{\omega}_c(t) = \frac{\partial\bar{\omega}_c}{\partial\omega_b}\,\delta\omega_b(t) \approx \frac{g^2}{\Delta_{bc}^2}\,\delta\omega_b(t), \tag{6.54}$$

thereby producing cavity dephasing. This is summarized in Fig. 6.1(b).

6.5.2 Driven sweet spots

FTT frequency fluctuations $\delta\omega_b(t)$ are converted into cavity-frequency fluctuations $\delta\bar{\omega}_c(t)$ through the Lamb shift, producing cavity dephasing. This dephasing channel can be suppressed by driving the FTT [Fig. 6.1(a)], which induces a photon-number–dependent energy shift that can be tuned to cancel the Lamb-shift fluctuations.

In the frame rotating at the drive frequency ω_d, the driven dispersive Hamiltonian reads

$$H_R = \Delta_{bd}\, b^\dagger b + \frac{K}{2}\, b^{\dagger 2} b^2 + \overline{\omega}_c\, c^\dagger c + \chi\, b^\dagger b\, c^\dagger c + \Omega_0\,(b + b^\dagger), \tag{6.55}$$

where $\Delta_{bd} = \bar{\omega}_b - \omega_d$ is the FTT–drive detuning. The drive couples $|g, m\rangle$ and $|e, m\rangle$ at fixed cavity photon number m, producing level repulsion that lowers the energy of $|g, m\rangle$. The dispersive interaction offsets the detuning by $m\chi$, so the relevant detuning is $\Delta_m \equiv \Delta_{bd} + m\chi$ and the drive-induced energy shift becomes photon-number dependent. Examples for $m = 0, 1$ are shown in Fig. 6.1(c), and this m-dependent shift can be tuned to cancel fluctuations of the static Lamb shift [Fig. 6.1(d)].

An analytic expression follows in the weak-drive regime, assuming (i) $\Omega_0 \ll \min_m |\Delta_m|$ over the photon numbers of interest, so that the drive does not strongly hybridize $|g, m\rangle$ and $|e, m\rangle$, and (ii) $|\Omega_0| \ll |K + 2\Delta_{bd}|$, so that hybridization with the state $|f, m\rangle$, remains weak. Under these conditions, a Schrieffer–Wolff transformation (Appendix 6.3) diagonalizes H_R. Retaining terms up to $\mathcal{O}(\Omega_0^2/\Delta_m^2)$ and projecting onto the $\{|g, m\rangle, |e, m\rangle\}$ subspace gives

$$\begin{aligned} H_d = \widetilde{\Delta}_{bd}\, \sigma^+\sigma^- + \bar{\omega}_c\, c^\dagger c - \sum_m \frac{\Omega_0^2}{\Delta_m} |m\rangle\langle m| \\ + \sigma^+\sigma^- \left[\chi\, c^\dagger c + \sum_m \frac{2\Omega_0^2}{\Delta_m} |m\rangle\langle m| \right], \end{aligned} \tag{6.56}$$

where $\sigma^+\sigma^- \equiv |e\rangle\langle e|$ and $|m\rangle$ denotes a cavity Fock state. The drive renormalizes the FTT detuning, $\widetilde{\Delta}_{bd} \equiv \Delta_{bd} - \Omega_0^2/K$, and generates an m-dependent shift of the cavity transition frequency together with an additional Stark shift. The essential point is that this drive-induced shift can be tuned to cancel the flux-sensitive static Lamb shift, thereby creating a driven sweet spot.

To quantify the resulting cavity-frequency fluctuation, we define the net cavity transition frequency between adjacent cavity levels as

$$\widetilde{\omega}_{c,m\to m+1} = \bar{\omega}_c - \Omega_0^2/\Delta_{m+1} + \Omega_0^2/\Delta_m. \tag{6.57}$$

We use a bar for Lamb-shifted frequencies and a tilde for drive-renormalized transition frequencies.

FTT frequency fluctuation $\delta\omega_b(t)$ then induces

$$\delta\widetilde{\omega}_{c,m\to m+1}(t) = D_{m\to m+1}\,\delta\omega_b(t) \tag{6.58}$$

where $D_{m\to m+1} \equiv \frac{\partial\widetilde{\omega}_{c,m\to m+1}}{\partial\omega_b}$ is the susceptibility of the net cavity frequency to FTT-frequency fluctuation. It has two contributions: a bare term inherited from the Lamb-shift dependence $\bar{\omega}_c(\omega_b)$, and a drive-induced term from the dressing shift through $\Delta_m = \Delta_{bd} + m\chi$. A sweet spot occurs when these two terms cancel. Specifically, the physics of the cancellation differs in two regimes.

Unselective Regime. In this regime, the drive detuning $\Delta_{m^*} \equiv \Delta_{bd} + m^*\chi$ satisfies $|\Delta_{m^*}| \gg |\chi|$, so the drive couples $|g, m^*\rangle \leftrightarrow |e, m^*\rangle$ and neighboring photon-number pairs with comparable strength. As a result, the effective cavity nonlinearity near m^* is suppressed. For $m^* = 0$, expanding the denominators $\Delta_m^{-1} = (\Delta_{bd} + m\chi)^{-1}$ to leading order in $|\chi/\Delta_{bd}|$ yields the approximate effective Hamiltonian

$$H_d \approx \widetilde{\Delta}_{bd}\,\sigma^+\sigma^- + \widetilde{\omega}_{c,0\to 1}\,c^\dagger c + \chi\left(1 - \frac{2\Omega_0^2}{\Delta_{bd}^2}\right)\sigma^+\sigma^-\,c^\dagger c, \tag{6.59}$$

up to constant energy offsets, where

$$\widetilde{\omega}_{c,0\to 1} = \bar{\omega}_c + \frac{\Omega_0^2\chi}{\Delta_{bd}^2}. \tag{6.60}$$

The corresponding susceptibility of the net cavity frequency is

$$D_{0\to 1} \approx \frac{g^2}{\Delta_{bc}^2}\left[1 - \frac{4\Omega_0^2 K}{\Delta_{bd}^3}\right]. \tag{6.61}$$

The far-detuned sweet spot is therefore set by $D_{0\to1} = 0$, which gives the single solution

$$\Delta_{bd} = \sqrt[3]{4\Omega_0^2 K}. \tag{6.62}$$

This prediction is confirmed by Fig. 6.2(b). Because the rotating-frame spectrum of H_R approximates the Floquet quasienergies (defined modulo ω_d; see App. C.1), the analytical susceptibility is compared against a Floquet calculation using the full-model Hamiltonian (App. 7.2). For $\Omega_0/2\pi = 10$ MHz, $D_{0\to1}$ approaches the bare Lamb-shift value $g^2/\Delta_{bc}^2 \approx 0.01$ when the drive is far detuned from the FTT frequency $\bar{\omega}_b$. As ω_d is tuned toward $\bar{\omega}_b$, a single sweet spot appears on the $\omega_d > \bar{\omega}_b$ side, consistent with the negative anharmonicity of the FTT. The characteristic detuning is consistent with Eq. (6.62), which predicts $\Delta_{bd}/2\pi \sim -34$ MHz for $K/2\pi \approx -0.1$ GHz used here. The residual offset between the analytical and numerical dip positions at this drive strength arises because higher-order corrections to the Hamiltonian Eq. (6.56) become non-negligible. Closer to resonance, $D_{0\to1}$ rises again as the drive-induced energy shift dominates the susceptibility.

Selective Regime. In this regime, the drive is closest to resonance for a particular photon number m^*, so that $|\Delta_{m^*}| \ll \chi$. Consequently, only the hybridization between $|g, m^*\rangle$ and $|e, m^*\rangle$ contributes appreciably to the cavity frequency, giving

$$\widetilde{\omega}_{c,m^*\to m^*+1} \approx \bar{\omega}_c + \frac{\Omega_0^2}{\Delta_{m^*}}. \tag{6.63}$$

Differentiating with respect to ω_b yields

$$D_{m^*} \approx \frac{g^2}{\Delta_{bc}^2} - \frac{\Omega_0^2}{\Delta_{m^*}^2}. \tag{6.64}$$

Setting $D_{m^*} = 0$ gives the sweet-spot condition

$$\Delta_{m^*} = \pm \left| \frac{\Delta_{bc}}{g} \Omega_0 \right| \Longleftrightarrow \Delta_{bd} = \pm \left| \frac{\Delta_{bc}}{g} \Omega_0 \right| - m^* \chi. \tag{6.65}$$

This prediction is confirmed by Fig. 6.2(a), where two sweet spots appear as symmetric dips on either side of the resonance. Their characteristic detuning is consistent with Eq. (6.65), which predicts $|\Delta_{bd}| \sim (|\Delta_{bc}|/g)\Omega_0 \approx 0.5$ MHz for the parameters used here ($g/|\Delta_{bc}| \simeq 0.1$). Overall, the agreement between the analytics and exact numerics shows that in the selective regime a low-power drive cancels the bare Lamb-shift susceptibility.

Sweet-Spot Map and Regime Crossover. Fig. 6.2(c) shows $D_{0\to1}$ as a function of drive amplitude Ω_0 and detuning Δ_{bd}. The key point is that the selective and unselective sweet spots are not distinct phenomena, but rather different parts of a continuous contour defined by $D_{0\to1} = 0$. At small Ω_0, the selective-regime condition in Eq. (6.65) predicts the linear scaling $|\Delta_{bd}| \propto \Omega_0$, visible as two symmetric sweet spots on either side of the effective resonance. As Ω_0 increases, the inner branch, closer to resonance and at smaller $|\Delta_{bd}|$, enters a nonperturbative regime in which the weak-drive approximation underlying Eq. (6.56) no longer applies. The remaining perturbative branch then crosses over toward the unselective regime and approaches the cubic scaling predicted by Eq. (6.62). At still larger Ω_0, the Stark-shift contribution broadens the detuning range over which $D_{0\to1} \approx 0$, improving robustness against flux noise by maintaining susceptibility cancellation over a wider range of fluctuations, albeit closer to the limits of perturbation theory. Higher-order drive effects generate additional sweet spots; see App. D.1. The same mechanism is not restricted to the FTT and is illustrated for a SNAIL-based circuit in App. E.1. Finally, although the sweet-spot condition does not in general suppress sensitivity to drive-amplitude noise, for AWG noise levels typical of experiments the associated dephasing is negligible, see App. 8.5.

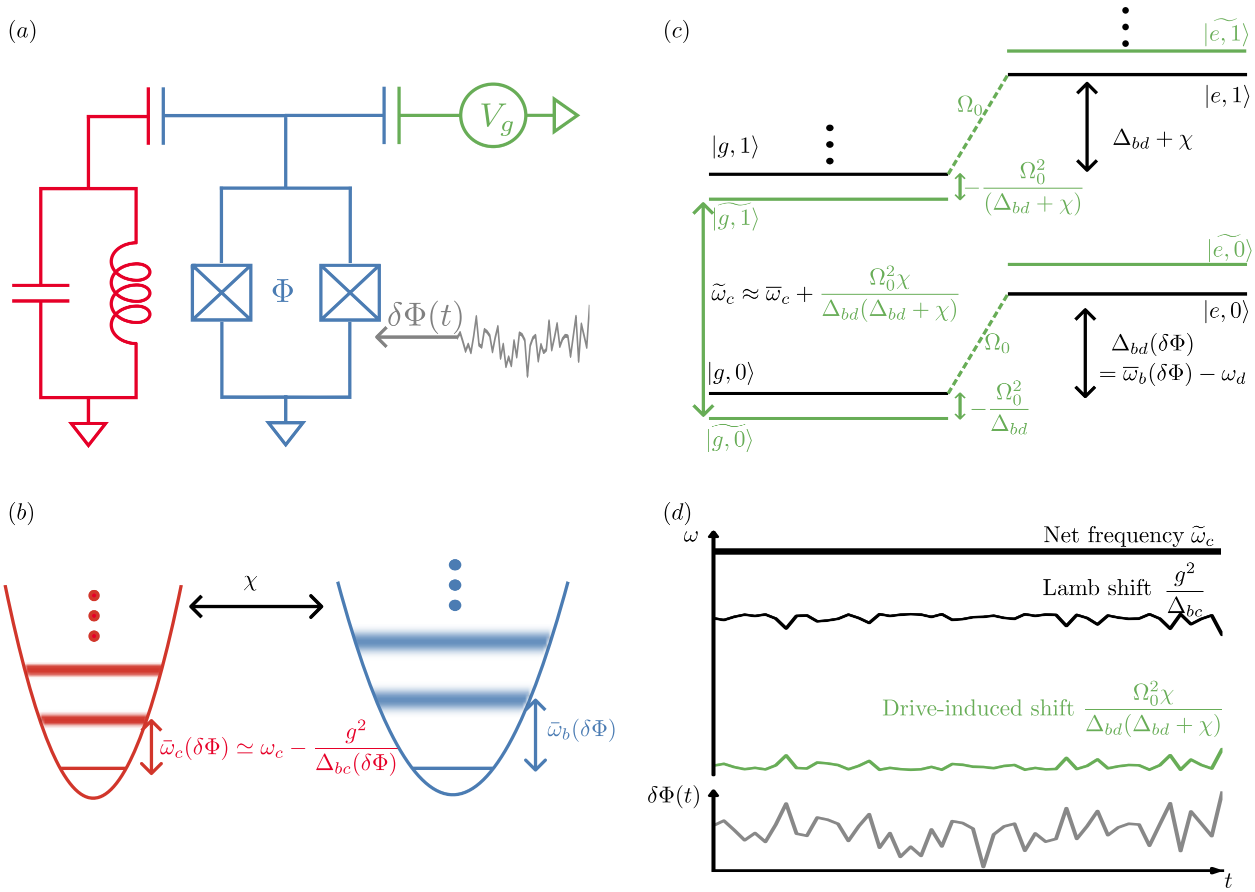

Figure 6.1: **Scheme for engineering a cavity frequency at a driven sweet spot.** **(a)** Circuit schematic of a flux-tunable transmon (FTT) dispersively coupled to a superconducting cavity and driven by a coherent voltage source V_g. Flux noise $\delta\Phi(t)$ modulates the FTT frequency and provides the dominant noise channel. **(b)** Noise transduction via the Lamb shift. The bare modes have frequencies ω_c and $\omega_b(\delta\Phi)$ with flux-dependent detuning $\Delta_{bc}(\delta\Phi) = \omega_b(\delta\Phi) - \omega_c$. The coupling g hybridizes the modes, yielding dressed frequencies $\bar{\omega}_{b/c}(\delta\Phi)$; the cavity frequency acquires a Lamb shift $-g^2/\Delta_{bc}$ and therefore inherits sensitivity to $\delta\Phi(t)$. **(c)** Drive-induced, photon-number–dependent shift in the frame rotating at ω_d. A coherent drive of amplitude Ω_0 hybridizes the manifolds $|g,m\rangle$ and $|e,m\rangle$, producing level repulsion (green) that renormalizes the cavity transition frequency in an m-dependent manner. **(d)** First-order noise cancellation at the driven sweet spot. In the presence of $\delta\Phi(t)$ (gray), the static Lamb-shift contribution to the cavity frequency (black) is compensated by the drive-induced shift (green) at an appropriate drive, yielding a net cavity frequency $\tilde{\omega}_c$ (thick black) that is first-order insensitive to flux fluctuations (a flux-insensitive point in the sense of vanishing first-order susceptibility).

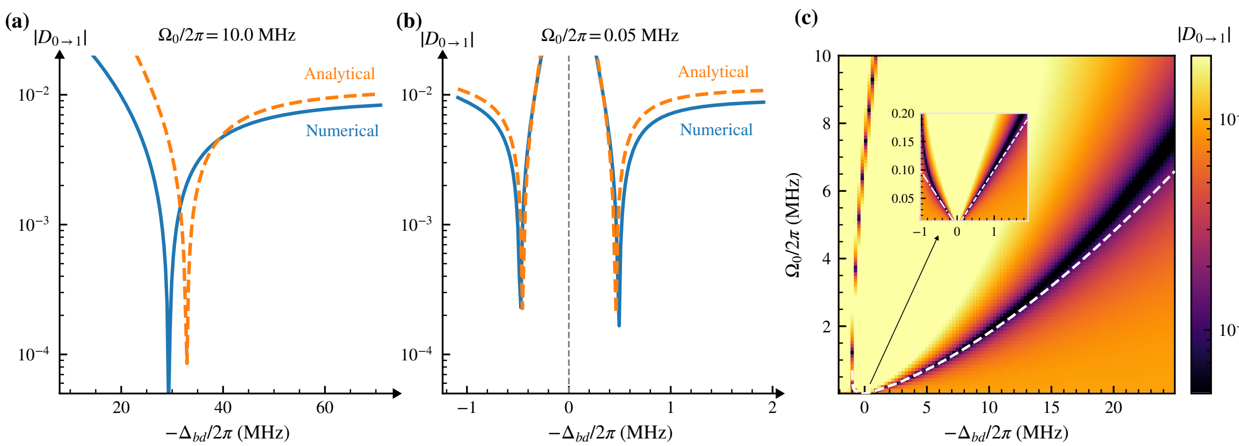

Figure 6.2: Drive-induced flux sweet spots, defined by $D_{0\to 1} = 0$. (a) Unselective regime at $\Omega_0/2\pi = 10$ MHz, showing a single sweet spot. (b) Selective regime at $\Omega_0/2\pi = 0.05$ MHz, showing two symmetric sweet spots about the effective resonance. (c) Two-dimensional map of $D_{0\to 1}$ versus drive amplitude Ω_0 and detuning Δ_{bd}; Dashed curves denote analytical sweet-spot conditions from Eqs. (6.57) and (6.64) for the main panel and the inset, respectively.

CHAPTER 7

OVERALL DEPHASING RATE AND PERFORMANCE ANALYSIS

7.1 Overall dephasing rate

7.1.1 Total dephasing in the unselective regime

In the unselective regime $|\Delta_{bd}| \gg |\chi|$, the master equation can be expanded to first order in $|\chi/\Delta_{bd}|$ about the sweet-spot condition (Eq. (6.62)). This yields the Lindblad master equation for the density matrix in the effective drive frame ρ_d,

$$\dot{\rho}_d = -i\big[H_d + H^d_{\text{noise}}, \rho_d\big] + \widetilde{\gamma}^{\text{un}}_{b\downarrow}\mathcal{D}[\sigma^-]\rho_d + \gamma^{\text{un}}_{b\uparrow}\mathcal{D}[\sigma^+]\rho_d$$
$$+2\gamma^{\text{un}}_{c\phi}\mathcal{D}[c^\dagger c]\rho_d + \widetilde{\gamma}^{\text{un}}_{c\downarrow}\mathcal{D}[c]\rho_d, \tag{7.1}$$

See App. 6.4.4 for the derivation. The effective rates $\widetilde{\gamma}^{\text{un}}_{b\downarrow}$, $\gamma^{\text{un}}_{b\uparrow}$, $\gamma^{\text{un}}_{c\phi}$, and $\widetilde{\gamma}^{\text{un}}_{c\downarrow}$ are given below. Here, H_d is the effective driven Hamiltonian Eq. (6.59), and H^d_{noise} is the noise Hamiltonian in the effective-drive frame. In the unselective limit it takes the approximate form

$$H^d_{\text{noise}} \approx \delta\omega_b\Bigg[\Big(1-\frac{\Delta_{bd}}{2K}\Big)\sigma^+\sigma^- + \frac{2g^2}{\Delta_{bc}^2}\left(1-\frac{2K}{\Delta_{bc}}\right)\sigma^+\sigma^-c^\dagger c$$
$$-\frac{1}{2}\sqrt{\frac{\Delta_{bd}}{K}}\sigma_x\Bigg]. \tag{7.2}$$

Although the sweet-spot condition cancels the leading direct cavity-dephasing contribution ($D_{0\to1} = 0$), it does not eliminate all noise channels in the effective-drive frame, which are discusseed below.

Interpretation of the dissipators. The dissipators in Eq. (7.1) represent (i) drive-renormalized FTT relaxation, $\widetilde{\gamma}_{b\downarrow}^{\mathrm{un}}\mathcal{D}[\sigma^-]$; (ii) drive-induced FTT excitation, $\gamma_{b\uparrow}^{\mathrm{un}}\mathcal{D}[\sigma^+]$, which produces photon-shot-noise dephasing of the cavity via the dispersive coupling [169, 170]; (iii) cavity pure dephasing, $2\gamma_{c\phi}^{\mathrm{un}}\mathcal{D}[c^\dagger c]$ (the prefactor 2 is conventional); and (iv) renormalized cavity decay, $\widetilde{\gamma}_{c\downarrow}^{\mathrm{un}}\mathcal{D}[c]$. At the sweet-spot condition of Eq. (6.62), the corresponding rates are

$$\begin{aligned}
\widetilde{\gamma}_{b\downarrow}^{\mathrm{un}} &= \gamma_{b\downarrow}\left(1-\frac{\Delta_{bd}}{K}\right),\\
\gamma_{b\uparrow}^{\mathrm{un}} &= \frac{\gamma_{b\downarrow}}{16}\left(\frac{\Delta_{bd}}{K}\right)^2\left(1-\frac{2\Delta_{bd}}{K}\right),\\
\gamma_{c\phi}^{\mathrm{un}} &= \frac{1}{2}\gamma_{b\downarrow}\left(\frac{g}{\Delta_{bc}}\right)^4\frac{K}{\Delta_{bd}}, \qquad \widetilde{\gamma}_{c\downarrow}^{\mathrm{un}} = \gamma_{c\downarrow}\left(1+\frac{g^2}{\Delta_{bc}^2}\right).
\end{aligned} \tag{7.3}$$

The cavity-decay renormalization scales as g^2/Δ_{bc}^2 and remains small (e.g., ~ 0.01) even in the relatively strong dispersive-coupling regime.

Interpretation of the noise Hamiltonian. At the sweet spot, cavity dephasing $c^\dagger c$ in Eq. (7.2) vanishes by construction , suppressing direct cavity-frequency noise. Also, AC-Stark-shift-induced cavity-frequency fluctuations are neglected because the FTT is barely excited if it is initialized in the ground state. Residual dephasing is dominated by a transverse component $\propto \sigma_x$, which induces FTT depolarization under $1/f$ flux noise. The transverse-noise-induced excitation rate is

$$\gamma_{b\uparrow}^{1/f,\mathrm{un}} = \left(\frac{\Omega_0}{\Delta_{bd}}\frac{\partial\omega_b}{\partial\Phi}\right)^2\frac{S_0^2}{|\Delta_{bd}/2\pi|} \overset{(6.62)}{=} \frac{1}{4}\left(\frac{\partial\omega_b}{\partial\Phi}\right)^2\frac{S_0^2}{|K/2\pi|} \tag{7.4}$$

at the sweet-spot condition.

Total cavity dephasing rate. In the unselective regime, evaluated at the sweet spot, the total cavity dephasing rate is the sum of the three contributions in

$$\begin{aligned}\Gamma_{\phi,\mathrm{total}} &= \gamma_{b\Uparrow}^{1/f,\mathrm{un}} + \gamma_{b\Uparrow}^{\mathrm{un}} + \gamma_{c\phi}^{\mathrm{un}} \\ &= \left(\frac{\partial \omega_b}{\partial \Phi}\right)^2 \frac{S_0^2}{4K} + \frac{\gamma_{b\downarrow}}{16}\left(\frac{\Delta_{bd}}{K}\right)^2 \left(1 - \frac{2\Delta_{bd}}{K}\right) \\ &+ \frac{\gamma_{b\downarrow}}{2}\left(\frac{g}{\Delta_{bc}}\right)^4 \frac{K}{\Delta_{bd}}. \end{aligned} \tag{7.5}$$

We approximate the photon-shot-noise contribution to cavity dephasing by the FTT heating rate $\gamma_{b\Uparrow}^{\mathrm{un}}$ in the limit $\chi \gg \widetilde{\gamma}_{b\downarrow}$ [171], because a single photon event is sufficient to entirely destroy phase coherence before the photon decays. Under this approximation, each contribution in Eq. (7.5) agrees well with numerical simulations, as verified in App. F.2. We also emphasize that the drive induces only a small correction to $\gamma_{c\downarrow}$, scaling as $g^2/\Delta_{bc}^2 \ll 1$; see App. F.2.

7.1.2 Total dephasing in the selective regime

Operating at the selective-regime sweet spot protects the cavity states $|g, m^\star\rangle$–$|g, m^\star + 1\rangle$ from flux-noise dephasing. Under drive, however, the residual dephasing budget in the selective regime is set by two contributions: (i) cavity pure dephasing at a rate $\gamma_{c\phi}^{\mathrm{sel}} = \left(g^2/\Delta_{bc}^2\right)\gamma_{b\downarrow}$, arising from the Purcell-like mixing of cavity and FTT decay channels, and (ii) FTT heating at a rate $\gamma_{b\Uparrow}^{\mathrm{sel}} \propto 1/|\Delta_{bd}|$ induced by photon shot noise (see App. 6.4.4 for derivations). Because the shot-noise contribution scales unfavorably as $1/|\Delta_{bd}|$, operating at larger detuning—i.e., in the unselective regime—yields a lower total dephasing rate.

7.1.3 Numerical evaluation of the total dephasing rate

Full-system simulations are used to extract the cavity dephasing rate under drive; the numerical procedure and fitting protocol are summarized in App. F.1. The noise amplitude S_0 and the detuning Δ_{bd} are sampled over the parameter range of interest. For each Δ_{bd}, the drive amplitude is chosen to satisfy the sweet-spot condition, and the corresponding total dephasing rate $\Gamma_{\phi,\mathrm{total}}$ is extracted.

Fig. 7.1(a) summarizes the results. For $S_0 = 10^{-5}/\Phi_0$, the drive-induced contribution $\gamma_{b\uparrow}^{1/f,\mathrm{un}}$ dominates $\Gamma_{\phi,\mathrm{total}}$ over the scanned detuning range. At large detuning, $|\Delta_{bd}| > |\chi| \simeq 2$ MHz, the total rate is well described by $\gamma_{b\uparrow}^{1/f,\mathrm{un}}$, which is independent of the detuning. At smaller detuning, i.e. $-\Delta_{bd}/2\pi = 1\ \mathrm{MHz} < |\chi/2\pi|$, the rate increases as the dynamics enters the selective regime, consistent with the $1/|\Delta_{bd}|$ scaling.

For smaller noise amplitude, e.g. $S_0 = 10^{-6}/\Phi_0$, the S_0-independent drive-induced contributions $\gamma_{b\uparrow}^{\mathrm{un}}$ and $\gamma_{c\phi}^{\mathrm{un}}$ dominate $\Gamma_{\phi,\mathrm{total}}$. At large detuning ($-\Delta_{bd}/2\pi = 19$ MHz), $\gamma_{b\uparrow}^{\mathrm{un}} \propto (\Delta_{bd}/K)^2$ dominates while $\gamma_{c\phi}^{\mathrm{un}} \sim K/\Delta_{bd}$ is suppressed; the opposite hierarchy holds at small detuning ($-\Delta_{bd}/2\pi = 1$ MHz). Consequently, $-\Delta_{bd}/2\pi = 6$ MHz yields the minimal $\Gamma_{\phi,\mathrm{total}}$ within the scanned range. In the undriven case, the dephasing time is ~ 0.1–1 ms as S_0 varies from $10^{-5}/\Phi_0$ to $10^{-6}/\Phi_0$. With $\Delta_{bd}/2\pi \simeq -6$ MHz and sweet-spot driving, the dephasing time improves by up to $\sim 20\times$ across the considered S_0 range, confirming the overall dephasing enhancement.

To visualize the dephasing enhancement directly, we perform time-domain simulations that include relaxation and stochastic flux noise with $S_0 = 10^{-5}/\Phi_0$. The initial state is $|\psi_0\rangle = \frac{1}{\sqrt{2}}\big(\,|g,0\rangle + |g,1\rangle\,\big)$. With drive, the state is adiabatically mapped onto the target Floquet states using a shaped pulse $2\Omega_0(t)\cos\big(\omega t + \phi(t)\big)$. In the rotating frame this yields the I and Q quadratures

$$\Omega_I(t) = \Omega(t)\cos\phi(t), \qquad \Omega_Q(t) = \Omega(t)\sin\phi(t). \tag{7.6}$$

We implement a DRAG correction, $\Omega_Q(t) = -\dot{\Omega}_I(t)/\Delta_{bd}$, which accelerates the adiabatic transfer relative to $\Omega_I(t)$-only ramps (see App. 7.4). A pulse ramp with $\tau = 75$ ns is shown in Fig. 7.1(b).

The dephasing is quantified by the coherence operator

$$X(t) = |\phi_{g,1}(t)\rangle\langle\phi_{g,0}(t)| + \mathrm{H.c.}, \tag{7.7}$$

$$P_X(t) = \mathrm{Tr}[\rho(t)\, X(t)]\,, \tag{7.8}$$

where $|\phi_{g,1}(t)\rangle$ and $|\phi_{g,0}(t)\rangle$ are adiabatically mapped from the bare states. Without drive, $1/f$ noise produces sub-Gaussian free-induction decay. At the driven sweet spot, the dominant residual channel is Markovian, arising from FTT depolarization, and yields the exponential decay observed in Fig. 7.1(c). The dephasing time T_2 is extracted from an exponential fit to $P_X(t)$ (see Appendix I). Under sweet-spot driving, the dephasing time reaches ~ 2 ms, compared with ~ 100 μs in the undriven case.

7.2 Exact calculation of $D_{0\to 1}$

The driven circuit is simulated using the full system Hamiltonian H_{sys} in Eq. (6.5), without further approximations. The parameters are $E_{J1} = E_{J2} = 30.19\,\mathrm{GHz}$, $E_C = 0.10\,\mathrm{GHz}$, $\omega_c/2\pi = 5.226\,\mathrm{GHz}$, $g'/2\pi = 0.050\,\mathrm{GHz}$, and $\Phi = 0.20\,\Phi_0$. For these values, the effective parameters appearing in Eq. (6.49) are $\omega_b/2\pi = 6.16\,\mathrm{GHz}$, $K/2\pi = -0.10\,\mathrm{GHz}$, and $g/2\pi = 0.10\,\mathrm{GHz}$.

The basis is obtained by diagonalizing the static part of H_{sys}, i.e., $H_{\mathrm{sys}}(t)$ with $\Omega_0 = \delta\Phi(t) = 0$.

In the dispersive regime, these eigenstates are labeled by the corresponding bare basis states. The FTT (cavity) Hilbert space is truncated to 3 (4) levels. Increasing the truncation changes the results only marginally and does not affect the conclusions.

Floquet simulations are performed directly with $H_{\rm sys}(t)$ to obtain time-periodic Floquet modes $|\phi_\alpha(t)\rangle$ and quasienergies ϵ_α, with $|\phi_\alpha(t+T)\rangle = |\phi_\alpha(t)\rangle$ and $T = 2\pi/\omega_d$. To assign each Floquet mode a label in a chosen bare basis $\{|j\rangle\}$, the time-averaged overlap weight

$$W_{\alpha j} = \frac{1}{T}\int_0^T dt\ |\langle j \mid \phi_\alpha(t)\rangle|^2 \tag{7.9}$$

is evaluated. Each Floquet mode α is labeled by the basis state $|j\rangle$ with the largest weight,

$$\text{label}(\alpha) = \arg\max_j W_{\alpha j}. \tag{7.10}$$

This procedure yields a relabeling $\alpha \mapsto (n, m)$, where n and m denote the FTT and cavity excitation numbers, respectively, along with the corresponding quasienergies ϵ_{nm}.

Using the labeled quasienergies, the flux sensitivity of the $(g, 1)$–$(g, 0)$ quasienergy splitting is evaluated and normalized by the flux dispersion of ω_b. Specifically,

$$D_{0\to 1} = \frac{\partial}{\partial \Phi}(\epsilon_{g1} - \epsilon_{g0}) \bigg/ \frac{\partial \omega_b}{\partial \Phi}, \tag{7.11}$$

which yields the exact $D_{0\to 1}$ reported in Fig. 6.2.

7.3 Numerical details of obtaining dephasing rates

This section details the numerical procedure used to extract the dephasing rates in Sec. 7.1.3. The dynamics are simulated with the full Hamiltonian $H_{\rm full}(\delta\Phi)$ represented in a truncated oscillator

(Fock) basis and evolved under the Lindblad master equation

$$\dot{\rho} = -i\big[\, H_{\mathrm{full}}\big(\delta\Phi(t)\big),\ \rho\,\big] + \gamma_{b\downarrow}\, \mathcal{D}[b]\rho, \tag{7.12}$$

where $\mathcal{D}[b]\rho = b\rho b^\dagger - \frac{1}{2}\{b^\dagger b, \rho\}$.

Generation of $1/f$ noise. Flux noise is incorporated through stochastic trajectories $\delta\Phi(t)$ whose spectrum follows a $1/f$ power spectral density. For each trajectory, Eq. (F.1) is integrated to obtain $\rho(t)$, and all reported curves are computed from ensemble averages of the corresponding observables. The noise realizations $\delta\Phi(t)$ are generated using the frequency-domain synthesis method of Ref. [172]: Fourier components are assigned random phases and Gaussian-distributed amplitudes with variances proportional to the target $1/f$ spectrum, followed by an inverse Fourier transform to obtain $\delta\Phi(t)$ in the time domain. Fig. F.1 shows the resulting power spectral densities for 100 representative trajectories. This ensemble size is sufficient for the quantities of interest to converge, as indicated by the small error bars in Fig. 7.1(a) and Fig. F.2(b).

The total simulated evolution time is $T \sim 10\ \mu\mathrm{s}$, and the infrared cutoff is set to $\omega_{\mathrm{IR}}/2\pi = 10\ \mathrm{kHz}$ (equivalently $f_{\min} = 10^{-5}\,\mathrm{GHz}$). This choice satisfies $f_{\min} \ll 1/T \approx 100\ \mathrm{kHz}$, capturing the relevant slow-noise dynamics within the simulated window while avoiding the infrared divergence of an ideal $1/f$ spectrum. The ultraviolet cutoff is set to $f_{\max} = 1\ \mathrm{GHz}$; the extracted dephasing rates are insensitive to further increases of $f_{\max}$.

Extraction of dephasing time. The states with enhanced dephasing discussed in main text are proxies for the corresponding Floquet states, as shown in Appendix C.1. In the numerics, Floquet modes are labeled by their correspondence to bare states as described in Appendix 7.2. The two Floquet states used for the cavity dephasing analysis in Sec. 7.1.3 are denoted $|\phi_{g,0}(t)\rangle$ and

$|\phi_{g,1}(t)\rangle$.

Relaxation and dephasing are extracted from dynamics evolving from two initial states. Relaxation is characterized by preparing

$$\rho(0) = |\phi_{g,1}(0)\rangle\langle\phi_{g,1}(0)|$$

and monitoring the return toward $|\phi_{g,0}(t)\rangle$. Dephasing is characterized by preparing the equal superposition

$$\rho(0) = |+\rangle\langle+| \,, \qquad |+\rangle = \frac{1}{\sqrt{2}}\Big(|\phi_{g,0}(0)\rangle + |\phi_{g,1}(0)\rangle \Big),$$

and tracking the decay of the corresponding coherence. The relevant quantities are

$$\mathrm{Tr}[\rho(t)\, P_{g,0}(t)] \quad \text{and} \quad \mathrm{Tr}[\rho(t)\, X(t)] \,,$$

where

$$P_{g,0}(t) = |\phi_{g,0}(t)\rangle\langle\phi_{g,0}(t)| \,, X(t) = |\phi_{g,1}(t)\rangle\langle\phi_{g,0}(t)| + \text{H.c.}$$

Relaxation and dephasing times are obtained by fitting ensemble-averaged signals. Specifically, T_1 is extracted from the rise of the ground-state population by fitting $\mathrm{Tr}[P_{g,0}(t)\,\overline{\rho}(t)]$ to $1 - e^{-t/T_1}$, where $\overline{\rho}(t)$ denotes the ensemble-averaged density matrix over different flux noise trajectories. The coherence time T_2 is extracted from the decay of the superposition signal by fitting $\mathrm{Tr}[P_+(t)\,\overline{\rho}(t)]$ to e^{-t/T_2} (with the appropriate normalization convention for the plotted data). The pure dephasing time then follows from

$$\frac{1}{T_\phi} = \frac{1}{T_2} - \frac{1}{2T_1}, \tag{7.13}$$

and we report the total dephasing rate as $\Gamma_{\phi,\mathrm{total}} = 1/T_\phi$. The data in Fig. 7.1 use the same system parameters and level truncations as Appendix 7.2. Unless otherwise stated, the fitting windows are $1\ \mu$s for T_1 and $5\ \mu$s for T_2; extending these windows does not change the extracted rates.

7.4 Adiabatic transition

In the adiabatic frame, we introduce a time-dependent unitary transformation $U(\Omega_t)$ that diagonalizes the instantaneous Hamiltonian $H(\Omega_t)$. The corresponding effective Hamiltonian is

$$\begin{aligned} H_{\mathrm{eff}}(\Omega_t) &= U^\dagger(\Omega_t)\left(H(\Omega_t) - i\hbar\frac{\partial}{\partial t}\right)U(\Omega_t) \\ &= H_d(\Omega_t) + V(\Omega_t). \end{aligned} \tag{7.14}$$

The diagonal part H_d contains the instantaneous energy eigenvalues $E_j(\Omega_t)$:

$$H_d(\Omega_t) = \sum_j E_j(\Omega_t)\,|j\rangle\langle j|. \tag{7.15}$$

The coupling term V originates from the induced gauge contribution $i\hbar\,U^\dagger\dot{U}$ and captures the effect of the time-dependent driving. Splitting V into off-diagonal and diagonal components gives

$$\begin{aligned} V = &\underbrace{-i\hbar\sum_{j\neq j'}\frac{\langle j'|\,U^\dagger\dot{H}\,U\,|j\rangle}{E_j - E_{j'}}\,|j'\rangle\langle j|}_{V_{od}} \\ &+ \underbrace{-i\hbar\sum_j\langle j|\,U^\dagger\dot{U}\,|j\rangle\,|j\rangle\langle j|}_{V_d}. \end{aligned} \tag{7.16}$$

where the diagonal terms contribute to the Berry phase, while the off-diagonal terms drive transitions.

In our case of interest, the goal is to adiabatically evolve into the desired Floquet states. When counter-rotating terms are negligible, this evolution is equivalent to adiabatic dynamics governed by the rotating-frame Hamiltonian

$$\begin{aligned} H_R(t) = \Delta_{bd}\, b^\dagger b + \frac{K}{2}\, b^{\dagger 2} b^2 + \overline{\omega}_c\, c^\dagger c \\ + \chi\, b^\dagger b\, c^\dagger c + \Omega_I(t)\,(b + b^\dagger). \end{aligned} \tag{7.17}$$

Under the drive $\Omega_I(t)$, the Hamiltonian in the instantaneous eigenbasis can be written as $H_{\text{eff}} = H_d(\Omega_I) + V(\Omega_I)$. For the driven operating regime of interest, the diagonal contribution is well approximated by Eq. (6.59)

$$\begin{aligned} H_d(\Omega_I) \approx \left(\Delta_{bd} - \frac{\Omega_I^2}{\kappa}\right) \sigma^+\sigma^- + \left(\tilde{\omega}_c + \frac{\Omega_I^2 \chi}{\Delta_{bd}^2}\right) c^\dagger c \\ + \chi \left(1 - 2\frac{\Omega_I^2}{\Delta_{bd}^2}\right) \sigma^+\sigma^-\, c^\dagger c. \end{aligned} \tag{7.18}$$

Nonadiabatic leakage is generated by the time dependence of the drive through the off-diagonal term V_{od}. In particular,

$$U^\dagger \dot{H}_R U = \dot{\Omega}_I\, U^\dagger (b^\dagger + b) U \overset{(6.33)}{\approx} \dot{\Omega}_I\, \sigma_x + \mathcal{O}\!\left(\frac{\dot{\Omega}_I}{\Delta_{bd}}\right). \tag{7.19}$$

which yields the effective coupling between $|g\rangle$ and $|e\rangle$,

$$V_{od}(\Omega_I) \approx \frac{\dot{\Omega}_I}{\Delta_{bd}}\, \sigma_y. \tag{7.20}$$

Suppressing leakage requires the nonadiabatic coupling to remain small compared to the instantaneous gap,

$$\frac{\dot{\Omega}_I}{\Delta_{bd}} \ll \Delta_{bd}. \tag{7.21}$$

For a ramp of characteristic amplitude Ω_0 executed over a ramp time τ (so that $\dot{\Omega}_I \sim \Omega_0/\tau$), this implies the lower bound

$$\tau \gg \frac{\Omega_0}{\Delta_{bd}^2} \overset{(6.62)}{=} \sqrt{\frac{1}{4K\Delta_{bd}}}. \tag{7.22}$$

Increasing Δ_{bd} can reduce τ to obtain the desired state.

To bypass this speed limit, we implement a shortcut to adiabaticity using an active quadrature control field Ω_Q which is DRAG-style. Specifically, we augment the drive Hamiltonian by

$$H_{\text{total}} = H_R - i\Omega_Q(b - b^\dagger), \tag{7.23}$$

and choose the quadrature amplitude to cancel the nonadiabatic coupling,

$$\Omega_Q = -\frac{\dot{\Omega}_I}{\Delta_{bd}}. \tag{7.24}$$

The effective Hamiltonian in the adiabatic frame then becomes

$$\begin{aligned} &U^\dagger \left(H_R - i\Omega_Q(b - b^\dagger) - i\frac{\partial}{\partial t} \right) U \\ &\approx H_d + V_d + \frac{\dot{\Omega}_I}{\Delta_{bd}}\sigma_y - \frac{\dot{\Omega}_I}{\Delta_{bd}}\sigma_y \\ &= H_d + V_d, \end{aligned} \tag{7.25}$$

so the evolution remains diagonal in the instantaneous eigenbasis even for fast ramps.

Numerical simulations confirm the qualitative analysis above. We compute the state-preparation infidelity at the end of the ramp as a function of the ramp time τ for several detunings Δ_{bd}. The infidelity is defined as the average loss of overlap with the target Floquet states,

$$\mathcal{I}(\tau) = 1 - \frac{1}{2}\left[\mathrm{Tr}\big(P_{g0}(\tau)\,\bar{\mathcal{E}}_\tau(\rho_{0,g0})\big) + \mathrm{Tr}\big(P_{g1}(\tau)\,\bar{\mathcal{E}}_\tau(\rho_{0,g1})\big)\right]. \tag{7.26}$$

where $P_{g0}(\tau) = |\phi_{g0}(\tau)\rangle\langle\phi_{g0}(\tau)|$ and $P_{g1}(\tau) = |\phi_{g1}(\tau)\rangle\langle\phi_{g1}(\tau)|$ project onto the desired Floquet states at the end of the pulse. The map $\bar{\mathcal{E}}_\tau(\cdot) = \langle\mathcal{E}_\tau^{\delta\Phi}(\cdot)\rangle_{\delta\Phi}$ denotes the evolution averaged over flux trajectories $\delta\Phi(t)$, with each trajectory generated by the master equation Eq. (F.1) The initial states are $\rho_{0,g0} = |\bar{g}0\rangle\langle\bar{g}0|$ and $\rho_{0,g1} = |\bar{g}1\rangle\langle\bar{g}1|$, where the overbar denotes eigenstates of the static system Hamiltonian in the dispersive regime.

Results are shown in Fig. 7.3. For a fixed detuning, the infidelity initially decreases with increasing τ because a slower ramp improves adiabaticity and suppresses coherent leakage, but it eventually increases as the longer evolution accumulates more decoherence error. For a fixed ramp time (e.g., $\tau = 50\,\mathrm{ns}$), larger detuning yields lower infidelity, due to larger instantaneous gap. Finally, for the same detuning and ramp time, DRAG outperforms a Gaussian ramp because the added Q-quadrature control enhances adiabaticity by suppressing unwanted transitions.

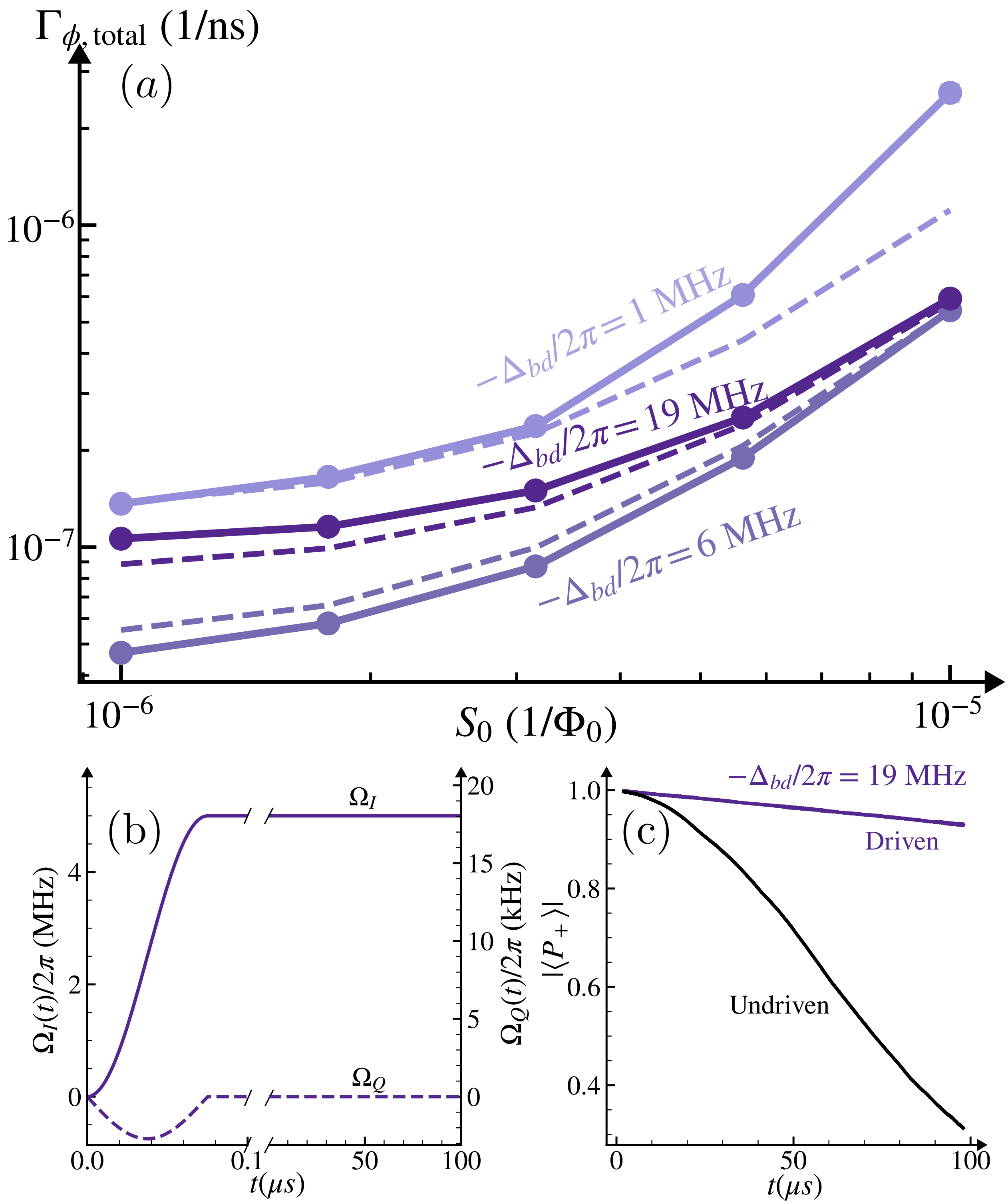

Figure 7.1: Numerical evaluation and time-domain validation of the total cavity dephasing rate under sweet-spot driving. (a) Total dephasing rate $\Gamma_{\phi,\mathrm{total}}$ extracted from full-system simulations as a function of detuning Δ_{bd}, with the drive amplitude chosen at each detuning to satisfy the sweet-spot condition. S_0 denotes the amplitude of $1/f$ flux noise. (b) DRAG-style adiabatic ramp pulse. (c) Time-domain decay of $P_X(t) = \mathrm{Tr}[\rho(t)\, X(t)]$, with $X(t) = |\phi_{g,1}(t)\rangle\langle\phi_{g,0}(t)| + \mathrm{H.c.}$. The time-dependent Floquet states are labeled by their bare-state correspondence. The driven system exhibits a cavity dephasing time of ~ 2 ms, compared to ~ 100 μs in the undriven case.

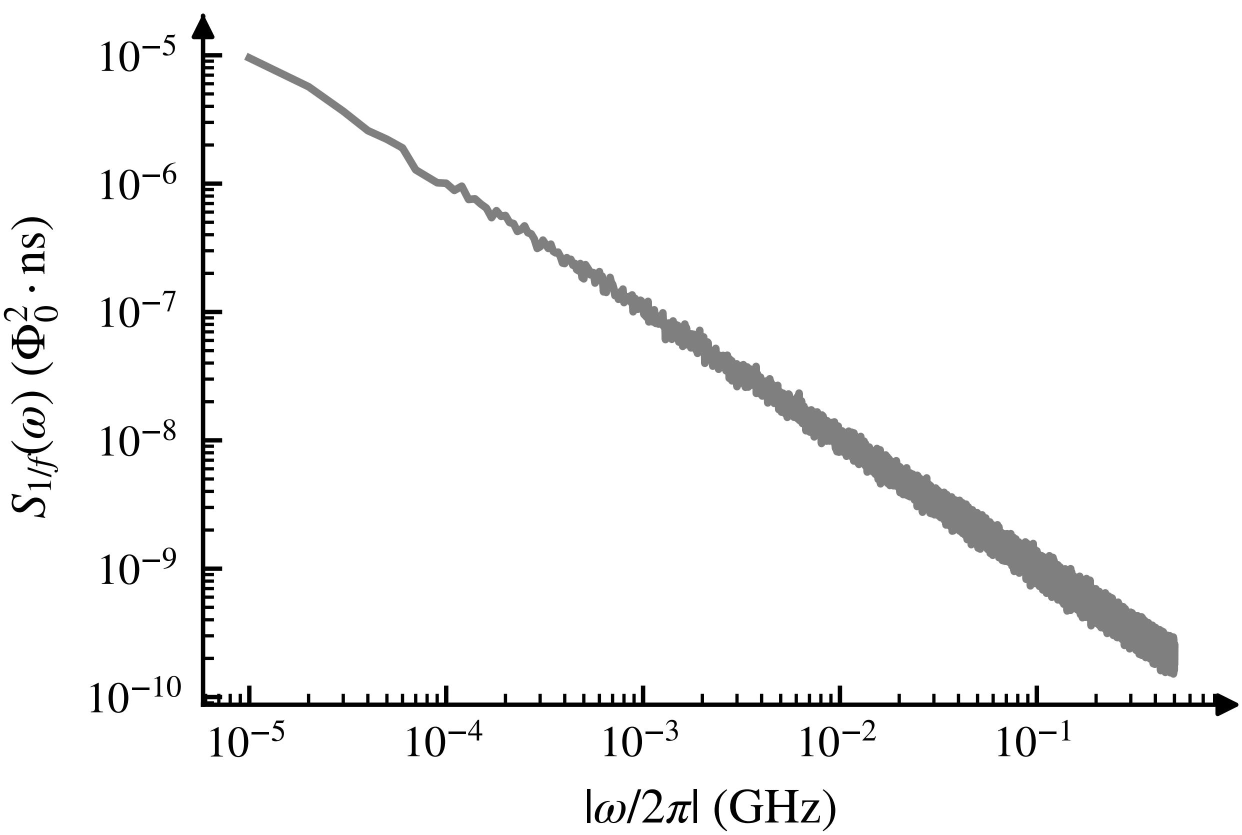

Figure 7.2: Power spectral density of $\delta\Phi(t)$ for 100 representative $1/f$ noise trajectories generated with the cutoffs specified below.

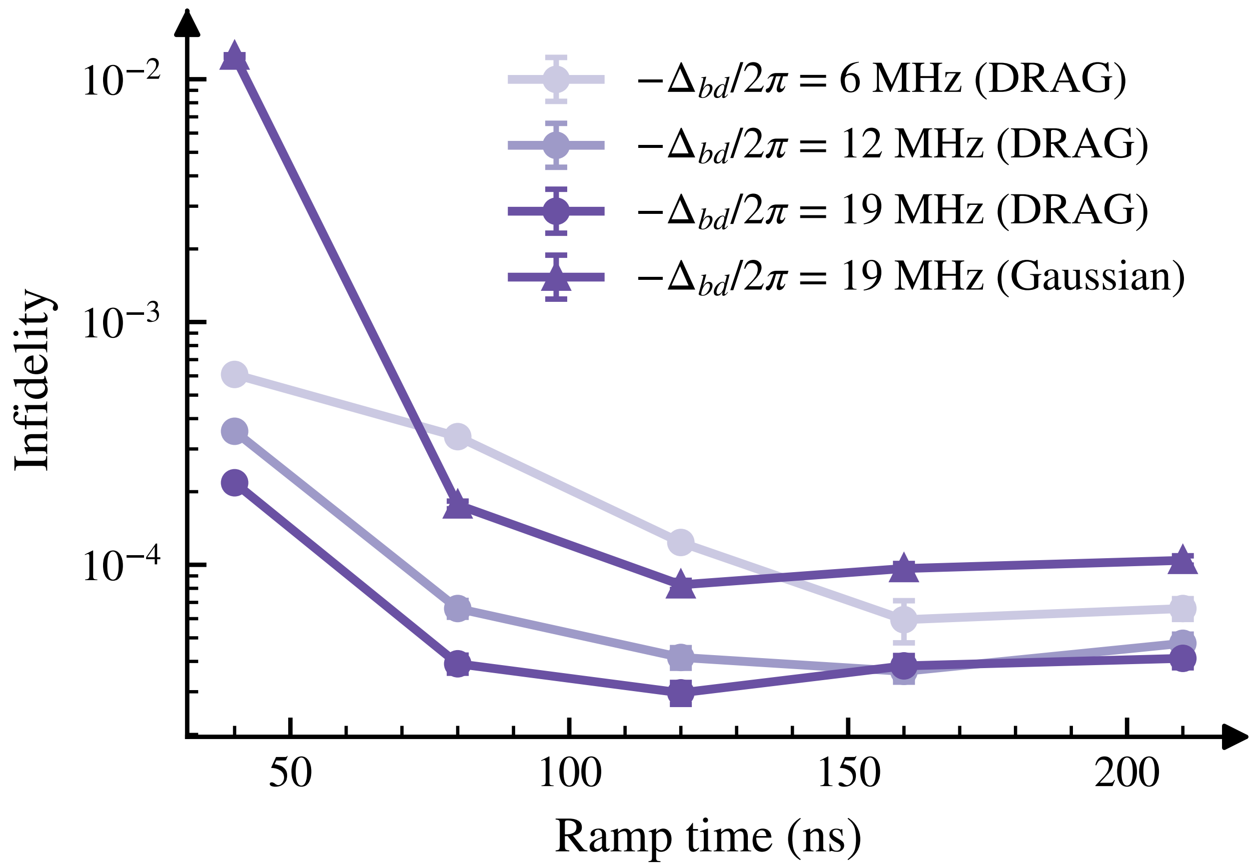

Figure 7.3: State-preparation infidelity $\mathcal{I}(\tau)$ versus ramp time τ for several detunings Δ_{bd}, comparing Gaussian and DRAG ramps. The non-monotonic dependence reflects the trade-off between coherent leakage from unwanted transitions (suppressed for longer τ) and decoherence (which accumulates for longer τ).

CHAPTER 8
APPLICATIONS TO BOSONIC ERROR-CORRECTION CODES

8.1 Application to Bosonic Encodings

In the previous sections, we showed that operating at a driven sweet spot can extend the cavity dephasing time by more than an order of magnitude. Here we apply the same strategy to bosonic quantum error correction (QEC). The central observation is that, in the unselective regime, the drive does not resolve individual photon-number states, so a single choice of drive parameters simultaneously suppresses the dephasing rates of all relevant cavity transitions induced by FTT-frequency fluctuations. This property is precisely what is needed for bosonic encodings, which store quantum information in superpositions of multiple Fock states.

Throughout this section, we fix the drive amplitude at $\Omega_0/2\pi = 10$ MHz and sweep the detuning Δ_{bd} to identify the operating point that simultaneously minimizes the relevant dephasing rates. The metrics below—transition dephasing rates and logical-subspace coherence proxies—quantify the suppression of inherited dephasing rather than a full logical error rate, which would additionally require modeling the complete QEC cycle.

8.1.1 Binomial encoding

As a concrete example, consider the minimal binomial code [62]

$$|0_L\rangle = \frac{|0\rangle + |4\rangle}{\sqrt{2}}, \qquad |1_L\rangle = |2\rangle, \tag{8.1}$$

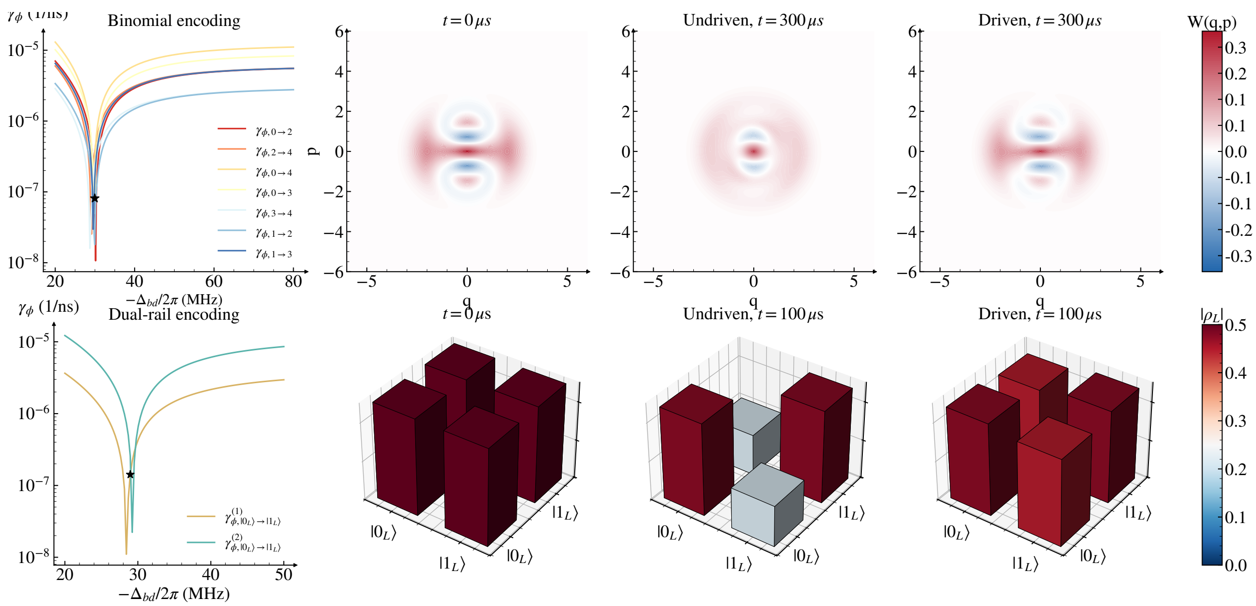

Figure 8.1: **Driven sweet-spot protection for bosonic encodings** ($\Omega_0/2\pi = 10\,\mathrm{MHz}$ throughout). *Binomial code*—**(a)** Dephasing rate $\gamma_{\phi,n\to m}$ of cavity transitions due to FTT-frequency fluctuations as a function of Δ_{bd}; the star marks $\Delta_{bd}/2\pi = -30\,\mathrm{MHz}$, where all relevant $\gamma_{\phi,n\to m}$ are suppressed by at least $\sim 13\times$. **(b)** Wigner function of $|+_L\rangle$ at $t = 300\,\mu\mathrm{s}$: the driven case retains the interference fringes that are lost without the drive. *Dual-rail encoding*—**(c)** Dephasing rate $\gamma^{(i)}_{\phi,0\to1}$ for each cavity as a function of Δ_{bd}; both cavities exhibit $\sim 10\times$ lower dephasing near $-\Delta_{bd}/2\pi \sim 29\,\mathrm{MHz}$. **(d)** Logical-subspace tomography for one dual-rail qubit: the drive suppresses dephasing, while leakage from single-photon loss remains unchanged.

the simplest binomial encoding that corrects a single photon-loss event. The drive is not intended to eliminate all dephasing; rather, it suppresses the specific transition-frequency fluctuations that dominate the logical-phase diffusion and recovery-sensitive coherences for this encoding. We now identify these transitions explicitly.

For the logical basis state $|0_L\rangle \propto |0\rangle + |4\rangle$, fluctuations of the $0 \to 4$ splitting, $\delta\widetilde{\omega}_{c,0\to4}(t)$, induce phase diffusion between $|0\rangle$ and $|4\rangle$ and therefore deform $|0_L\rangle$. Because this code does not protect against pure dephasing, the coherence of an arbitrary superposition $\alpha|0_L\rangle + \beta|1_L\rangle$ is carried by the bare-state coherences $|0\rangle \leftrightarrow |2\rangle$ and $|2\rangle \leftrightarrow |4\rangle$. Suppressing the corresponding dephasing rates $\gamma_{\phi,0\to2}$ and $\gamma_{\phi,2\to4}$ therefore directly improves logical coherence.

After a single photon loss, the code space maps as $|0_L\rangle \to |3\rangle$ and $|1_L\rangle \to |1\rangle$, so the logical information temporarily resides in the $|1\rangle \leftrightarrow |3\rangle$ coherence prior to correction. Suppressing $\gamma_{\phi,1\to3}$ is thus essential for high-fidelity recovery. During the recovery itself, intermediate superpositions populate the $\{|0\rangle, |3\rangle, |4\rangle\}$ manifold and the $\{|1\rangle, |2\rangle\}$ manifold, making the corresponding transition dephasing rates—e.g., $\gamma_{\phi,0\to3}$, $\gamma_{\phi,3\to4}$, and $\gamma_{\phi,1\to2}$—additional targets for suppression.

To quantify the improvement, we sweep the detuning Δ_{bd} at fixed $\Omega_0/2\pi = 10\,\mathrm{MHz}$ and evaluate the dephasing rate due to flux noise,

$$\gamma_{\phi,n\to m} = S_0 \left| \frac{\partial \widetilde{\omega}_{c,n\to m}}{\partial \Phi} \right| \sqrt{2\ln(|\omega_{\mathrm{ir}} t|)}. \tag{8.2}$$

Here, ω_{ir} is the infrared cutoff of the $1/f$ noise spectrum (set by the inverse of the total measurement time) and t is the experimental timescale. An earlier experiment reported $\ln(|\omega_{\mathrm{ir}} t|) \approx 4$ [173], and we use this value in the following numerics. The results are shown in Fig. 8.1(a). The minima of several dephasing rates overlap near $-\Delta_{bd}/2\pi \simeq 30\,\mathrm{MHz}$, consistent with the unselective regime $\Delta_{bd} \gg |\chi|$ (here $|\chi|/2\pi \sim 0.5\,\mathrm{MHz}$).

Choosing $\Delta_{bd}/2\pi = -30\,\mathrm{MHz}$ yields a worst-case reduction of the dephasing rate by a factor of ~ 13 across the relevant transitions. This improvement is confirmed by Wigner tomography [Fig. 8.1(b)], which serves as a direct proxy for the preservation of coherence within the logical subspace. For the initial state $|+_L\rangle = (|0_L\rangle + |1_L\rangle)/\sqrt{2}$, the undriven Wigner function reduces to a mixture of Fock states after $300\,\mu\mathrm{s}$, indicating complete loss of logical coherence. In the driven case, the characteristic interference fringes are largely retained, confirming that the drive preserves the intra-code-word coherence required by the binomial encoding.

8.1.2 Dual-rail encoding

Dual-rail encoding is widely used because the dominant single-photon-loss error maps to a detectable erasure [155, 174]. A single dual-rail qubit is encoded in two cavities with logical states $|0_L\rangle = |01\rangle$ and $|1_L\rangle = |10\rangle$. The fault-tolerance advantage of this encoding relies on the error hierarchy in which single-photon loss dominates over dephasing; inherited cavity dephasing from a flux-tunable coupler can degrade this bias. To implement two-qubit gates, one approach is to couple two dual-rail qubits via an FTT and exploit its three-wave-mixing interaction.

In this architecture, the FTT (mode b) couples to two cavities labeled by i. Flux fluctuations $\delta\Phi(t)$ modulate the FTT frequency, and this modulation enters the cavity spectrum through the Lamb shift, producing cavity-frequency fluctuations

$$\delta\widetilde{\omega}_c^{(i)}(t) = \left(\frac{g^{(i)}}{\Delta_{bc}^{(i)}}\right)^2 \delta\omega_b(t), \tag{8.3}$$

where $g^{(i)}$ and $\Delta_{bc}^{(i)}$ are the coupling strength and detuning between the FTT and cavity i, respectively. Because each cavity forms one rail of a dual-rail qubit, these fluctuations contribute directly to logical dephasing. By appropriately driving the coupler, the dephasing rates of both cavities due to inherited FTT-frequency noise can be reduced simultaneously.

Fig. 8.1(c) shows the numerically computed dephasing rate $\gamma_{\phi,0\to1}^{(i)}$ for each cavity as Δ_{bd} is swept. Choosing $-\Delta_{bd}/2\pi \sim 29\,\mathrm{MHz}$ yields $\sim 10\times$ lower dephasing rates for both cavities than in the undriven case. We then simulate the dynamics of a single dual-rail qubit and show the state tomography in the logical subspace in Fig. 8.1(d). The simulation confirms that the drive largely suppresses pure dephasing within the logical subspace, while single-photon-loss-induced leakage out of the logical subspace remains nearly unchanged between the driven and undriven cases. This demonstrates that the driven sweet spot acts as a bias-preserving knob: it mitigates inherited dephasing while leaving the dominant single-photon-loss channel—and hence the favorable error hierarchy of the dual-rail encoding—intact.

8.2 Conclusion

In summary, we have introduced a hardware-efficient protocol to dynamically suppress cavity dephasing caused by flux noise inherited from a coupled nonlinear element. By applying a continuous, off-resonant microwave drive to the FTT, we generate a photon-number-dependent AC-Stark shift that cancels the cavity frequency's first-order susceptibility to coupler-frequency fluctuations. Our theoretical analysis and Monte Carlo simulations show that operating at this driven sweet spot can extend the cavity dephasing time by more than an order of magnitude, achieving a $\sim 20\times$ improvement from ~ 100 μs to ~ 2 ms.

While the continuous drive inevitably introduces secondary decoherence channels, such as photon-shot-noise dephasing from drive-induced FTT excitation, we show that operating in the unselective regime at larger detunings parametrically suppresses these effects. In this regime, the primary flux-noise cancellation dominates, resulting in a net enhancement of coherence. Crucially, this mitigation strategy is highly hardware-efficient: it circumvents the need for fast flux tuning during idling, thereby avoiding unwanted transitions that can occur when the coupler frequency is swept past cavity resonances.

Furthermore, we have shown that the mechanism is not limited to the FTT: because it is rooted in drive-induced energy shifts, it extends to other flux-dependent nonlinear elements such as SNAIL-based couplers. We also showed that the protocol is directly relevant for bosonic quantum error correction, where a single drive setting can simultaneously suppress the flux sensitivity of multiple critical transitions in binomial and dual-rail encodings. Overall, the approach provides a practical route toward preserving the noise bias needed for cavity-based QEC at non-sweet-spot

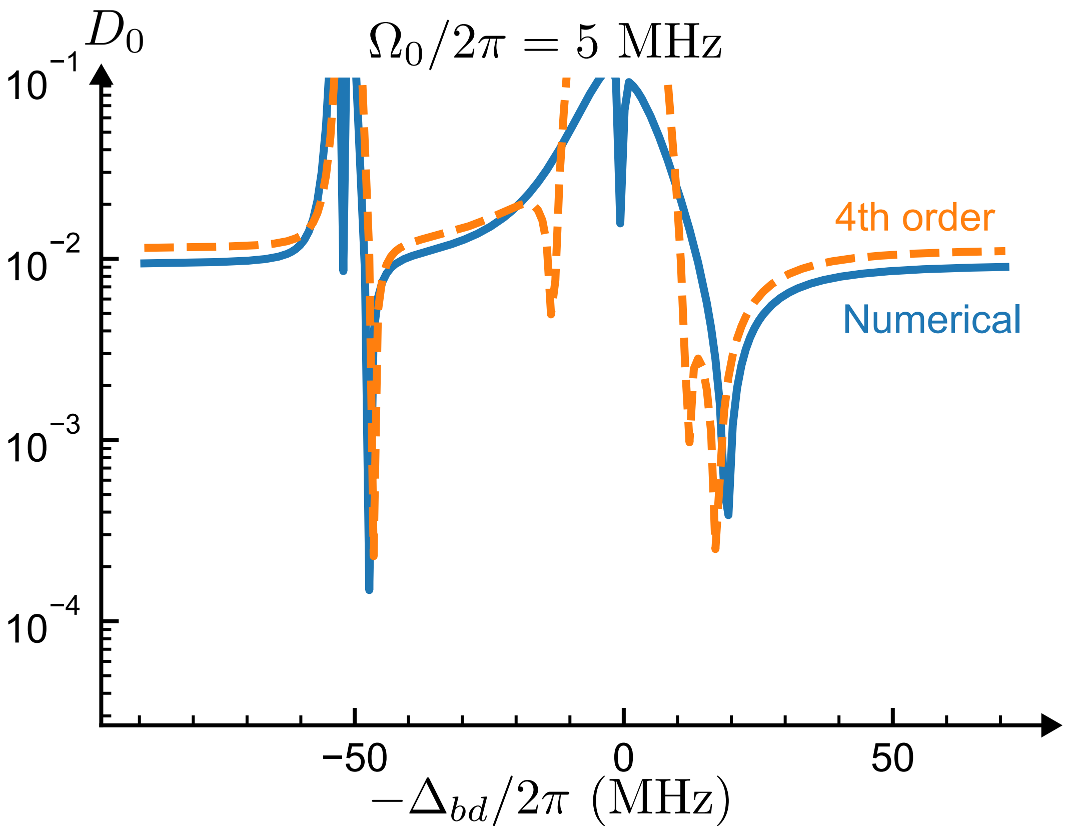

Figure 8.2: Susceptibility D_0 including higher-order corrections in the Schrieffer–Wolff expansion. An additional sweet spot appears near $-\Delta_{bd} \simeq K/2$, arising from hybridization between $|g\rangle$ and $|f\rangle$ states

flux operating points.

8.3 Sweet spots from higher-order corrections

Higher-order, drive-induced terms in the Schrieffer–Wolff expansion modify the cavity susceptibility and can generate additional flux sweet spots. To illustrate this effect, we consider the vacuum transition frequency $\widetilde{\omega}_{c,0\to1}$. Retaining terms up to fourth order yields the drive-induced correction

$$\begin{aligned}\widetilde{\omega}_{c,0\to1} = \overline{\omega}_c + \frac{\Omega_0^2}{-\Delta_{bd}-\chi} + \frac{\Omega_0^2}{\Delta_{bd}} - \frac{\Omega_0^4}{(-\Delta_{bd}-\chi)^3} \\ + \frac{2\Omega_0^4}{(-\Delta_{bd}-\chi)^2(-2\Delta_{bd}-K-2\chi)} - \frac{\Omega_0^4}{\Delta_{bd}^3} \\ - \frac{2\Omega_0^4}{\Delta_{bd}^2(-2\Delta_{bd}-K)}.\end{aligned} \tag{8.4}$$

The resulting susceptibility D_0 is shown in Fig. D.1. In addition to the sweet spots predicted at leading order, an extra sweet spot appears near $-\Delta_{bd} \simeq K/2$.

This feature originates from near-resonant hybridization between $|g, m\rangle$ and the higher FTT level $|f, m\rangle$. At this detuning, the drive couples states separated by approximately the anharmonicity K, producing an additional Stark shift that alters the cancellation condition for D_0. Operation at this sweet spot may therefore be more sensitive to drive-induced decoherence channels and should be assessed accordingly.

8.4 Example of a SNAIL coupler

The superconducting nonlinear asymmetric inductive element (SNAIL) is a three-wave-mixing circuit element widely used for parametric amplification and gate operations. When coupled capacitively to a cavity, the system Hamiltonian can be written as

$$\begin{aligned}H = \omega_s s^\dagger s + \sum_j g_j (s+s^\dagger)^j + \omega_c c^\dagger c \\ + g(s+s^\dagger)(c+c^\dagger) + 2\Omega_0 \cos(\omega_d t)(s+s^\dagger).\end{aligned} \tag{8.5}$$

Here ω_s is the SNAIL mode frequency, g_j are the nonlinear coefficients, g is the SNAIL–cavity coupling strength, s (c) is the SNAIL (cavity) mode operator, and ω_d is the drive frequency. The

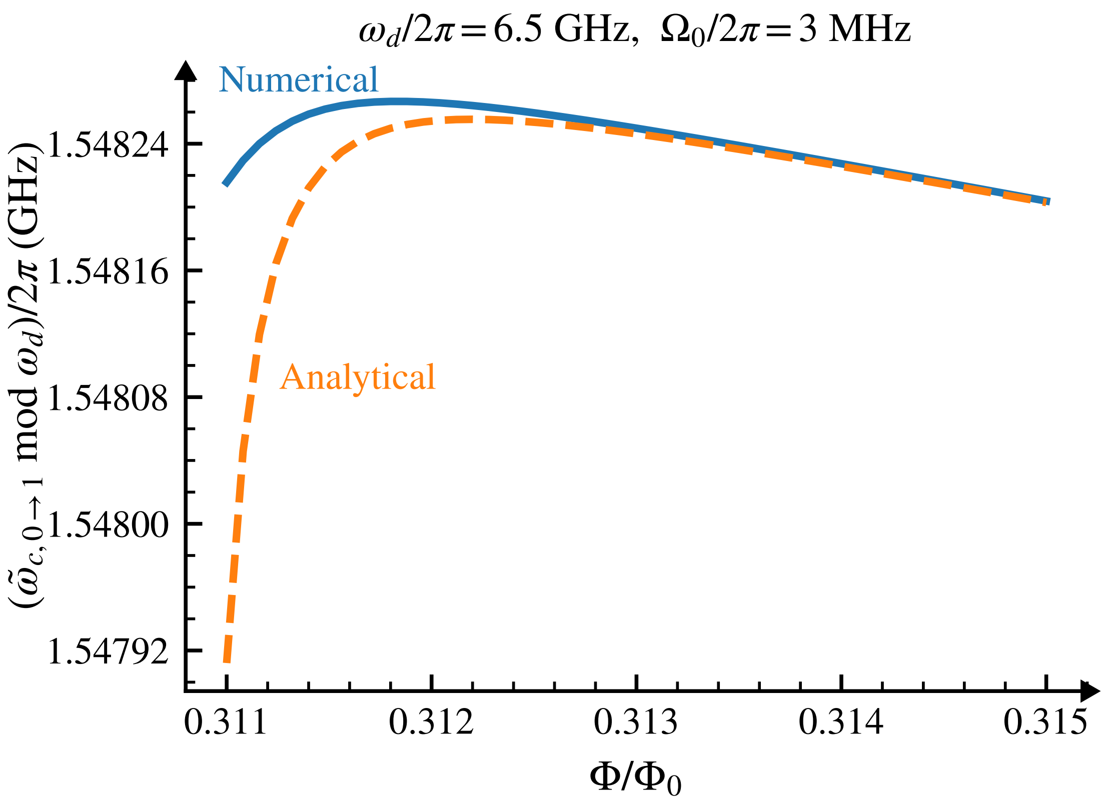

Figure 8.3: Drive-induced sweet spot in a SNAIL–cavity system. Under continuous drive, the cavity transition energy exhibits a maximum near $\Phi \simeq 0.311\,\Phi_0$, indicating reduced first-order flux susceptibility. Solid lines show numerical results and dashed lines show the perturbative prediction from Eq. (6.57).

parameters ω_s, g_j, g, and Ω_0 are flux dependent through Φ.

To illustrate the generality of the drive-induced cancellation mechanism, the SNAIL is biased at $\Phi = 0.3\,\Phi_0$, where the cubic nonlinearity g_3 enables three-wave mixing. At this operating point, $g_3/2\pi \approx 67$ MHz, $\omega_s/2\pi = 6.59$ GHz, and $\omega_c/2\pi = 5$ GHz.

A coherent drive with amplitude $\Omega_0 = 3$ MHz at $\omega_d = 6.5$ GHz produces a flux sweet spot in the cavity spectrum. As shown in Fig. E.1, the cavity transition energy, extracted using the procedure in App. 7.2, develops a local maximum near $\Phi \simeq 0.311\,\Phi_0$, consistent with vanishing first-order flux susceptibility. The perturbative expression in Eq. (6.57) reproduces this sweet spot quantitatively, demonstrating that the drive-induced cancellation mechanism extends beyond the FTT to more general nonlinear, flux-dependent modes.

8.5 Amplitude-noise-induced dephasing

With sweet spot drive, the net cavity frequency

$$\widetilde{\omega}_{c,0\to 1} = \bar{\omega}_c + \frac{\Omega_0^2 \chi}{\Delta_{bd}^2}. \tag{8.6}$$

is insensitive to flux noise, but it is sensitive to drive-amplitude noise.

We consider an amplitude-noise spectrum containing both white-noise and $1/f$ components,

$$S_\Omega(\omega) = (S_0^{\mathrm{w}})^2 + \frac{(S_0^{1/f})^2}{|\omega|/2\pi}, \tag{8.7}$$

where S_0^{w} and $S_0^{1/f}$ denote the amplitudes of the white-noise and $1/f$ noise contributions, respectively. Within leading-order perturbation theory, the corresponding cavity dephasing rate is

$$\frac{1}{T_\phi} = (S_0^{\mathrm{w}})^2 \left|\frac{\partial \widetilde{\omega}_{c,0\to1}}{\partial \Omega}\right|^2 + S_0^{1/f} \left|\frac{\partial \widetilde{\omega}_{c,0\to1}}{\partial \Omega}\right| \sqrt{2\ln\big(|\omega_{\mathrm{ir}} t|\big)}. \tag{8.8}$$

We evaluate this expression for a specific hardware platform to estimate the corresponding dephasing time. For the Zurich Instruments HDAWG, the white-noise floor is $25\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$, and the $1/f$ corner frequency is approximately $100\,\mathrm{kHz}$. These specifications imply a $1/f$ voltage-noise spectral density of approximately $7.9\,\mu\mathrm{V}/\sqrt{\mathrm{Hz}}$ at $1\,\mathrm{Hz}$.

To convert these voltage-noise specifications into Rabi-frequency noise amplitudes, we use the linear relation between the drive voltage V and the induced Rabi frequency Ω. Because cryogenic attenuation suppresses the signal and the amplitude noise by the same factor, it does not by itself change the fractional noise, so that $\delta\Omega/\Omega = \delta V/V$. Consequently, for a fixed target drive strength Ω, the effective drive-noise amplitudes depend on the AWG output voltage used to generate that drive. If the dominant instrument noise is set by an approximately additive voltage floor, then using a larger V_{AWG} and compensating with additional attenuation reduces the fractional noise $\delta V/V_{\mathrm{AWG}}$. The white-noise and $1/f$ amplitudes are therefore

$$S_0^{\mathrm{w}} = \Omega \times \frac{25\,\mathrm{nV}/\sqrt{\mathrm{Hz}}}{V_{\mathrm{AWG}}}, \qquad S_0^{1/f} = \Omega \times \frac{7.9\,\mu\mathrm{V}/\sqrt{\mathrm{Hz}}}{V_{\mathrm{AWG}}}, \tag{8.9}$$

where V_{AWG} is the peak output voltage of the instrument. As a representative operating point, for a target drive strength of $\Omega/2\pi = 10\,\mathrm{MHz}$ and $V_{\mathrm{AWG}} = 0.5\,\mathrm{V}$, we obtain $S_0^{\mathrm{w}} \approx 0.5\sqrt{\mathrm{Hz}}$ and $S_0^{1/f} \approx 158\sqrt{\mathrm{Hz}}$.

Using the parameters at the sweet spot in Fig. 6.2(a), we numerically evaluate

$$\left|\frac{\partial \widetilde{\omega}_{c,0\to1}}{\partial \Omega}\right| = 0.044. \tag{8.10}$$

Assuming a typical measurement timescale such that $\sqrt{2\ln\big(|\omega_{\mathrm{ir}}t|\big)} \approx 4$ [173], we obtain a pure dephasing rate of

$$\frac{1}{T_\phi} \approx 35\,\mathrm{s}^{-1}, \tag{8.11}$$

which corresponds to an amplitude-noise-limited dephasing time of

$$T_\phi \approx 29\,\mathrm{ms}. \tag{8.12}$$

CHAPTER 9

CONCLUSION AND FUTURE WORK

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APPENDIX A

SCALING OF THE NUMBER OF MATRIX-VECTOR MULTIPLICATIONS μ

The asymptotic behavior of runtime, discussed in Chapter 4, is governed by the scaling of μ (3.52) with the dimension d of the Hilbert space. The dependence $\mu(d)$ is specific to the matrix A to be exponentiated (which is proportional to the Hamiltonian). This prevents a general scaling relation from being stated. I thus turn to a discussion of several concrete examples.

I first consider the example of the transmon–cavity system, see Eq. (4.1), where d is increased via the cavity dimension d_c while leaving the transmon cutoff fixed. The corresponding Hamiltonian H_1, written in the bare product basis, then has a constant maximum number of non-zero elements in each row. (I.e., the parameter σ' (3.42) is independent of d.) Numerically, I find in this case that μ scales linearly with $||A||_1$ $\big(\mu \sim \Theta(||A||_1)\big)$, and this linearity holds for a wide range of τ, see Fig. A.1(a). In this example, two ingredients lead to the scaling $||A||_1 \sim \Theta(d^{1/2})$: (1) d is a constant multiple of d_c, and (2) the cavity ladder operators in H_1 which have a one-norm scaling $\Theta(d_c^{1/2})$. Thus, the overall scaling $\mu \sim \Theta(d^{1/2})$ is obtained, see Fig. A.1(e). In the second example, I consider the system of three coupled transmons, see Eq. (4.2). The Hamiltonian H_2 is written in the harmonic oscillator basis and has a constant σ', which leads to $\mu \sim \Theta(||A||_1)$ [Fig. A.1(b)]. The one-norm of operator $(b^\dagger b)^2$ in H_2 has quadratic scaling with each transmon dimension. Since d is increased by enlarging the dimensions of all three transmons simultaneously, $||A||_1 \sim \Theta(d^{2/3})$. This scaling results in $\mu \sim \Theta(d^{2/3})$, see Fig. A.1(f).

For other Hamiltonian matrices where σ' depends on d, the scaling $\mu \sim \Theta(||A||_1)$ may still hold. In the following, I show two concrete examples—one where linearity holds, and another where it breaks. I first consider a driven system of N_q qubits with nearest-neighbor σ_z coupling.

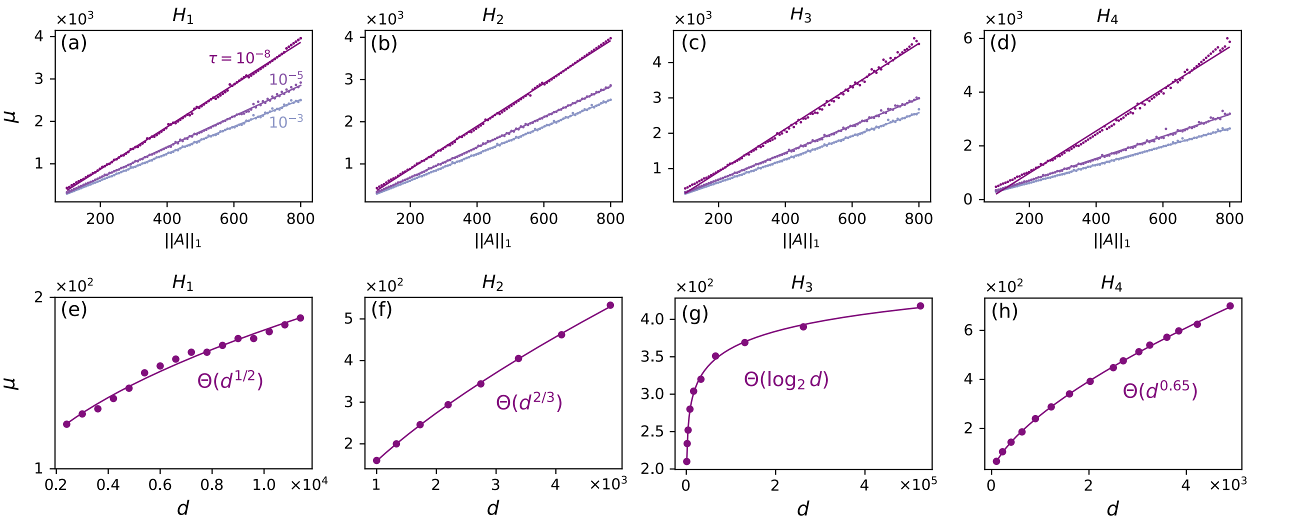

Figure A.1: Scaling of μ, the number of matrix-vector multiplications. (a)–(d) The scaling of μ with $||A||_1$ for Hamiltonians H_j, $j = 1, 2, 3, 4$, when considering several error tolerances τ. (e)–(h) The scaling of μ with dimension d when considering $\tau = 10^{-8}$ for each H_j, respectively. In all figures, dots are sampling data and solid lines are fitting curves. The parameter setup of H_1, H_2 is the same as those given in the captions of Figs. 4.1 and 4.2. Parameters of H_3 and H_4 are as follows: for H_3, $E_{Ci}/2\pi = 2.5\,\mathrm{GHz}$, $E_{Ji}/2\pi = 8.9\,\mathrm{GHz}$, $E_{Li}/2\pi = 0.5\,\mathrm{GHz}$, $g/2\pi = 0.1\,\mathrm{GHz}$, and $\phi = 0.33$; for H_4, $a_x^{(\nu)}/2\pi = a_y^{(\nu)}/2\pi = 0.5\,\mathrm{GHz}$ and $g/2\pi = 0.1\,\mathrm{GHz}$.

In the rotating frame, the system is described by

$$H_3(t) = \sum_{\nu=1}^{N_q} [a_x^{(\nu)}(t)\sigma_x^{(\nu)} + a_y^{(\nu)}(t)\sigma_y^{(\nu)}] + \sum_{\nu=1}^{N_q-1} g\sigma_z^{(\nu)}\sigma_z^{(\nu+1)}. \tag{A.1}$$

Each qubit is controlled via drive fields $a_x^{(\nu)}$ and $a_y^{(\nu)}$. For this example, the numerical results show that the scaling $\mu \sim \Theta(||A||_1)$ still holds, see Fig. A.1(c). Since d is increased by enlarging N_q, $||A||_1 \sim \Theta(N_q) \sim \Theta(\log_2 d)$. This scaling results in $\mu \sim \Theta(\log_2 d)$, see Fig. A.1(g). For the second example, I consider two capacitively coupled fluxonium qubits with Hamiltonian

$$H_4(t) = \sum_{i=1}^{2} \left(4E_{Ci}n_i^2 - E_{Ji}\cos(\varphi_i) + \frac{E_{Li}}{2}[\varphi_i - \phi_i(t)]^2\right) + gn_1n_2. \tag{A.2}$$

Here, E_{Ci}, E_{Ji}, E_{Li} denote the charging, inductive, and Josephson energies of two fluxonium qubits, and g is the interaction strength. φ and n are the phase and charge number operators, defined in the usual way. The Hamiltonian is represented in the harmonic oscillator basis. Fig. A.1(d) shows that the scaling of μ is superlinear with $||A||_1$ for $\tau = 10^{-8}$. I thus settle for a numerical determination of the scaling relation between μ and d for $\tau = 10^{-8}$ by increasing the dimensional cutoff for both fluxonium qubits simultaneously. The observed scaling is $\mu \sim \Theta(d^{0.65})$, see Fig. A.1(h).

APPENDIX B

ALTERNATIVE HARD-CODED GRADIENT SCHEME: NUMERICAL IMPLEMENTATION AND SCALING

The numerical implementation of the hard-coded gradient scheme requires evaluation of the short-time propagator and its derivative. Under certain conditions, it turns out to be advantageous to numerically evaluate the short-time propagator by employing matrix diagonalization. In this appendix, I first discuss the method to compute the propagator derivative and then analyze the scaling of HG.

B.1 Evaluation of propagator derivative

The operator $-iH\Delta t$ is anti-Hermitian, and hence can be diagonalized and exponentiated according to

$$A = SDS^\dagger, \tag{B.1}$$

$$e^A = Se^D S^\dagger. \tag{B.2}$$

Here, S is unitary and D diagonal. As a result, the propagator derivative can be written as

$$\frac{de^A}{da} = Se^D \frac{dD}{da} S^\dagger + \frac{dS}{da} e^D S^\dagger + Se^D \frac{dS^\dagger}{da}. \tag{B.3}$$

Applying the inverse unitary transformation to $\frac{de^A}{da}$ (B.3) yields

$$S^\dagger \frac{de^A}{da} S = e^D \frac{dD}{da} + S^\dagger \frac{dS}{da} e^D + e^D \frac{dS^\dagger}{da} S. \tag{B.4}$$

Based on $\frac{dS^\dagger}{da} S + S^\dagger \frac{dS}{da} = 0$, Eq. (B.4) can be rewritten as

$$S^\dagger \frac{de^A}{da} S = e^D \frac{dD}{da} + E \circ (S^\dagger \frac{dS}{da}) \tag{B.5}$$

with $E_{ij} := e^{D_{jj}} - e^{D_{ii}}$. Here, $\circ$ denotes the Hadamard product (element-wise multiplication). To obtain dD/da and dS/da, one takes the derivative of Eq. (B.1) and repeats the same steps as for Eqs. (B.4) and (B.5), leading to

$$S^\dagger \frac{dA}{da} S = \frac{dD}{da} + E' \circ (S^\dagger \frac{dS}{da}), \tag{B.6}$$

where $E'_{ij} := D_{jj} - D_{ii}$. Since the diagonal elements of E' are zero, it follows that

$$\frac{dD}{da} = I \circ (S^\dagger \frac{dA}{da} S). \tag{B.7}$$

For off-diagonal elements of the operator in Eq. (B.6):

$$F \circ (S^\dagger \frac{dA}{da} S) + G = S^\dagger \frac{dS}{da}, \tag{B.8}$$

where

$$F_{ij} := \begin{cases} \frac{1}{D_{jj}-D_{ii}}, & \text{if } D_{jj}-D_{ii} \neq 0. \\ 0, & \text{otherwise.} \end{cases} \tag{B.9}$$

$$G_{ij} := \begin{cases} 0, & \text{if } D_{jj}-D_{ii} \neq 0. \\ (S^\dagger \frac{dS}{da})_{ij}, & \text{otherwise.} \end{cases}$$

Inserting Eqs. (B.7) and (B.8) into Eq. (B.5) gives

$$\frac{de^A}{da} = S\big((e^D + E \circ F) \circ (S^\dagger \frac{dA}{da} S)\big)S^\dagger, \tag{B.10}$$

where $E \circ G = 0$ has been used.

B.2 Scaling analysis

The gradients of the contributions to the cost function (for the case of state transfer) are evaluated by the forward-backward propagation scheme. In this scheme, the storage required for states does not scale with the number of intermediate time steps. The matrices that must be stored are the system and control Hamiltonians as well as the dense matrices S, E, and E'. Thus, the total memory usage scales as $O(d^2)$. The runtime of the scheme is dominated by $O(N)$ evaluations of the propagator and its derivative. Employing matrix diagonalization for the propagator evaluation requires a runtime scaling $\sim O(d^3)$ [175]. Calculation of the propagator derivative involves matrix-matrix multiplication and Hadamard product with runtime scaling $O(d^3)$ and $O(d^2)$, respectively. Thus, the overall runtime of the HG scheme scales as $O(Nd^3)$.

The gradients of the contributions to the gate-operation cost function are evaluated in an analogous manner, resulting in the same runtime and memory scaling.

APPENDIX C

FLOQUET FORMALISM AND JUSTIFICATION OF THE RWA

C.1 Floquet Formalism

In this Appendix, we justify the use of the static rotating-frame Hamiltonian H_R [Eq. (6.55)] to approximate the quasi-energies and Floquet states of the driven dispersive Hamiltonian.

C.1.1 Floquet Theorem

Consider a Hilbert space $\mathcal{H}$ of dimension N and a time-dependent Hamiltonian $H(t)$ with period τ, satisfying $H(t+\tau) = H(t)$. According to Floquet's theorem, there exists a τ-periodic orthonormal basis $\{|\psi_n(t)\rangle\}$ and a set of quasienergies $\{\varepsilon_n\}$ such that the solutions to the Schrödinger equation, $|\phi_n(t)\rangle$, can be written as:

$$|\phi_n(t)\rangle = e^{-i\varepsilon_n t} |\psi_n(t)\rangle . \tag{C.1}$$

C.1.2 Construction of the Solution

To acquire a differential equation that relates the periodic modes $|\psi_n(t)\rangle$ and quasienergies ε_n to the Hamiltonian $H(t)$, we substitute the ansatz solution into the time-dependent Schrödinger equation:

$$i\frac{\partial}{\partial t} |\Psi(t)\rangle = H(t)\ |\Psi(t)\rangle . \tag{C.2}$$

Substituting $|\Psi(t)\rangle = e^{-i\varepsilon_n t} |\psi_n(t)\rangle$ yields the Floquet eigenvalue equation:

$$\big(H(t) - i\partial_t\big)\ |\psi_n(t)\rangle = \varepsilon_n\ |\psi_n(t)\rangle . \tag{C.3}$$

Since the quantities on both sides of Eq. (C.3) are periodic, we can treat the time dependence using Fourier analysis. A concise way to formalize this is to introduce the *Extended Hilbert space* $\mathcal{F}$, also known as Sambe space:

$$\mathcal{F} = \mathrm{span}\big\{|\,\mu, n\rangle\!\rangle \;\big|\; \mu \in \mathbb{Z},\ n = 1, \ldots, N\big\}. \tag{C.4}$$

We use the double ket notation $|\,\cdot\rangle\!\rangle$ to denote a basis vector in this extended space. Each basis vector corresponds to a function mapping time $t \in \mathbb{R}$ to a state in the physical space $\mathcal{H}$:

$$|\,\mu, n\rangle\!\rangle : \mathbb{R} \to \mathcal{H}. \tag{C.5}$$

Specifically, these basis states are defined as:

$$|\,\mu, n\rangle\!\rangle(t) = |n\rangle\ e^{i\mu\omega_d t}, \qquad \omega_d \equiv \frac{2\pi}{\tau}, \tag{C.6}$$

where $\{|n\rangle\}$ is an orthonormal basis in $\mathcal{H}$. The inner product on $\mathcal{F}$ is defined by averaging over one period:

$$\begin{aligned}
\langle\!\langle \mu', n' \,|\, \mu, n\rangle\!\rangle &\equiv \frac{1}{\tau}\int_0^\tau \big[|\,\mu', n'\rangle\!\rangle(t)\big]^\dagger |\,\mu, n\rangle\!\rangle(t)\, dt \\
&= \frac{1}{\tau}\int_0^\tau e^{-i\mu'\omega_d t}\,\langle n'|n\rangle\, e^{i\mu\omega_d t}\, dt \\
&= \delta_{n,n'}\frac{1}{\tau}\int_0^\tau e^{i(\mu-\mu')\omega_d t}\, dt \\
&= \delta_{\mu,\mu'}\,\delta_{n,n'}.
\end{aligned} \tag{C.7}$$

To rewrite Eq. (C.3) in the basis $|\,\mu, n\rangle\!\rangle$, we define the quasienergy operator $\overline{Q} : \mathcal{F} \to \mathcal{F}$ such that its action in the time domain is:

$$\big(\overline{Q}\,|\,f\rangle\!\rangle\big)(t) = \big(H(t) - i\partial_t\big)\ |f(t)\rangle\,. \tag{C.8}$$

We define the vector $|\,\psi_n\rangle\!\rangle \in \mathcal{F}$ corresponding to the periodic function $|\psi_n(t)\rangle$ as:

$$|\,\psi_n\rangle\!\rangle(t) = |\psi_n(t)\rangle\,. \tag{C.9}$$

Consequently, Eq. (C.3) is represented as an eigenvalue problem in the extended Hilbert space:

$$\overline{Q}\,|\,\psi_n\rangle\!\rangle = \varepsilon_n\,|\,\psi_n\rangle\!\rangle. \tag{C.10}$$

To compute the eigenvalues and eigenvectors, we find the matrix representation of $\overline{Q}$ in the basis $|\,\mu, n\rangle\!\rangle$:

$$\overline{Q} = \sum_{\mu',n'}\sum_{\mu,n} \bar{q}_{\mu'n',\mu n}\ |\,\mu', n'\rangle\!\rangle\langle\!\langle\mu, n\,|, \tag{C.11}$$

with the corresponding matrix elements:

$$\begin{aligned}\bar{q}_{\mu'n',\mu n} &= \langle\!\langle\mu', n'\,|\,\overline{Q}|\mu, n\rangle\!\rangle \\ &= \frac{1}{\tau}\int_0^\tau \big[|\,\mu', n'\rangle\!\rangle(t)\big]^\dagger\,\big(\overline{Q}\,|\,\mu, n\rangle\!\rangle(t)\big)\,dt.\end{aligned} \tag{C.12}$$

Substituting the explicit form $|\,\mu, n\rangle\!\rangle(t) = |n\rangle\, e^{i\mu\omega_d t}$:

$$
\begin{aligned}
\bar{q}_{\mu' n', \mu n} &= \frac{1}{\tau}\int_0^\tau e^{-i\mu'\omega_d t}\,\langle n'|\,\big(H(t) - i\partial_t\big)\big(|n\rangle\, e^{i\mu\omega_d t}\big)\, dt \\
&= \frac{1}{\tau}\int_0^\tau \langle n'|H(t)|n\rangle\, e^{i(\mu-\mu')\omega_d t}\, dt \;+\; \mu\,\omega_d\,\delta_{\mu'\mu}\,\delta_{n'n} \\
&= h_{n'n}^{(\mu-\mu')} \;+\; \mu\,\omega_d\,\delta_{\mu'\mu}\,\delta_{n'n},
\end{aligned}
\tag{C.13}
$$

where $h_{n'n}^{(k)}$ are the Fourier components of the Hamiltonian:

$$
H^{(k)} = \frac{1}{\tau}\int_0^\tau H(t)\, e^{ik\omega_d t}\, dt, \qquad h_{n'n}^{(k)} = \langle n'|H^{(k)}|n\rangle. \tag{C.14}
$$

Thus, the operator $\overline{Q}$ can be written as a block matrix with indices μ running (for example) from -2 to 2:

$$
\bar{q} = \begin{pmatrix}
\ddots & & & & & \\
& h^{(0)} - 2\omega_d I & h^{(1)} & h^{(2)} & h^{(3)} & h^{(4)} \\
& h^{(-1)} & h^{(0)} - \omega_d I & h^{(1)} & h^{(2)} & h^{(3)} \\
& h^{(-2)} & h^{(-1)} & h^{(0)} & h^{(1)} & h^{(2)} \\
& h^{(-3)} & h^{(-2)} & h^{(-1)} & h^{(0)} + \omega_d I & h^{(1)} \\
& h^{(-4)} & h^{(-3)} & h^{(-2)} & h^{(-1)} & h^{(0)} + 2\omega_d I \\
& & & & & \ddots
\end{pmatrix}, \tag{C.15}
$$

where I is the identity matrix. Note that the off-diagonal blocks depend on the difference in indices, $H^{(\mu-\mu')}$. Finally, the quasienergies are found by solving the characteristic equation:

$$
\det\big(\bar{q} - \varepsilon \bar{I}\big) = 0, \tag{C.16}
$$

where $\bar{I}$ is the identity operator in the extended Hilbert space.

C.1.3 Quasienergy and Floquet States under H_{disp}

In this subsection we relate an approximation of quasienergy and Floquet states to a rotating-wave approximation.

C.1.3 Quasienergy Operator for the Driven Dispersive Hamiltonian

The driven dispersive Hamiltonian is:

$$\begin{aligned} H(t) = \bar{\omega}_b b^\dagger b + \frac{K}{2} b^{\dagger 2} b^2 + \bar{\omega}_c c^\dagger c \\ + \chi b^\dagger b c^\dagger c + 2\Omega_0 \cos(\omega_d t)(b + b^\dagger). \end{aligned} \tag{C.17}$$

Following the recipe in the previous section, the quasienergy operator in the extended Hilbert space $\mathcal{F}$ is written as

$$\overline{Q} = \overline{Q}_0 + \overline{V}. \tag{C.18}$$

A convenient basis is $|\,\mu, (n,m)\rangle\!\rangle$, where $|\,\mu, (n,m)\rangle\!\rangle(t) = e^{i\mu\omega_d t}|n,m\rangle$ maps time to basis of the static dispersive Hamiltonian. The unperturbed component $\overline{Q}_0$ is diagonal in the basis $|\,\mu, (n,m)\rangle\!\rangle$:

$$\overline{Q}_0 = \sum_{\mu,n,m} \left(\bar{\omega}_b n + \frac{K}{2} n(n-1) + \bar{\omega}_c m + \chi n m + \mu\omega_d \right) |\,\mu, (n,m)\rangle\!\rangle\langle\!\langle \mu, (n,m)\,|, \tag{C.19}$$

$$\overline{V} = \Omega_0 \sum_{\mu,n,m} \sqrt{n+1}\,\left(|\,\mu+1, (n+1,m)\rangle\!\rangle\langle\!\langle \mu, (n,m)\,| + |\,\mu-1, (n+1,m)\rangle\!\rangle\langle\!\langle \mu, (n,m)\,|\right) + \text{h.c.} \tag{C.20}$$

The perturbation $\overline{V}$ arises from the periodic drive,

$$2\Omega_0 \cos(\omega_d t) = \Omega_0\left(e^{i\omega_d t} + e^{-i\omega_d t}\right), \tag{C.21}$$

which mixes adjacent photon sectors. In the extended (Floquet) Hilbert space, the Fourier components $e^{\pm i\omega_d t}$ connect adjacent sectors $\mu \to \mu \mp 1$, while the drive excites FTT from $n \to n+1$. These processes have detunings

$$\Delta E_1 = (\bar{\omega}_b - \omega_d)n + (n-1)K + \chi m, \tag{C.22}$$

$$\Delta E_2 = (\bar{\omega}_b + \omega_d)n + (n-1)K + \chi m, \tag{C.23}$$

respectively. For our parameter regime of interest $|\bar{\omega}_b - \omega_d| \ll \omega_d$, we have parameters hierarchy

$$\Omega_0 < \Delta E_1 \ll \Delta E_2. \tag{C.24}$$

so that $\overline{V}$ can be treated perturbatively.

C.1.3 Approximation of Quasienergy and Floquet States

To show eigenvalue of H_R can approximate quasienergies, we block-diagonalize the operator $\bar{Q}$ via a Schrieffer–Wolff transformation within the near-resonant subspace selected by the hierarchy in Eq. (C.24). Specifically, we define the projectors

$$\bar{P}_\mu = \sum_{n,m} |\mu - n, n, m\rangle \langle \mu - n, n, m| \,, \tag{C.25}$$

which collect the states coupled most strongly by the drive. Since $\bar{P}_\mu$ mixes different μ indices, it is convenient to relabel the basis with the unitary

$$\bar{R} = \sum_{\mu,n,m} |\mu, n, m\rangle \langle \mu - n, n, m| \,. \tag{C.26}$$

In this relabeled frame,

$$\bar{P}^R_\mu = \bar{R}^\dagger \bar{P}_\mu \bar{R} = \sum_{n,m} |\mu, n, m\rangle \langle \mu, n, m| \,, \tag{C.27}$$

$$\bar{Q}_R = \bar{R}^\dagger \bar{Q}\, \bar{R}, \tag{C.28}$$

so that $\bar{Q}_R$ acquires a block structure in the fixed-μ sectors:

$$\bar{q}_R = \begin{pmatrix} \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \\ \cdots & [H_R - 2\omega_d I] & 0 & [b] & 0 & 0 & \cdots \\ \cdots & 0 & [H_R - \omega_d I] & 0 & [b] & 0 & \cdots \\ \cdots & [b^\dagger] & 0 & [H_R] & 0 & [b] & \cdots \\ \cdots & 0 & [b^\dagger] & 0 & [H_R + \omega_d I] & 0 & \cdots \\ \cdots & 0 & 0 & [b^\dagger] & 0 & [H_R + 2\omega_d I] & \cdots \\ \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \tag{C.29}$$

where $[O]$ denotes the matrix of O in the $\{|n, m\rangle\}$ basis. To first order, the effective quasienergy operator is given by the block-diagonal part,

$$\sum_\mu \bar{P}^R_\mu \, \bar{Q}_R \, \bar{P}^R_\mu. \tag{C.30}$$

It therefore suffices to diagonalize a single sector (e.g., $\mu = 0$), denoted by H_R:

$$H_R |\psi_j\rangle = E_j |\psi_j\rangle, \qquad |\psi_j\rangle = \sum_{n,m} c^j_{nm} |n, m\rangle. \tag{C.31}$$

The quasienergy is then

$$\varepsilon_{j,\mu} = E_j + \mu\omega_d, \tag{C.32}$$

with corresponding Floquet states (in the relabeled frame)

$$|\psi_{j,\mu}\rangle_R = \sum_{n,m} c^j_{nm} |\mu, n, m\rangle. \tag{C.33}$$

C.1.3 Connection with the RWA

The relabeling $\bar{R}$ in extended Hilbert space implements the same rotation as a standard rotating-frame transformation in the physical Hilbert space. Acting on a Floquet basis state and evaluating its time dependence gives

$$\begin{aligned} \big(\bar{R} |\mu, n, m\rangle\big)(t) &= |\mu + n, n, m\rangle (t) = |n, m\rangle \; e^{i(\mu+n)\omega_d t} \\ &= \left(e^{in\omega_d t} |n, m\rangle\right) e^{i\mu\omega_d t}, \end{aligned} \tag{C.34}$$

so $\bar{R}$ amounts to multiplying $|n, m\rangle$ by the phase $e^{in\omega_d t}$. This is precisely the action of the rotating-frame unitary

$$U_R(t) = e^{i\omega_d t\, b^\dagger b}, \tag{C.35}$$

which yields the effective Hamiltonian

$$H_{\text{eff}}(t) = U_R^\dagger(t) H(t) U_R(t) - i\, U_R^\dagger(t) \dot{U}_R(t). \tag{C.36}$$

In this frame, terms oscillating at $\pm 2\omega_d$ are rapidly varying; neglecting them under the rotating-wave approximation produces a time-independent Hamiltonian H_R.

APPENDIX D
HIGHER-ORDER SWEET SPOTS

D.1 Sweet spots from higher-order corrections

Higher-order, drive-induced terms in the Schrieffer–Wolff expansion modify the cavity susceptibility and can generate additional flux sweet spots. To illustrate this effect, we consider the vacuum transition frequency $\widetilde{\omega}_{c,0\to1}$. Retaining terms up to fourth order yields the drive-induced correction

$$\begin{aligned}\widetilde{\omega}_{c,0\to1} = \overline{\omega}_c &+ \frac{\Omega_0^2}{-\Delta_{bd}-\chi} + \frac{\Omega_0^2}{\Delta_{bd}} - \frac{\Omega_0^4}{(-\Delta_{bd}-\chi)^3} \\ &+ \frac{2\Omega_0^4}{(-\Delta_{bd}-\chi)^2(-2\Delta_{bd}-K-2\chi)} - \frac{\Omega_0^4}{\Delta_{bd}^3} \\ &- \frac{2\Omega_0^4}{\Delta_{bd}^2(-2\Delta_{bd}-K)}.\end{aligned} \tag{D.1}$$

The resulting susceptibility D_0 is shown in Fig. D.1. In addition to the sweet spots predicted at leading order, an extra sweet spot appears near $-\Delta_{bd} \simeq K/2$.

This feature originates from near-resonant hybridization between $|g, m\rangle$ and the higher FTT level $|f, m\rangle$. At this detuning, the drive couples states separated by approximately the anharmonicity K, producing an additional Stark shift that alters the cancellation condition for D_0. Operation at this sweet spot may therefore be more sensitive to drive-induced decoherence channels and should be assessed accordingly.

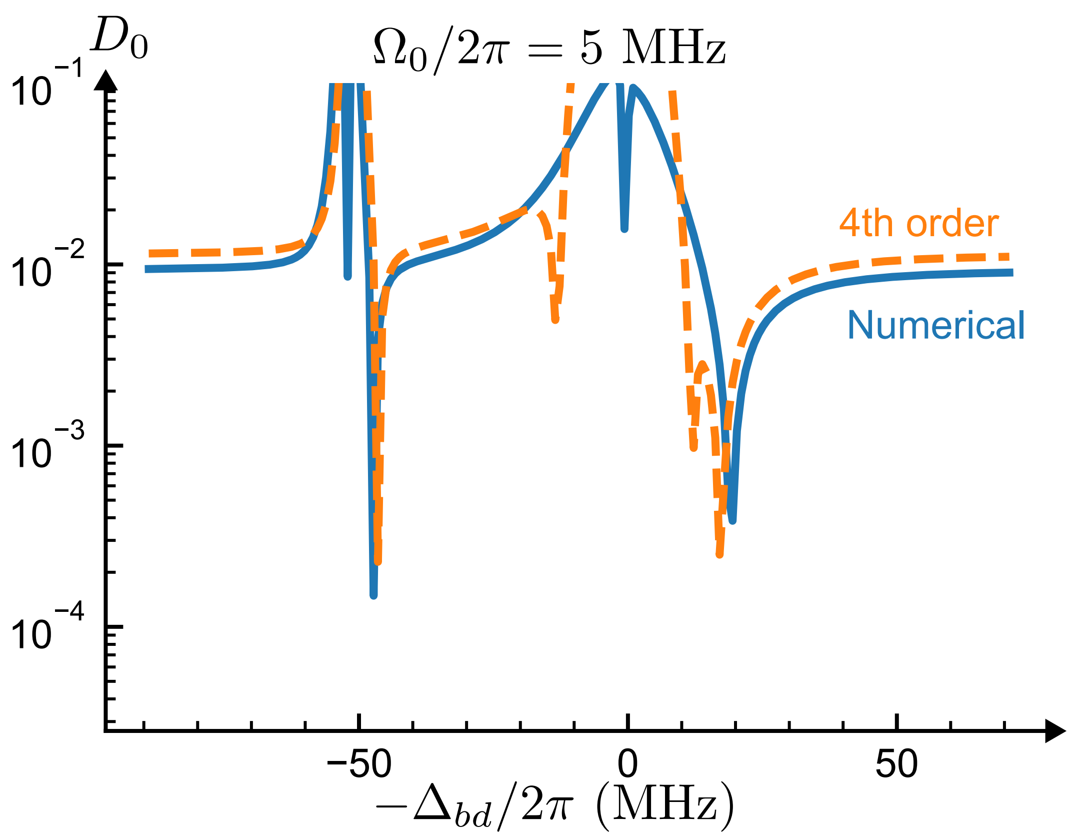

Figure D.1: Susceptibility D_0 including higher-order corrections in the Schrieffer–Wolff expansion. An additional sweet spot appears near $-\Delta_{bd} \simeq K/2$, arising from hybridization between $|g\rangle$ and $|f\rangle$ states

APPENDIX E
SNAIL COUPLER GENERALIZATION

E.1 Example of a SNAIL coupler

The superconducting nonlinear asymmetric inductive element (SNAIL) is a three-wave-mixing circuit element widely used for parametric amplification and gate operations. When coupled capacitively to a cavity, the system Hamiltonian can be written as

$$
\begin{aligned}
H = \omega_s s^\dagger s + \sum_j g_j (s + s^\dagger)^j + \omega_c c^\dagger c \\
+ g(s + s^\dagger)(c + c^\dagger) + 2\Omega_0 \cos(\omega_d t)(s + s^\dagger).
\end{aligned}
\tag{E.1}
$$

Here ω_s is the SNAIL mode frequency, g_j are the nonlinear coefficients, g is the SNAIL–cavity coupling strength, s (c) is the SNAIL (cavity) mode operator, and ω_d is the drive frequency. The parameters ω_s, g_j, g, and Ω_0 are flux dependent through Φ.

To illustrate the generality of the drive-induced cancellation mechanism, the SNAIL is biased at $\Phi = 0.3\,\Phi_0$, where the cubic nonlinearity g_3 enables three-wave mixing. At this operating point, $g_3/2\pi \approx 67$ MHz, $\omega_s/2\pi = 6.59$ GHz, and $\omega_c/2\pi = 5$ GHz.

A coherent drive with amplitude $\Omega_0 = 3$ MHz at $\omega_d = 6.5$ GHz produces a flux sweet spot in the cavity spectrum. As shown in Fig. E.1, the cavity transition energy, extracted using the procedure in App. 7.2, develops a local maximum near $\Phi \simeq 0.311\,\Phi_0$, consistent with vanishing first-order flux susceptibility. The perturbative expression in Eq. (6.57) reproduces this sweet spot quantitatively, demonstrating that the drive-induced cancellation mechanism extends beyond the FTT to more general nonlinear, flux-dependent modes.

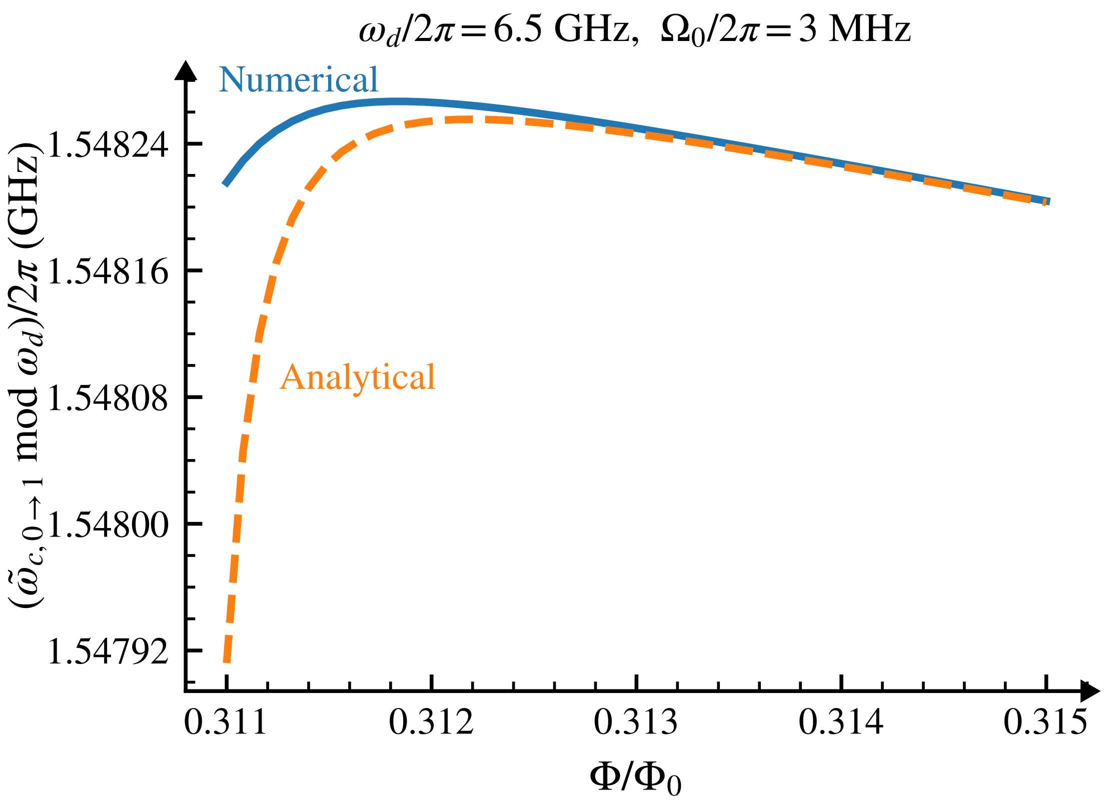

Figure E.1: Drive-induced sweet spot in a SNAIL–cavity system. Under continuous drive, the cavity transition energy exhibits a maximum near $\Phi \simeq 0.311\,\Phi_0$, indicating reduced first-order flux susceptibility. Solid lines show numerical results and dashed lines show the perturbative prediction from Eq. (6.57).

APPENDIX F
NUMERICAL DETAILS

F.1 Numerical details of obtaining dephasing rates

This section details the numerical procedure used to extract the dephasing rates in Sec. 7.1.3. The dynamics are simulated with the full Hamiltonian $H_{\rm full}(\delta\Phi)$ represented in a truncated oscillator (Fock) basis and evolved under the Lindblad master equation

$$\dot{\rho} = -i\big[\, H_{\rm full}\big(\delta\Phi(t)\big),\ \rho\,\big] + \gamma_{b\downarrow}\, \mathcal{D}[b]\rho, \tag{F.1}$$

where $\mathcal{D}[b]\rho = b\rho b^\dagger - \frac{1}{2}\{b^\dagger b, \rho\}$.

Generation of $1/f$ noise. Flux noise is incorporated through stochastic trajectories $\delta\Phi(t)$ whose spectrum follows a $1/f$ power spectral density. For each trajectory, Eq. (F.1) is integrated to obtain $\rho(t)$, and all reported curves are computed from ensemble averages of the corresponding observables. The noise realizations $\delta\Phi(t)$ are generated using the frequency-domain synthesis method of Ref. [172]: Fourier components are assigned random phases and Gaussian-distributed amplitudes with variances proportional to the target $1/f$ spectrum, followed by an inverse Fourier transform to obtain $\delta\Phi(t)$ in the time domain. Fig. F.1 shows the resulting power spectral densities for 100 representative trajectories. This ensemble size is sufficient for the quantities of interest to converge, as indicated by the small error bars in Fig. 7.1(a) and Fig. F.2(b).

The total simulated evolution time is $T \sim 10\ \mu$s, and the infrared cutoff is set to $\omega_{\rm IR}/2\pi = 10$ kHz (equivalently $f_{\rm min} = 10^{-5}$ GHz). This choice satisfies $f_{\rm min} \ll 1/T \approx 100$ kHz, cap-

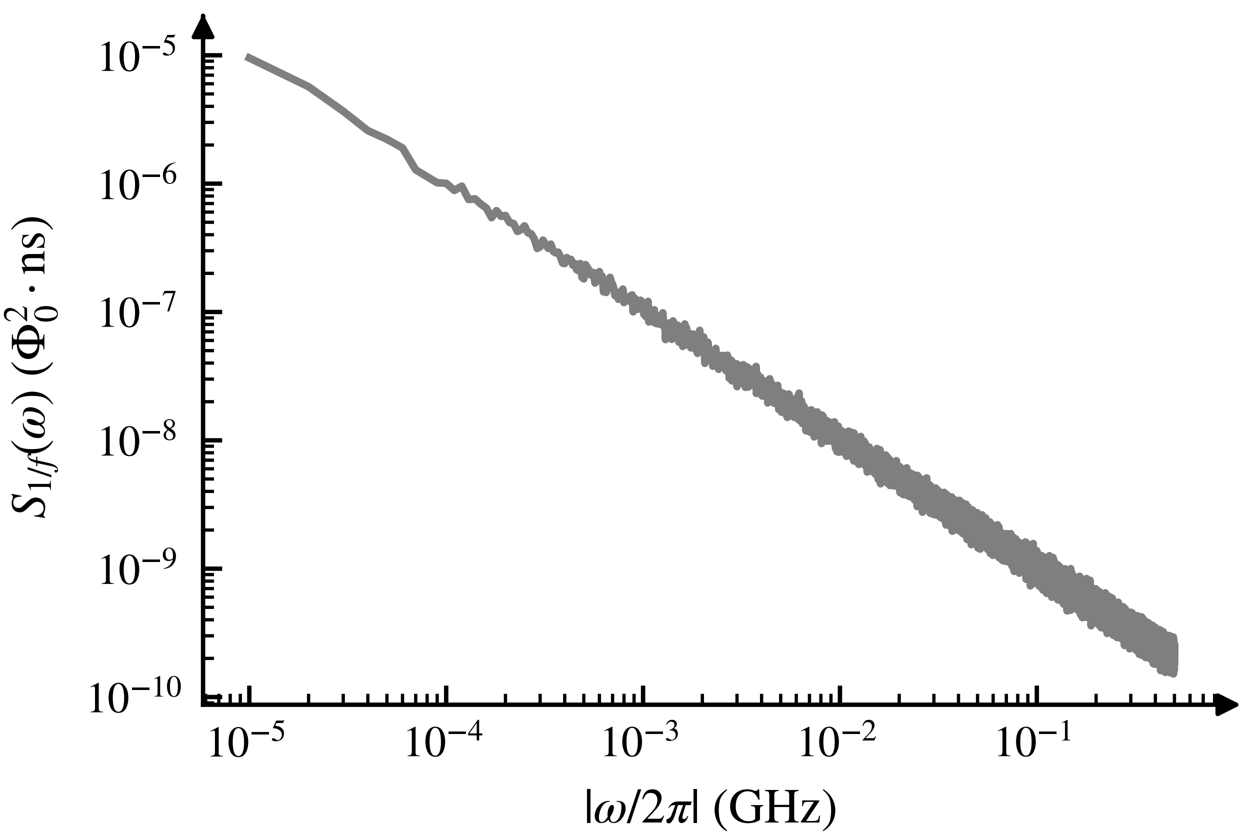

Figure F.1: Power spectral density of $\delta\Phi(t)$ for 100 representative $1/f$ noise trajectories generated with the cutoffs specified below.

turing the relevant slow-noise dynamics within the simulated window while avoiding the infrared divergence of an ideal $1/f$ spectrum. The ultraviolet cutoff is set to $f_{\max} = 1$ GHz; the extracted dephasing rates are insensitive to further increases of $f_{\max}$.

Extraction of dephasing time. The states with enhanced dephasing discussed in main text are proxies for the corresponding Floquet states, as shown in Appendix C.1. In the numerics, Floquet modes are labeled by their correspondence to bare states as described in Appendix 7.2. The two Floquet states used for the cavity dephasing analysis in Sec. 7.1.3 are denoted $|\phi_{g,0}(t)\rangle$ and $|\phi_{g,1}(t)\rangle$.

Relaxation and dephasing are extracted from dynamics evolving from two initial states. Relaxation is characterized by preparing

$$\rho(0) = |\phi_{g,1}(0)\rangle\langle\phi_{g,1}(0)|$$

and monitoring the return toward $|\phi_{g,0}(t)\rangle$. Dephasing is characterized by preparing the equal superposition

$$\rho(0) = |+\rangle\langle+| \,, \qquad |+\rangle = \frac{1}{\sqrt{2}}\Big(\, |\phi_{g,0}(0)\rangle + |\phi_{g,1}(0)\rangle \,\Big),$$

and tracking the decay of the corresponding coherence. The relevant quantities are

$$\mathrm{Tr}[\rho(t)\, P_{g,0}(t)] \quad \text{and} \quad \mathrm{Tr}[\rho(t)\, X(t)] \,,$$

where

$$P_{g,0}(t) = |\phi_{g,0}(t)\rangle\langle\phi_{g,0}(t)| \,, X(t) = |\phi_{g,1}(t)\rangle\langle\phi_{g,0}(t)| + \text{H.c.}$$

Relaxation and dephasing times are obtained by fitting ensemble-averaged signals. Specifically,

T_1 is extracted from the rise of the ground-state population by fitting $\mathrm{Tr}[P_{g,0}(t)\,\overline{\rho}(t)]$ to $1-e^{-t/T_1}$, where $\overline{\rho}(t)$ denotes the ensemble-averaged density matrix over different flux noise trajectories. The coherence time T_2 is extracted from the decay of the superposition signal by fitting $\mathrm{Tr}[P_+(t)\,\overline{\rho}(t)]$ to e^{-t/T_2} (with the appropriate normalization convention for the plotted data). The pure dephasing time then follows from

$$\frac{1}{T_\phi} = \frac{1}{T_2} - \frac{1}{2T_1}, \tag{F.2}$$

and we report the total dephasing rate as $\Gamma_{\phi,\mathrm{total}} = 1/T_\phi$. The data in Fig. 7.1 use the same system parameters and level truncations as Appendix 7.2. Unless otherwise stated, the fitting windows are $1\ \mu$s for T_1 and $5\ \mu$s for T_2; extending these windows does not change the extracted rates.

F.2 Rates in the unselective regime

F.2.1 Dephasing rates

This subsection benchmarks the analytical dephasing rates in Eq. (7.5) against numerical simulations. The numerical reference rate for $\gamma_{b\uparrow}^{\mathrm{un}} + \gamma_{c\phi}^{\mathrm{un}}$ is extracted using the procedure in App. F.1, with the flux-noise drive set to $\delta\Phi(t) = 0$.

In the large-detuning regime, the reference rate is well captured by the analytical approximation in Fig. F.2(a):

$$\gamma_{b\uparrow}^{\mathrm{un}} + \gamma_{c\phi}^{\mathrm{un}} \tag{F.3}$$

$$= \frac{\gamma_{b\downarrow}}{16}\left(\frac{\Delta_{bd}}{K}\right)^2\left(1-\frac{2\Delta_{bd}}{K}\right) + \frac{\gamma_{b\downarrow}}{2}\left(\frac{g}{\Delta_{bc}}\right)^4\frac{K}{\Delta_{bd}}. \tag{F.4}$$

In this limit, dephasing is dominated by photon-shot noise from heating, i.e., $\gamma_{b\uparrow}^{\mathrm{un}}$, while $\gamma_{c\phi}^{\mathrm{un}}$ is parametrically suppressed at large detuning, consistent with Eq. (F.4).

As the detuning decreases, the numerical and analytical results deviate because Eq. (F.4) assumes $|\Delta_{bd}/\chi| \gg 1$ (here $\chi/2\pi \simeq -2\,\mathrm{MHz}$). In this regime, the total dephasing rate increases and $\gamma_{c\phi}^{\mathrm{un}}$ becomes the dominant contribution.

Finally, the flux-noise-induced depolarization rate is benchmarked numerically against

$$\gamma_{b\uparrow}^{1/f,\mathrm{un}} = \left(\frac{\partial \omega_b}{\partial \Phi}\right)^2 \frac{S_0^2}{4K}. \tag{F.5}$$

The simulation is initialized in the Floquet ground state $|\phi_{g0}(0)\rangle$, and the excitation rate is extracted by fitting the average overlap $\mathrm{Tr}[\rho\,|\phi_{g0}(0)\rangle\langle\phi_{g0}(0)|]$ to $\frac{1}{2}\left(1 - e^{-2\gamma_{b\uparrow}^{1/f,\mathrm{un}} t}\right)$. Sweeping the noise amplitude S_0 yields a quadratic scaling, and the numerical results follow the analytical prediction (black dashed line) closely [Fig. F.2(b)]. The small error bars in the figure indicate statistical convergence for the chosen ensemble size ($N = 100$ trajectories).

F.2.2 Decay rates

This subsection shows that the cavity decay rate is only weakly affected by the drive. For the parameters used to generate Fig. 7.1, the undriven decay rate is $4.75 \times 10^{-7}\,\mathrm{ns}^{-1}$. Under drive, the decay rate remains nearly unchanged, as shown in Fig. F.2(c).

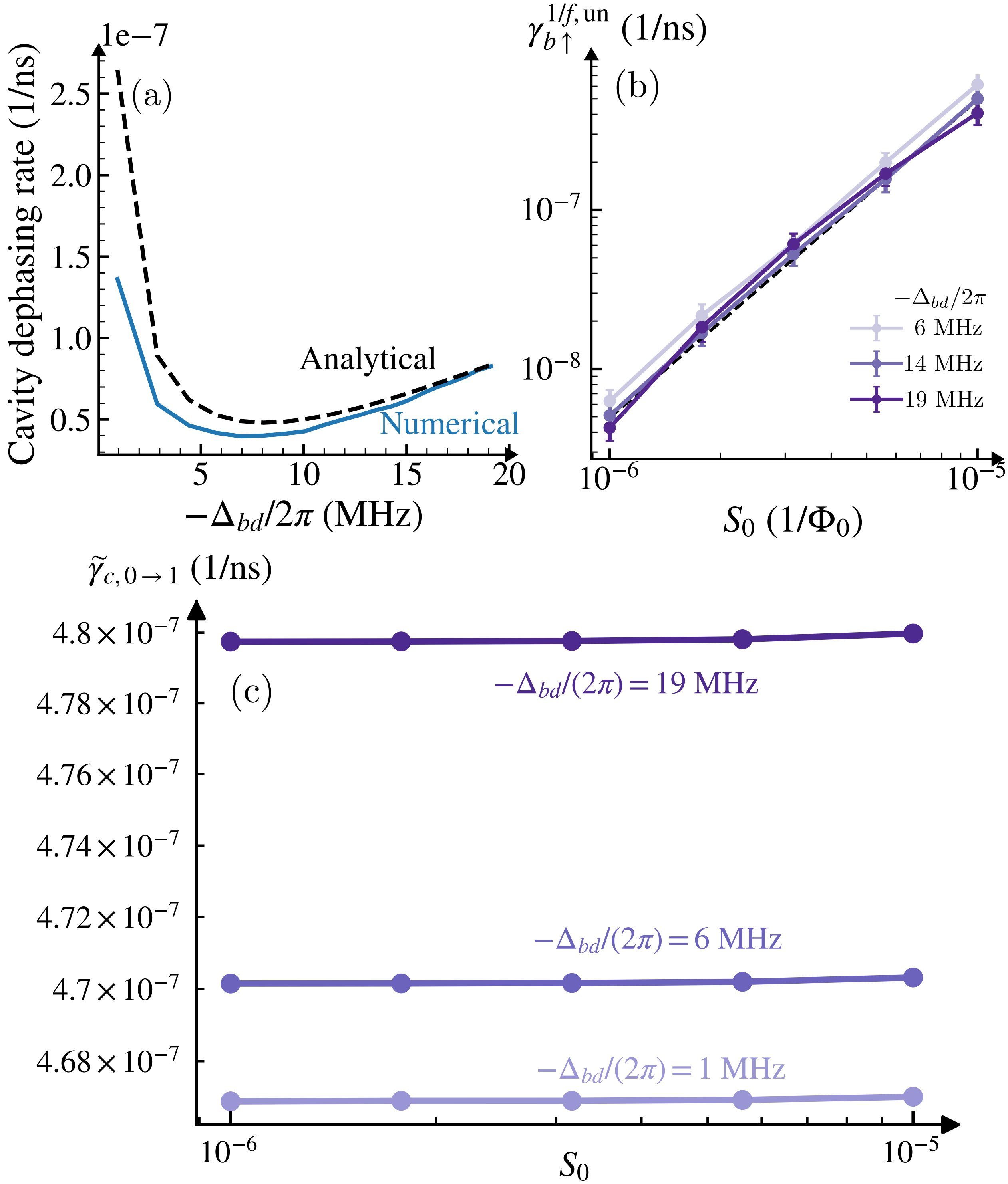

Figure F.2: Numerical validation of rates in the unselective regime. (a) Reference dephasing rate $\gamma_{b\uparrow}^{\mathrm{un}} + \gamma_{c\phi}^{\mathrm{un}}$ extracted from the procedure in App. F.1 with $\delta\Phi(t) = 0$, compared with the analytical approximation in Eq. (F.4). (b) Flux-noise-induced depolarization rate $\gamma_{b\uparrow}^{1/f,\mathrm{un}}$ versus noise amplitude S_0. The quadratic scaling in S_0 and agreement with $\gamma_{b\uparrow}^{1/f,\mathrm{un}} = \left(\frac{\partial\omega_b}{\partial\Phi}\right)^2 \frac{S_0^2}{4K}$ (black dashed line) are observed. (c) Cavity decay rate with sweet-spot drive.